



SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)

Well File No. 25759

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☐ Notice of Intent	Approximate Start Date
☒ Report of Work Done	Date Work Completed December 29, 2017
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☒ Other	NDAC 43-02-03-49 Compliance

Well Name and Number See Attached					
Footages		Qtr-Qtr	Section	Township	Range
F	L	F	L	N	W
Field		Pool		County	

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Continental Resources, Inc. has inspected all the sites listed on the attached list and all are in compliance with NDAC 43-02-03-49. All sites are in compliance based on the fact that our oil storage tanks, flow-through process vessels, recycle pumps, and load lines are all located within secondary containment.

confirmed per RSD

Company Continental Resources, Inc.		Telephone Number (405) 234-9020	
Address PO Box 268870			
City Oklahoma City		State OK	Zip Code 73126
Signature <i>Robert Sandbo</i>	Printed Name Robert Sandbo		
Title Regulatory Compliance Supervisor	Date February 8, 2018		
Email Address robert.sandbo@clr.com			

FOR STATE USE ONLY	
☐ Received	☒ Approved
Date 3/22/18	
By <i>W. J. Anderson</i>	
Title <i>Int. Supervisor</i>	



Oil and Gas Division

Lynn D. Helms - Director

Bruce E. Hicks - Assistant Director

Department of Mineral Resources

Lynn D. Helms - Director

North Dakota Industrial Commission

www.dmr.nd.gov/oilgas

August 29, 2017

CONTINENTAL RESOURCES
ATTENTION: BOB SANDBO
P.O. BOX 268870
OKLAHOMA CITY, OK 73126

RE: North Dakota Administrative Code (NDAC) Section 43-02-03-49
Perimeter Berm Requirement

NDIC # Please see attached list of 103 Facilities

Dear Bob Sandbo:

Please be advised that the attached list of facilities require a perimeter berm to be constructed within 180 days of this notice because they have:

1. Storage tanks;
2. Daily throughput of more than one hundred barrels of fluid per day; and
3. Includes production equipment or load lines that are not contained within secondary containment dikes

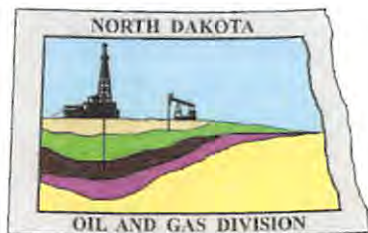
The berm must be at least six inches in height, constructed of sufficiently impermeable material to provide emergency containment, and must be maintained until the facility is either granted a waiver or the site is reclaimed.

Pursuant to NDAC Section 43-02-03-49 - Within one hundred eighty days from the date the operator is notified by the Commission, a perimeter berm, at least six inches in height, must be constructed of sufficiently impermeable material to provide emergency containment and to divert surface drainage away from the site around all storage facilities and production sites that include storage tanks, have a daily throughput of more than one hundred barrels of fluid per day, and include production equipment or load lines that are not contained within secondary containment dikes. The Director may consider an extension of time to implement these requirements if conditions prevent timely construction, or modification of these requirements if other factors are present that provide sufficient protection from environmental impacts.

This perimeter berm requirement may be modified or waived if the operator can demonstrate that other factors are present that provide sufficient protection from environmental impacts. A Sundry Notice (Form 4) outlining any engineering controls or other factors must be submitted to the Commission for approval of this modification or waiver. Should you have any questions regarding this matter, feel free to contact me at 701-770-3554.

Sincerely,


Richard Dunn
Field Inspector



Oil and Gas Division

Lynn D. Helms - Director

Bruce E. Hicks - Assistant Director

Department of Mineral Resources

Lynn D. Helms - Director

North Dakota Industrial Commission

www.dmr.nd.gov/oilgas

NDIC #	Facility Name	TB/CTB #	NDIC #	Facility Name	TB/CTB #
19126	LANSING 1-25H		24834	DURHAM 3X-2H	
19578	MISSOULA 1-21H		24837	WAHPETON 2-16H2	
19740	SYRACUSE 1-23H		24840	WAHPETON 4-16H1	
19858	JAMESTOWN 1-17H		24842	WAHPETON 5-16H2	
19915	STEELE 1-24H		24843	WAHPETON 6-16H	
19918	CHARLOTTE 1-22H		24844	WAHPETON 7-16H3	
20566	MONTPELIER 1-14H		24908	CHARLOTTE 6-22H2	
20629	PATTERSON 1-13H		25116	BJARNE 2-29H	
20638	KUHN 1-12H		25117	BJARNE 3-29H	
20676	NORFOLK 1-1H		25156	COLUMBUS FEDERAL 1-16H	
21128	CHARLOTTE 2-22H		25157	TALLAHASSEE 3-16H	
21511	BOULDER 1-4H		25158	TALLAHASSEE 2-16H	
21600	ROCHESTER 1-24H		25159	COLUMBUS FEDERAL 2-16H	
22155	LANSING 2-25H		25160	COLUMBUS FEDERAL 3-16H	
22158	KUHN 2-12H		25189	PATTERSON FEDERAL 2-13H	
22273	STEELE 2-24H		25190	PATTERSON FEDERAL 3-13H	
22375	CHICAGO 2-26H		25826	AKRON 6-34H1	
22891	MONROE 1-2H		25827	AKRON 5-34H1	
23048	CHICAGO 3-26H		26190	MONTPELIER 4-14H	
23049	CHICAGO 4-26H		26191	MONTPELIER 3-14H1	
23050	SYRACUSE 3-23H		26420	MONTPELIER 2-14H	
23051	SYRACUSE 4-23H		26476	ROCHESTER FEDERAL 6-24H	
23086	NORFOLK 2-1H		26477	ROCHESTER FEDERAL 7-24H1	
23087	NORFOLK 3-1H		26525	JERRY 2-8H	
23351	MISSOULA 2-21H		26526	JERRY 3-8H	
23352	MISSOULA 3-21H		26530	JERRY 5-8H	
23427	MISSOULA 7-21H		26531	JERRY 4-8H	
23428	MISSOULA 6-21H		26535	JERRY 7-8H	
23429	MISSOULA 5-21H		26536	JERRY 6-8H	
23430	MISSOULA 4-21H		27418	GARFIELD FEDERAL 7-5H1	
23477	DURHAM 2-2H		27419	GARFIELD FEDERAL 6-5H	
23608	CHARLOTTE 5-22H		27420	GARFIELD FEDERAL 5-5H1	
23609	AKRON 3-27AH		27421	GARFIELD 4-5H	
23610	AKRON 2-27AH		27694	BERLAIN 3-30H	
23611	AKRON 4-34H		27695	BERLAIN 2-30H	
23612	CHARLOTTE 4-22H		28202	JAMESTOWN FEDERAL 2-17H	
23664	CHARLOTTE 3-22H		28203	JAMESTOWN FEDERAL 3-17H1	
23747	ROCHESTER 3-24H		28405	HARRISBURG 1-34H	
23748	ROCHESTER 2-24H		28604	JAMESTOWN FEDERAL 6-17H	
23749	ROCHESTER 5-24H1		28605	JAMESTOWN FEDERAL 7-17H	
23750	ROCHESTER 4-24H		28735	DURHAM 7-2H	
24490	DURHAM 4-2H		28736	DURHAM 6-2H1	
24491	DURHAM 5-2H		28737	UHLMAN 1-7H	
24507	NORFOLK 5-1H		28999	NORFOLK 6-1H1	
24508	NORFOLK 4-1H		29000	NORFOLK 7-1H	
24804	WAHPETON 14-16H2	224837-01	31508	AKRON FEDERAL 7-27H	
24805	WAHPETON 13-16H		31838	CHARLOTTE 7X-22H	
24806	WAHPETON 12-16H3		32033	NORFOLK 11-1H	
24807	WAHPETON 11-16H1		32034	NORFOLK 10-1H1	
24808	WAHPETON 10-16H2		32035	NORFOLK 9-1H	
24809	WAHPETON 9-16H		32036	NORFOLK 8-1H1	
24810	WAHPETON 8-16H1				

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)

Received

OCT 05 2016

ND Oil & Gas Division

Well File No.

25159

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☐ Notice of Intent	Approximate Start Date	☐ Drilling Prognosis	☐ Spill Report
☒ Report of Work Done	Date Work Completed July 23, 2016	☐ Redrilling or Repair	☐ Shooting
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date	☐ Casing or Liner	☐ Acidizing
		☐ Plug Well	☐ Fracture Treatment
		☐ Supplemental History	☒ Change Production Method
		☐ Temporarily Abandon	☐ Reclamation
		☐ Other	

Well Name and Number Columbus Federal 2-16H					
Footages		Qtr-Qtr	Section	Township	Range
2508 F N L 222 F E L		SENE	16	153 N	101 W
Field	Pool	County			
Baker	Bakken	McKenzie			

24-HOUR PRODUCTION RATE			
Before		After	
Oil	97 Bbls	Oil	82 Bbls
Water	123 Bbls	Water	153 Bbls
Gas	80 MCF	Gas	76 MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Continental Resources, Inc. requests a change in production method on the above mentioned well. The well went from flowing to Rod Pump on 7/23/2016. New Tubing: 2 1/2 Depth: 9940

Company Continental Resources		Telephone Number (405) 234-9688	
Address P.O. Box 268870			
City Oklahoma City		State OK	Zip Code 73126
Signature 	Printed Name Zach Green		
Title Regulatory Specialist	Date September 30, 2016		
Email Address Zach.Green@clr.com			

FOR STATE USE ONLY

☒ Received	☐ Approved
Date 11-2-2016	
By 	
Title TAYLOR ROTH Engineering Technician	

North Dakota Industrial Commission Follow-up Spill Report

API Number 33 - 053 - 04855					Well File or Facility No. 25159				
Operator Continental Resources, Inc.					Telephone Number (406)433-3006				
Address P.O Box 268835					City Oklahoma City		State OK		Zip Code 73126
Well Name and Number or Facility Name Columbus Federal 2-16H					Field Baker				
Location of Well or Facility		Footages F L F L		Qtr-Qtr SE-NE	Section 16	Township 153 N	Range 101 W	County McKenzie	
Description of Spill Location if not on Well or Facility Site and/or Distance and Direction from Well or Facility									
Directions to Site									
Release Discovered By Rig Supervisor		Date Release Discovered February 27, 2014		Time Release Discovered 11 : 15 PM		Date Release Controlled February 27, 2014		Time Release Controlled :	
Company Personnel Notified Patrick Skillestad		How Notified Phone		Date Notified February 28, 2014		Time Notified :			
Type of Incident Tank Overflow			Root Cause of Release Human Error			Date Clean up Activities Concluded February 28, 2014			
Distance to Nearest Residence or Occupied Building			Distance to Nearest Fresh Water Well						
Piping Specifics (If Applicable)		Size (Decimal Format) "	Type			Location of Piping			
Volume of Release		Oil		Saltwater		Other 15.00 Barrels			
Volume of Release Recovered		Oil		Saltwater		Other 15.00 Barrels			
Was Release Contained Within Dike No		If No, Was Release Contained on Well Site Yes		If No, Was Release Contained on Facility Site or Pipeline ROW					
Areal Extent of Release if not Within Dike			Affected Medium Well/Facility Soil			General Land Use Well/Facility Site			
Describe Cause of Release or Fire and Other Type of Incidents, Root Causes of Release, Land Uses, and Released Substances While transferring fresh water into frac tanks, employee overfilled the tank.									
Action Taken to Control Release and Clean Up Action Undertaken All standing fluid was sucked up and returned to tank on-site.									
Potential Environmental Impacts Location soil									
Planned Future Action and/or Action Taken to Prevent Reoccurrence									
Where Were Recovered Liquids Disposed Tank on-site					Where Were Recovered Solids Disposed N/A				
Weather Conditions	Wind Speed MPH	Wind Direction	Temperature ° F	Skies	Estimated Cleanup Cost \$		Damage Value \$		
Regulatory Agencies/Others Notified NDIC/NDDH		Person Notified Online report		Date Notified February 28, 2014		Time Notified 4 : 50 PM		Notified By Patrick Skillestad	
Fee Surface Owner						:			
Local Fire Department						:			
Federal Agency Lease Number						:			
BLM						:			
USFS						:			
Report Originator Nate Stannope			Title Assoc. Environmental Specialist			Date September 26, 2016			
Signature <i>Nate Stannope</i>						Date <i>9/26/16</i>			

Industrial Commission of North Dakota

Oil and Gas Division

Spill / Incident Report

Date/Time Reported : Feb 28 2014 / 16:48

State Agency person :

Responsible Party : CONTINENTAL RESOURCES, INC.

Well Operator : CONTINENTAL RESOURCES, INC.

Date/Time of Incident : 2/27/2014 12:00:00 AM

NDIC File Number : 25159

Facility Number :

Well or Facility Name : COLUMBUS FEDERAL 2-16H

Type of Incident : Tank Overflow

Field Name : BAKER

County : MCKENZIE

Section : 16

Township : 153

Range : 101

Quarter-Quarter :

Quarter :

Distance to nearest residence :

Distance to nearest water well :

Release Oil :

Release Brine :

Release Other : 15 Barrels

Recovered Oil :

Recovered Brine :

Recovered Other : 15 Barrels

Has/Will the incident be reported to the NRC? : No

Was release contained : Yes - On Constructed Well Site

Description of other released substance : Fresh water mixed with biocide.

Immediate risk evaluation : None

Followup Report Requested Y/N : Y

**AUTHORIZATION TO PURCHASE AND TRANSPORT OIL FROM LEASE - FORM 8**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5698 (03-2000)



Well File No.
25159

NDIC CTB No.
125159

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND FOUR COPIES.

Well Name and Number Columbus Federal 2-16H	Qtr-Qtr SENE	Section 16	Township 153 N	Range 101 W	County McKenzie
Operator Continental Resources Inc.	Telephone Number 405-234-9000		Field Baker		
Address P.O. Box 268870	City Oklahoma City		State OK	Zip Code 73126	

Name of First Purchaser Continental Resources, Inc.	Telephone Number 405-234-9000	% Purchased 100	Date Effective December 18, 2013	
Principal Place of Business P.O. Box 268870	City Oklahoma City	State OK	Zip Code 73126	
Field Address	City	State	Zip Code	

Name of Transporter Hiland Crude, LLC	Telephone Number 580-616-2053	% Transported	Date Effective December 18, 2013	
Address P.O. Box 3886	City Enid	State OK	Zip Code 73702	

The above named producer authorizes the above named purchaser to purchase the percentage of oil stated above which is produced from the lease designated above until further notice. The oil will be transported by the above named transporter.

Other First Purchasers Purchasing From This Lease	% Purchased	Date Effective
Other First Purchasers Purchasing From This Lease	% Purchased	Date Effective
Other Transporters Transporting From This Lease	% Transported	Date Effective
Other Transporters Transporting From This Lease	% Transported	Date Effective
Comments		

I hereby swear or affirm that the information provided is true, complete and correct as determined from all available records.		Date December 23, 2013
Signature <i>Becky Barnes</i>	Printed Name Becky Barnes	Title Regulatory Compliance Specialist
Above Signature Witnessed By		
Witness Signature <i>Christi Scritchfield</i>	Witness Printed Name Christi Scritchfield	Witness Title Regulatory Compliance Specialist

FOR STATE USE ONLY

Date Approved MAY 27 2014
By <i>Annita Taylor</i>
Title Oil & Gas Production Analyst



WELL COMPLETION OR RECOMPLETION REPORT - FORM 6

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 2468 (04-2010)

Well File No.
25159

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

Designate Type of Completion			
☒ Oil Well	☐ EOR Well	☐ Recompletion	☐ Deepened Well
☐ Gas Well	☐ SWD Well	☐ Water Supply Well	☐ Other:
Well Name and Number Columbus Federal 2-16H		Spacing Unit Description Sec 4,9,16 & 21-153N-101W	
Operator Continental Resources, Inc.		Telephone Number 405-234-9000	
Address P.O. Box 268870		Field Baker	
City Oklahoma City		State Ok	Zip Code 73126
		Permit Type ☐ Wildcat ☒ Development ☐ Extension	

LOCATION OF WELL

At Surface 2508 F N L	222 F E L	Qtr-Qtr SENE	Section 16	Township 153 N	Range 101 W	County McKenzie
Spud Date 8/12/2013	Date TD Reached 12/13/2013	Drilling Contractor and Rig Number Cyclone 4		KB Elevation (Ft) 1940	Graded Elevation (Ft) 1920	
Type of Electric and Other Logs Run (See Instructions) CBL/GR, Mud Logs, Geological Report						

CASING & TUBULARS RECORD (Report all strings set in well)

Well Bore	String Type	Size (Inch)	Top Set (MD Ft)	Depth Set (MD Ft)	Hole Size (Inch)	Weight (Lbs/Ft)	Anchor Set (MD Ft)	Packer Set (MD Ft)	Sacks Cement	Top of Cement
Lateral1	Conductor	16		100	20				8 yds	
	Surface	9 5/8		1965	13 1/2	36			868	
	Intermediate	7		10850	8 3/4	29-32			1087	1922
	Liner	4 1/2	10071	23274	6	11.6				
	Tubing	2 7/8			4	6.5				

PERFORATION & OPEN HOLE INTERVALS

Well Bore	Well Bore TD Drillers Depth (MD Ft)	Completion Type	Open Hole/Perforated Interval (MD,Ft) Top Bottom	Kick-off Point (MD Ft)	Top of Casing Window (MD Ft)	Date Perf'd or Drilled	Date Isolated	Isolation Method	Sacks Cement
Lateral1	13133	Perforations	10850 13133	10135	1922	2/27/2014			
Sidetrack1	23475	Open Hole	12488 23475	12488					

PRODUCTION

Current Producing Open Hole or Perforated Interval(s), This Completion, Top and Bottom, (MD Ft) Bakken 10,850'/23,475'						Name of Zone (If Different from Pool Name) Three Forks				
Date Well Completed (SEE INSTRUCTIONS) 5/3/2014			Producing Method Flowing		Pumping-Size & Type of Pump			Well Status (Producing or Shut-In) Producing		
Date of Test 5/4/2014	Hours Tested 24	Choke Size 16 /64	Production for Test		Oil (Bbls) 541	Gas (MCF) 387	Water (Bbls) 815	Oil Gravity-API (Corr.) 40.0 °	Disposition of Gas Sold	
Flowing Tubing Pressure (PSI) 1650		Flowing Casing Pressure (PSI) N/A		Calculated 24-Hour Rate	Oil (Bbls) 541	Gas (MCF) 387	Water (Bbls) 815	Gas-Oil Ratio 715		

GEOLOGICAL MARKERS

[illegible]

PLUG BACK INFORMATION

[illegible]

CORES CUT

Top (Ft)	Bottom (Ft)	Formation	Top (Ft)	Bottom (Ft)	Formation

Drill Stem Test

Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								

Well Specific Stimulations

Date Stimulated 2/27/2014	Stimulated Formation Three Forks	Top (Ft) 10850	Bottom (Ft) 23475	Stimulation Stages 43	Volume 85599	Volume Units Barrels
Type Treatment Sand Frac	Acid %	Lbs Proppant 3937842	Maximum Treatment Pressure (PSI) 8493		Maximum Treatment Rate (BBLS/Min) 31.0	
Details Pumped 305478# 40/70 mesh, 2539056# 20/40 sand and 1093308# 20/40 ceramic.						

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						

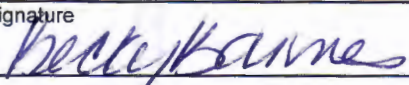
Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						

ADDITIONAL INFORMATION AND/OR LIST OF ATTACHMENTS

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I hereby swear or affirm that the information provided is true, complete and correct as determined from all available records.	Email Address becky.barnes@clr.com	Date 4/15/2014
	Signature 	Printed Name Becky Barnes

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SYNOPSIS

OPERATOR	Continental Resources Inc.
WELL NAME	Columbus Federal 2-16H
SURFACE LOCATION	2508' FNL & 222' FEL: Sec 16, T153N, R101W
BOTTOM HOLE LOCATION	236.1 FNL & 1348.94' FEL: Sec 4, T153N, R101W
FIELD	Williston
COUNTY/STATE	McKenzie Co., North Dakota
API NUMBER	33-053-04855
NorAm FILE NUMBER	NR2689
ELEVATIONS	G.L. = 1920' K.B. = 1940'
SPUD DATE	October 25, 2013
T.D. DATE	December 13, 2013
DRILLING CONTRACTOR	Cyclone 4
HOLE SIZE	10 3/4" to 1965', 8 3/4" to 10877' , 6" to 23475'
CASING SIZE & DEPTH	9 5/8" to 1965', 7" to 10847 , 5" to 23274'
DRILLING MUD COMPANY	GEO Drilling Fluids Inc
DRILLING FLUID TYPE	Invert OBM to 10877', Saltwater Brine to 23475'
DIRECTIONAL COMPANY	Leam Energy Services
WIRELINE LOGGING COMPANY	N/A
LOG RECORD	NorAm Wellsite Services
DRILLING SUPERVISION	Steve Northern, Biggens, Monty Harris
GEOLOGICAL SUPERVISION	Darcy Klessens, NorAm Wellsite Services
MUDLOGGING COMPANY	NorAm Wellsite Services
TOTAL DEPTH	23,475'

FORMATION TOPS (ft)**KB =1940'****GL =1920'**

FORMATION	PROGNOSIS			SAMPLES		
	MD	TVD	SS	MD	TVD	SS
Top Charles Salts	-	8329	-6389	8298	8296	-6356
Base Last Salt	-	9025	-7085	8299	8297	-7057
Mission Canyon	-	9250	-7310	9222	9219	-7279
Lodgepole	-	9817	-7877	9792	9790	-7850
False Bakken	-	-	-	10537	10483	-8543
Upper Bakken Shale	-	10505	-8565	10550	10490	-8550
Middle Bakken	-	10522	-8582	10576	10505	-8565
Lower Bakken Shale	-	10557	-8617	10641	10538	-8598
Three Forks	-	10583	-8643	10726	10568	-8628
7" Casing Point	-	-	-	10887	10586	-8646
End of Lateral	-	10603	-8663	10475	10556	-8616

DEVIATION SURVEY RECORD (Teledrift and Wireline Data)

Depth	Incl	Azim	TVD	VS	Coordinates		DLS
(ft)	(°)	(°)	(ft)	(ft)	N/S (ft)	E/W (ft)	(°/100')
1918	0.30	262.10	1917.98	0.75	0.75	-5.88	0.38
1983	0.70	256.10	1982.98	1.18	0.63	-6.43	0.62
2077	1.30	241.20	2076.96	0.66	-0.02	-7.93	0.69
2170	1.30	238.70	2169.94	-0.24	-1.08	-9.75	0.06
2263	1.60	243.70	2262.91	-1.18	-2.20	-11.82	0.35
2357	1.70	244.90	2356.87	-2.14	-3.37	-14.26	0.11
2450	2.10	226.30	2449.82	-3.68	-5.14	-16.74	0.78
2543	2.30	211.90	2542.75	-6.24	-7.90	-18.96	0.63
2638	2.20	188.10	2637.68	-9.54	-11.32	-20.22	0.98
2732	2.20	173.20	2731.61	-13.11	-14.90	-20.26	0.61
2825	1.80	154.00	2824.56	-16.25	-17.98	-19.41	0.83
2918	1.10	117.80	2917.53	-18.10	-19.71	-17.98	1.20
3011	1.40	63.50	3010.51	-18.16	-19.62	-16.17	1.26
3105	1.70	61.20	3104.47	-17.18	-18.44	-13.92	0.33
3198	1.60	64.70	3197.44	-16.17	-17.22	-11.54	0.15
3291	1.70	68.40	3290.40	-15.32	-16.16	-9.08	0.16
3384	1.60	72.10	3383.36	-14.63	-15.25	-6.57	0.16
3477	1.30	68.40	3476.33	-14.04	-14.46	-4.35	0.34
3571	1.10	65.10	3570.31	-13.42	-13.69	-2.54	0.23
3664	0.80	76.70	3663.30	-13.02	-13.16	-1.10	0.38
3758	0.60	77.60	3757.29	-12.86	-12.91	0.02	0.21
3851	0.50	74.90	3850.28	-12.73	-12.70	0.89	0.11
3944	0.60	78.40	3943.28	-12.60	-12.49	1.76	0.11
4038	0.40	74.40	4037.28	-12.48	-12.31	2.56	0.22
4132	0.40	89.50	4131.27	-12.44	-12.22	3.20	0.11
4225	0.40	91.40	4224.27	-12.51	-12.22	3.85	0.01
4319	0.40	103.70	4318.27	-12.65	-12.31	4.50	0.09
4412	0.30	102.50	4411.27	-12.82	-12.44	5.05	0.11
4505	0.30	147.30	4504.27	-13.11	-12.69	5.42	0.25
4599	0.40	191.60	4598.26	-13.64	-13.22	5.49	0.30
4692	0.40	214.70	4691.26	-14.20	-13.81	5.24	0.17
4785	0.50	181.80	4784.26	-14.86	-14.48	5.04	0.29
4879	0.50	145.60	4878.26	-15.62	-15.23	5.26	0.33
4972	0.30	118.90	4971.25	-16.11	-15.68	5.70	0.29
5066	0.60	36.80	5065.25	-15.88	-15.40	6.21	0.67
5159	0.90	61.40	5158.24	-15.22	-14.67	7.14	0.47
5253	0.90	67.50	5252.23	-14.70	-14.03	8.47	0.10
5347	1.00	83.00	5346.22	-14.45	-13.65	9.97	0.29
5440	1.20	76.90	5439.20	-14.28	-13.33	11.72	0.25
5533	1.40	65.40	5532.18	-13.76	-12.63	13.70	0.35
5627	1.80	55.40	5626.14	-12.64	-11.32	15.96	0.52

Depth	Incl	Azim	TVD	VS	Coordinates		DLS
(ft)	(°)	(°)	(ft)	(ft)	N/S (ft)	E/W (ft)	(°/100')
5720	1.80	46.60	5719.10	-11.01	-9.48	18.23	0.30
5813	2.00	56.10	5812.05	-9.32	-7.58	20.64	0.40
5907	1.80	57.50	5905.99	-7.84	-5.87	23.24	0.22
6000	1.00	48.00	5998.97	-6.67	-4.54	25.08	0.89
6094	0.90	44.00	6092.95	-5.69	-3.46	26.20	0.13
6187	0.10	18.10	6185.95	-5.14	-2.86	26.73	0.87
6281	0.40	343.70	6279.95	-4.74	-2.46	26.66	0.34
6375	0.30	36.10	6373.95	-4.23	-1.95	26.72	0.34
6469	0.30	336.70	6467.94	-3.82	-1.53	26.77	0.32
6562	0.50	318.70	6560.94	-3.26	-1.00	26.40	0.25
6655	0.40	312.80	6653.94	-2.69	-0.47	25.90	0.12
6749	0.10	343.50	6747.94	-2.37	-0.17	25.63	0.34
6843	0.30	69.10	6841.94	-2.22	0.00	25.84	0.33
6936	0.40	68.90	6934.94	-2.06	0.20	26.37	0.11
7030	0.50	325.90	7028.93	-1.61	0.66	26.44	0.75
7123	1.00	319.60	7121.93	-0.60	1.61	25.69	0.54
7216	1.00	319.80	7214.91	0.73	2.85	24.64	0.00
7309	0.90	312.80	7307.90	1.93	3.97	23.58	0.16
7403	0.80	301.90	7401.89	2.87	4.81	22.48	0.20
7497	0.90	287.30	7495.88	3.54	5.38	21.22	0.25
7590	0.60	274.80	7588.87	3.90	5.64	20.04	0.37
7683	0.60	270.70	7681.87	4.03	5.68	19.07	0.05
7777	0.50	267.80	7775.86	4.10	5.68	18.16	0.11
7870	0.70	252.30	7868.86	3.99	5.49	17.22	0.28
7963	0.80	262.70	7961.85	3.84	5.23	16.03	0.18
8056	0.40	255.60	8054.84	3.76	5.07	15.07	0.44
8149	0.40	252.50	8147.84	3.63	4.89	14.45	0.02
8243	0.30	237.90	8241.84	3.45	4.66	13.93	0.14
8336	0.40	248.60	8334.84	3.25	4.41	13.42	0.13
8429	0.60	252.30	8427.83	3.05	4.15	12.65	0.22
8522	0.60	273.40	8520.83	3.01	4.03	11.70	0.24
8616	0.70	285.00	8614.82	3.28	4.21	10.66	0.18
8709	0.70	297.60	8707.82	3.77	4.62	9.61	0.17
8803	0.80	289.40	8801.81	4.35	5.10	8.48	0.16
8896	1.00	280.10	8894.80	4.83	5.46	7.07	0.27
8990	0.60	286.20	8988.79	5.22	5.74	5.79	0.43
9083	0.40	301.10	9081.78	5.59	6.04	5.04	0.25
9176	0.40	291.30	9174.78	5.92	6.33	4.46	0.07
9270	0.40	288.70	9268.78	6.20	6.55	3.84	0.02
9363	0.40	277.10	9361.78	6.40	6.70	3.21	0.09
9457	0.40	260.40	9455.77	6.44	6.68	2.56	0.12
9550	0.40	253.20	9548.77	6.34	6.53	1.93	0.05

Depth	Incl	Azim	TVD	VS	Coordinates		DLS
(ft)	(°)	(°)	(ft)	(ft)	N/S (ft)	E/W (ft)	(°/100')
9643	0.50	253.50	9641.77	6.20	6.32	1.23	0.11
9736	0.30	222.60	9734.77	5.95	6.03	0.68	0.31
9830	0.30	244.20	9828.77	5.70	5.74	0.29	0.12
9923	0.30	256.70	9921.76	5.57	5.58	-0.16	0.07
10016	0.40	274.30	10014.76	5.59	5.55	-0.72	0.16
10081	0.40	256.90	10079.76	5.59	5.51	-1.17	0.19
10112	0.40	260.70	10110.76	5.57	5.47	-1.38	0.09
10143	2.00	317.50	10141.75	5.99	5.85	-1.86	5.85
10174	7.10	328.00	10172.65	8.13	7.88	-3.24	16.60
10236	16.70	325.40	10233.22	19.49	18.68	-10.24	14.97
10267	21.20	321.20	10262.53	28.02	26.72	-16.28	15.16
10298	24.40	318.70	10291.11	37.83	35.90	-24.02	10.78
10329	26.90	315.20	10319.05	48.37	45.69	-33.19	9.43
10361	29.80	312.00	10347.21	59.73	56.15	-44.21	10.23
10392	31.70	314.30	10373.85	71.53	66.99	-55.76	7.21
10423	36.60	313.80	10399.50	84.65	79.09	-68.27	15.83
10423	36.60	313.80	10399.50	84.65	79.09	-68.27	15.83
10454	40.30	312.40	10423.78	98.97	92.25	-82.35	12.26
10485	44.10	309.80	10446.74	113.94	105.92	-98.05	13.49
10516	47.80	307.80	10468.29	129.32	119.87	-115.41	12.80
10547	51.60	307.30	10488.34	145.28	134.27	-134.16	12.32
10578	56.30	308.40	10506.57	162.30	149.65	-153.94	15.43
10641	62.80	311.90	10538.30	200.81	184.73	-195.49	10.32
10672	66.50	313.10	10551.57	221.44	203.66	-216.14	12.44
10703	70.60	313.80	10562.90	243.00	223.50	-237.08	13.39
10734	75.30	313.50	10571.99	265.22	243.95	-258.52	15.19
10765	79.40	312.40	10578.78	287.64	264.55	-280.66	13.67
10796	84.20	311.00	10583.20	309.94	284.95	-303.56	16.12
10821	87.50	310.80	10585.00	327.82	301.28	-322.40	13.22
10873	89.00	310.80	10586.59	365.03	335.24	-361.75	2.88
10968	88.00	311.90	10589.08	433.65	397.98	-433.04	1.56
11063	89.70	314.70	10590.99	504.47	463.11	-502.15	3.45
11158	91.40	317.30	10590.07	578.20	531.43	-568.13	3.27
11253	92.70	321.40	10586.68	655.25	603.45	-629.97	4.52
11348	91.20	324.40	10583.44	735.60	679.16	-687.23	3.53
11443	90.90	325.90	10581.70	817.91	757.11	-741.50	1.61
11538	90.90	329.50	10580.21	902.25	837.38	-792.25	3.79
11633	90.20	332.30	10579.30	988.90	920.38	-838.45	3.04
11728	88.80	334.90	10580.13	1077.29	1005.46	-880.68	3.11
11823	88.50	338.40	10582.36	1167.38	1092.64	-918.32	3.70
11918	90.20	340.50	10583.44	1258.85	1181.58	-951.66	2.84
12013	90.30	343.00	10583.03	1351.28	1271.80	-981.41	2.63

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(ft)	(°)	(°)	(ft)	(ft)	N/S (ft)	E/W (ft)	(°/100')
12108	90.80	345.60	10582.12	1444.59	1363.24	-1007.11	2.79
12203	90.90	348.30	10580.71	1538.62	1455.77	-1028.56	2.84
12298	90.60	350.50	10579.46	1633.14	1549.13	-1046.03	2.34
12393	88.10	353.40	10580.54	1727.97	1643.17	-1059.33	4.03
12488	89.50	355.60	10582.53	1822.94	1737.71	-1068.43	2.74
12583	85.30	355.60	10586.84	1917.81	1832.31	-1075.71	4.42
12678	88.20	354.60	10592.22	2012.65	1926.80	-1083.82	3.23
12772	90.50	355.80	10593.29	2106.64	2020.45	-1091.68	2.76
12868	90.80	359.00	10592.20	2202.54	2116.34	-1096.03	3.35
12962	90.00	0.20	10591.54	2296.24	2210.33	-1096.69	1.53
13057	88.30	1.10	10592.95	2390.78	2305.31	-1095.61	2.02
13152	88.80	0.70	10595.36	2485.26	2400.27	-1094.12	0.67
13247	89.50	0.70	10596.77	2579.79	2495.25	-1092.96	0.74
13342	88.30	0.70	10598.59	2674.31	2590.22	-1091.80	1.26
13434	89.50	1.30	10600.36	2765.81	2682.19	-1090.19	1.46
13526	89.30	1.40	10601.32	2857.25	2774.16	-1088.03	0.24
13618	88.40	1.60	10603.17	2948.65	2866.11	-1085.62	1.00
13710	88.40	1.60	10605.74	3040.02	2958.03	-1083.05	0.00
13802	88.10	1.60	10608.54	3131.38	3049.96	-1080.48	0.33
13894	89.40	2.30	10610.55	3222.70	3141.88	-1077.35	1.60
13986	90.10	1.10	10610.95	3314.08	3233.83	-1074.63	1.51
14078	90.50	1.30	10610.47	3405.56	3325.81	-1072.70	0.49
14169	88.80	1.10	10611.03	3496.03	3416.79	-1070.79	1.88
14261	90.20	359.90	10611.83	3587.62	3508.78	-1069.99	2.00
14354	88.40	359.30	10612.97	3680.31	3601.76	-1070.64	2.04
14446	91.00	359.50	10613.45	3772.04	3693.75	-1071.60	2.83
14538	89.40	358.80	10613.13	3863.81	3785.73	-1072.97	1.90
14631	88.70	357.80	10614.67	3956.65	3878.68	-1075.73	1.31
14723	91.50	359.20	10614.51	4048.47	3970.64	-1078.13	3.40
14816	91.30	358.60	10612.24	4141.24	4063.59	-1079.92	0.68
14908	91.30	358.50	10610.15	4233.04	4155.54	-1082.25	0.11
15001	90.40	0.20	10608.77	4325.77	4248.52	-1083.30	2.07
15093	91.40	0.20	10607.32	4417.39	4340.50	-1082.98	1.09
15186	89.30	0.70	10606.76	4509.98	4433.49	-1082.25	2.32
15279	89.60	0.70	10607.65	4602.53	4526.48	-1081.11	0.32
15372	91.10	359.00	10607.08	4695.20	4619.47	-1081.36	2.44
15465	91.70	358.80	10604.81	4787.96	4712.43	-1083.14	0.68
15557	90.00	359.50	10603.44	4879.71	4804.40	-1084.51	2.00
15649	90.80	359.70	10602.80	4971.43	4896.40	-1085.15	0.90
15741	91.30	359.50	10601.12	5063.12	4988.38	-1085.79	0.59
15835	90.70	359.30	10599.47	5156.84	5082.36	-1086.77	0.67
15927	89.90	359.90	10598.99	5248.55	5174.36	-1087.42	1.09

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(ft)	(°)	(°)	(ft)	(ft)	N/S (ft)	E/W (ft)	(°/100')
16020	90.30	359.90	10598.83	5341.22	5267.36	-1087.58	0.43
16115	89.40	359.70	10599.08	5435.90	5362.35	-1087.91	0.97
16210	89.00	359.30	10600.41	5530.61	5457.34	-1088.74	0.60
16306	89.20	358.60	10601.91	5626.38	5553.31	-1090.50	0.76
16401	90.50	0.00	10602.16	5721.11	5648.30	-1091.66	2.01
16496	91.30	359.90	10600.67	5815.76	5743.29	-1091.74	0.85
16590	90.50	359.20	10599.19	5909.46	5837.27	-1092.48	1.13
16685	91.30	0.00	10597.70	6004.15	5932.26	-1093.14	1.19
16780	91.20	359.00	10595.63	6098.85	6027.23	-1093.97	1.06
16875	92.10	358.50	10592.89	6193.61	6122.17	-1096.04	1.08
16970	92.00	357.60	10589.50	6288.42	6217.05	-1099.27	0.95
17065	92.10	358.10	10586.10	6383.25	6311.92	-1102.84	0.54
17160	92.10	356.90	10582.62	6478.10	6406.77	-1106.98	1.26
17255	91.50	359.30	10579.63	6572.91	6501.66	-1110.12	2.60
17350	92.90	0.90	10575.98	6667.47	6596.58	-1109.96	2.24
17445	90.70	0.90	10573.00	6761.93	6691.52	-1108.47	2.32
17540	90.80	0.40	10571.76	6856.47	6786.50	-1107.39	0.54
17635	90.00	359.50	10571.09	6951.13	6881.50	-1107.47	1.27
17730	90.00	359.70	10571.09	7045.83	6976.50	-1108.14	0.21
17824	90.00	359.00	10571.09	7139.57	7070.49	-1109.20	0.74
17919	89.90	358.10	10571.18	7234.39	7165.46	-1111.61	0.95
18014	90.00	358.30	10571.26	7329.25	7260.41	-1114.59	0.24
18109	91.40	0.00	10570.10	7424.00	7355.39	-1116.00	2.32
18204	90.00	359.30	10568.94	7518.69	7450.38	-1116.58	1.65
18299	92.20	0.20	10567.11	7613.35	7545.35	-1116.99	2.50
18394	90.70	0.60	10564.71	7707.91	7640.32	-1116.33	1.63
18489	89.20	0.60	10564.79	7802.46	7735.31	-1115.34	1.58
18584	89.30	0.70	10566.04	7897.01	7830.29	-1114.26	0.15
18679	90.20	0.70	10566.45	7991.55	7925.29	-1113.10	0.95
18774	90.00	0.90	10566.29	8086.08	8020.28	-1111.77	0.30
18869	88.90	0.60	10567.20	8180.60	8115.26	-1110.53	1.20
18964	89.80	1.30	10568.28	8275.10	8210.24	-1108.95	1.20
19059	90.60	0.60	10567.94	8369.60	8305.23	-1107.38	1.12
19154	90.70	1.10	10566.87	8464.11	8400.21	-1105.97	0.54
19249	89.50	0.20	10566.70	8558.66	8495.20	-1104.89	1.58
19344	90.60	1.80	10566.62	8653.15	8590.18	-1103.23	2.04
19439	90.20	0.70	10565.95	8747.59	8685.15	-1101.16	1.23
19534	92.20	0.20	10563.96	8842.15	8780.13	-1100.42	2.17
19629	92.70	359.90	10559.90	8936.71	8875.04	-1100.33	0.61
19724	90.70	0.40	10557.09	9031.29	8969.99	-1100.08	2.17
19819	89.40	0.90	10557.00	9125.84	9064.98	-1099.01	1.47
19914	90.00	0.70	10557.50	9220.36	9159.97	-1097.68	0.67

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(ft)	(°)	(°)	(ft)	(ft)	N/S (ft)	E/W (ft)	(°/100')
20009	90.90	0.40	10556.75	9314.93	9254.96	-1096.77	1.00
20104	89.30	0.70	10556.59	9409.49	9349.96	-1095.86	1.71
20199	88.50	0.60	10558.41	9504.02	9444.93	-1094.78	0.85
20293	88.40	0.60	10560.95	9597.55	9538.89	-1093.79	0.11
20388	88.90	359.70	10563.19	9692.15	9633.86	-1093.55	1.08
20483	88.50	1.40	10565.35	9786.69	9728.83	-1092.63	1.84
20578	89.70	0.90	10566.84	9881.15	9823.80	-1090.73	1.37
20673	90.90	0.90	10566.34	9975.65	9918.78	-1089.24	1.26
20768	90.70	0.20	10565.02	10070.21	10013.77	-1088.32	0.77
20863	90.20	0.60	10564.27	10164.80	10108.76	-1087.66	0.67
20958	89.90	359.90	10564.19	10259.41	10203.76	-1087.25	0.80
21053	89.3	0.0001	10564.851	10354.07	10298.76	-1087.329	0.64031
21148	90.9	0.0001	10564.685	10448.71	10393.76	-1087.329	1.68421
21243	90.9	359.5	10563.193	10543.39	10488.74	-1087.744	0.52636
21338	90.5	359.3	10562.032	10638.11	10583.73	-1088.738	0.47074
21433	89.2	359.2	10562.281	10732.86	10678.72	-1089.982	1.37246
21528	90	359.2	10562.944	10827.61	10773.71	-1091.308	0.84211
21623	89.6	358.5	10563.276	10922.4	10868.69	-1093.215	0.84865
21718	90.00	0.20	10563.61	11017.13	10963.68	-1094.29	1.84
21813	89.30	0.40	10564.19	11111.74	11058.67	-1093.79	0.77
21908	90.70	0.20	10564.19	11206.34	11153.67	-1093.30	1.49
22003	90.70	359.50	10563.03	11301.00	11248.66	-1093.55	0.74
22098	89.70	359.50	10562.70	11395.72	11343.66	-1094.38	1.05
22193	89.50	359.50	10563.36	11490.43	11438.65	-1095.20	0.21
22288	89.4	359.5	10564.271	11585.15	11533.64	-1096.033	0.10526
22383	90.7	359.5	10564.188	11679.86	11628.64	-1096.862	1.36842
22478	91.1	359.3	10562.695	11774.58	11723.62	-1097.857	0.47074
22573	91.5	359	10560.54	11869.32	11818.58	-1099.266	0.52627
22668	91.5	357.6	10558.053	11964.13	11913.51	-1102.083	1.47318
22763	91.5	357.2	10555.566	12059.02	12008.38	-1106.391	0.42
22858	89.80	357.90	10554.49	12153.92	12103.28	-1110.45	1.94
22952	89.50	359.20	10555.06	12247.75	12197.25	-1112.83	1.42
23047	89.60	359.30	10555.81	12342.49	12292.23	-1114.07	0.15
23142	90.00	358.60	10556.14	12437.27	12387.22	-1115.81	0.85
23237	90.60	358.50	10555.64	12532.10	12482.18	-1118.22	0.64
23332	89.60	358.10	10555.48	12626.95	12577.14	-1121.04	1.13
23427	90.00	357.40	10555.81	12721.84	12672.07	-1124.77	0.85
23475	90.00	357.40	10555.81	12769.80	12720.02	-1126.94	0.00

BIT RECORD

BIT #	TYPE	SIZE (IN)	IN (FT)	OUT (FT)	TOTAL (FT)	HOURS DRLG	WOB (K)	RPM
1	Security MM65D	8.75	1975	10135	8170	68.5	20-30	50-70
2	Security MMD55M	8.75	10135	10877	742	23.5	25-80	0-35
3	Reed SKHE711M	6	10877	16299	5421	97.5	10-20	30-45
4	Reed SKHE711M	6	16299	23475	7178	107.5	15-30	35-50

DRILLING FLUID PARAMETERS

DATE	WT	VIS	PV	YP	PH	NaCL	Ca	% SOL
10/25/2013	9.1	43	5	8	-	35k	1.4	6.01
10/26/2013	9.6	43	7	8	-	36k	3.5	8.12
10/27/2013	9.7	55	23	9	-	41k	3.3	8.09
10/28/2013	10.0	52	15	14	-	45k	3.2	9.59
10/29/2013	10.1	53	14	11	-	41k	2.4	9.93
10/30/2013	10.5	53	16	11	-	40k	2.6	11.71
12/2/2013	9.5	32	1	1	8	138,000	32,000	0.54
12/4/2013	9.8	32	1	1	9	157,000	30,400	0.72
12/6/2013	9.9	33	1	1	8.5	171,000	30,200	0.19
12/7/2013	9.9	33	1	1	8.5	171,000	30,200	0.19
12/8/2013	9.9	33	1	1	8.5	171,000	30,200	0.19
12/9/2013	9.7	30	1	1	8.5	161,000	30,200	0.82
12/10/2013	9.7	30	1	1	8.5	161,000	30,200	0.82
12/11/2013	9.9	30	1	1	8.5	177,775	31,400	0.40
12/12/2013	9.9	30	1	1	8.5	177,775	31,400	0.40
12/13/2013	9.9	30	1	1	8.5	177,775	31,400	0.40

DAILY DRILLING CHRONOLOGY

DATE (m/d/y)	DEPTH @ 24.00	PROGRESS (ft/24 hours)	BREAKDOWN 00:00 – 24:00	RIG ACTIVITY
10/25/2013	1965'		00:00-06:00	BOP Testing, Casing pressure Fit test, Safety meetings, Pick up drill pipe, BHA, Tag Cement.
			06:00-07:30	Ream, Drill Float @ 1918' & Shoe @ 1965'
			07:30-08:00	Drill 10' of new hole, Fit test
			08:00-08:30	Drill 1975'-2302'
			08:30-09:30	Condition mud & circulate, Lost circulation
			09:30-17:00	Drill 2302'-4279'
			17:00-17:30	Rig Service
			17:30-24:00	Drill 4279'-5961'
10/26/2013	5961'	3996'	00:00-17:00	Drill 5961'-8017'
			17:00-17:30	Rig Service
			17:30-24:00	Drill 8017'-8554'
10/27/2013	8554'	2593'	00:00-02:00	Drill 8554'-8670'
			02:00-02:30	Rig Service
			02:30-15:30	Drill 8670'-9234'
			15:30-16:00	Rig Service
			16:00-24:00	Drill 9234'-9710'
10/28/2013	9710'	1156'	00:00-04:00	Drill 9710'-9884'
			04:00-04:30	Rig Service
			04:30-09:00	Drill 9884'-10135' (KOP)
			09:00-09:30	Pump Slug, Circulate
			09:30-13:30	TOOH for Curve Assembly
			13:30-15:30	BHA Operations
			15:30-17:00	TIH
			17:00-17:30	Rig Service
			17:30-22:30	TIH
			22:30-24:00	Drill/Slide 10135'-10153'
10/29/2013	10153'	443'	00:00-04:00	Drill/Slide 10153'-10261'
			04:00-04:30	Rig Service
			04:30-17:00	Drill/Slide 10261'-10715'
			17:00-17:30	Rig Service
			17:30-23:00	Drill/Slide 10715'-10877' (Land Curve)
			23:00-24:00	Circulate Bottoms Up
10/30/2013	10877'	364'	00:00-01:00	Wiper Trip
			01:00-03:00	Circulate hole clean
			03:00-04:00	Rig Service

DATE (m/d/y)	DEPTH @ 24.00	PROGRESS (ft/24 hours)	BREAKDOWN 00:00 – 24:00	RIG ACTIVITY
			04:00-08:30	TOOH
			08:30-09:30	BHA Operations - Lay Down Tools
			09:30-24:00	7" Intermediate Casing Operations
10/31/2013	10877'	0'	00:00-05:30	Cement Operations
			05:30-24:00	Prep Rig Skid
12/2/2013	10877'	0'	00:00-17:00	Walk Rig, Nipple Up BOP, Slip/Cut, Pick up Directional Tools, TIH to Cement
			17:00-19:30	Drill out cement, FIT test
			19:30-24:00	Drill/Slide 10877'-11150'
12/3/2013	11150'	273'	00:00-24:00	Drill/Slide 11150'-12717'
12/4/2013	12717'	1567'	00:00-06:00	Drill/Slide 12717'-13133'
			06:00-10:00	Lower Bakken Silt/Pronghorn Strike: Work Stuck Pipe, Circulate
			10:00-11:00	TOOH 6 Stands
			11:00-12:30	Trough 12544'-12560'
			12:30-24:00	Time Drill 12560'-12573'
12/5/2013	12573'		00:00-06:00	Time Drill 12573'-12596'
			06:00-24:00	Drill/Slide 12596'-13784'
12/6/2013	13784'	1211'	00:00-24:00	Drill/Slide 13784'-15260'
12/7/2013	15260'	1476'	00:00-18:00	Drill/Slide 15260'-16299'
			18:00-24:00	Circulate, TOOH for new BHA
12/8/2013	16299'	1039'	00:00-04:00	TOOH
			04:00-07:30	Directional Work
			07:30-18:00	TIH
			18:00-24:00	Drill/Slide 16299'-16558'
12/9/2013	16558'	259'	00:00-24:00	Drill/Slide 16558'-18508'
12/10/2013	18508'	1950'	00:00-24:00	Drill/Slide 18508'-20053'
12/11/2013	20053'	1545'	00:00-08:00	Drill/Slide 20053'-20547'
			08:00-09:00	CRI Safety Stand Down
			09:00-24:00	Drill/Slide 20547'-21506'
12/12/2013	21506'	1453'	00:00-24:00	Drill/Slide 21506'-22618'
12/13/2013	22618'	1112'	00:00-13:00	Drill/Slide 22618'-23475' (TD Well)
			13:00-24:00	Circulate, Wiper Trip, Circulate TOOH for Liner

LITHOLOGY

30' samples were caught in the vertical and curve by NorAm Wellsite Services. Logging began at 8250' MD on October 26, 2013 at 21:30 hours. 50' samples were caught in the Lateral by the Cyclone 4 Rig Crew.

Formation tops
Sample Interval
(in feet)

SAMPLE DESCRIPTION

- | | |
|-----------|---|
| 8220-8250 | LIMESTONE: red orange medium gray brown, buff, light to medium gray, firm to bright, microcrystalline, moderately silica, argillaceous, trace sandy material |
| 8220-8280 | LIMESTONE: red orange medium gray brown, light to medium gray, firm to bright, microcrystalline, moderately silica, argillaceous, trace sandy material
ANHYDRITE: white to gray, soft, amorphous |

CHARLES: 8298' MD, 8296' TVD (-6356)

- | | |
|-----------|---|
| 8280-8340 | SALT: clear, translucent trace LIMESTONE: light to medium gray, firm to bright, microcrystalline, moderately silica, argillaceous, trace sandy material |
| 8340-8400 | SALT: clear, translucent; trace LIMESTONE: light to medium gray, firm to bright, microcrystalline, moderately silica, argillaceous, trace sandy material, trace siltstone material; trace ANHYDRITE: cream to buff, |
| 8400-8520 | SALT: translucent to transparent, off white hard, crystalline, SILTSTONE: orange brown, soft, calcareous, argillaceous |
| 8520-8550 | SALT: clear, translucent; trace LIMESTONE: light to medium gray, firm to bright, microcrystalline, moderately silica, argillaceous, trace sandy material, trace siltstone material; trace ANHYDRITE: cream to buff, |
| 8550-8580 | ANHYDRITE: white to gray, soft, amorphous; LIMESTONE: light gray brown, light to medium gray, firm to bright, microcrystalline, moderately silty, argillaceous |
| 8580-8640 | LIMESTONE: light gray brown, light to medium gray, firm to bright, microcrystalline, moderately silty, argillaceous, ANHYDRITE: white, soft, amorphous; SALT: clear, translucent, hard |
| 8640-8700 | LIMESTONE: light gray brown, light to medium gray, firm to bright, microcrystalline, moderately silty, argillaceous, ANHYDRITE: white, soft, amorphous; trace SILTSTONE: red to orange, friable to firm, calcareous, material; SALT: clear, translucent, hard |

Formation tops
Sample Interval
(in feet)

SAMPLE DESCRIPTION

- 8700-8820 LIMESTONE: light gray brown, light to medium gray, firm to bright, microcrystalline, moderately silty, argillaceous, ANHYDRITE: white, soft, amorphous; SALT: clear, translucent, hard
- 8820-8880 SALT: clear, translucent, hard; LIMESTONE: light gray brown, light to medium gray, firm to bright, microcrystalline, moderately silty, argillaceous, ANHYDRITE: white to gray, soft, amorphous
- 8880-8910 ANHYDRITE: white to gray, soft, amorphous; LIMESTONE: light gray brown, light to medium gray, firm to bright, microcrystalline, moderately silty, argillaceous
- 8910-8940 LIMESTONE: light gray brown to tan, light gray, firm to bright, microcrystalline, moderately silty, argillaceous, ANHYDRITE: white to gray, soft, amorphous, trace SALT: clear, translucent, hard
- 8940-8970 SALT: clear, translucent, hard; trace LIMESTONE: light to medium gray, firm to hard, microcrystalline, moderately silty, trace argillaceous, mottled in part
- 8970-9000 SALT: clear, translucent, hard; trace LIMESTONE: light brown to tan, light to medium gray, firm to hard, microcrystalline, moderately silty, trace argillaceous, mottled in part

BASE LAST SALT: 8999' MD, 8997' TVD (-7057)

- 9000-9030 SALT: clear, translucent, hard; trace LIMESTONE: light to medium gray, tan to light brown, firm to hard, microcrystalline, moderately silty, trace argillaceous, mottled in part
- 9030-9060 LIMESTONE: light to medium gray, light brown, tan, off white in part, firm to hard, microcrystalline, silty, trace argillaceous, mottled in part
- 9060-9210 LIMESTONE: light to medium gray, light brown, tan, off white in part, firm to hard, microcrystalline, silty, trace argillaceous, mottled in part, trace ANHYDRITE: white to gray, soft, amorphous

MISSION CANYON: 9222' MD, 9219' TVD (-7279)

- 9210-9390 LIMESTONE: medium to light gray, light brown to gray, off white to tan in part, firm to hard, microcrystalline to cryptocrystalline, moderately silty, trace argillaceous, mottled in part

Formation tops
Sample Interval
(in feet)

SAMPLE DESCRIPTION

- 9390-9420 LIMESTONE: light to medium gray, light brown, tan, off white in part, firm to hard, microcrystalline, silty, trace argillaceous, mottled in part
- 9420-9510 LIMESTONE: medium to dark brown, light brown to tan in part, trace medium gray, mottled in part, firm, cryptocrystalline, earthy, slightly silty, argillaceous
- 9510-9690 LIMESTONE: medium gray brown, light to medium gray, firm to bright, microcrystalline, argillaceous, trace sandy material; ANHYDRITE: white, soft,
- 9690-9780 LIMESTONE: medium to dark brown, light brown to tan in part, trace medium gray, mottled in part, firm, cryptocrystalline, earthy, slightly silty, argillaceous

LODGEPOLE: 9792' MD, 9790' TVD (-7850)

- 9780-9870 LIMESTONE: light to medium gray to brown, light brown to tan in part, trace medium to dark gray, mottled in part, firm, microcrystalline to cryptocrystalline, earthy, slightly silty, argillaceous
- 9870-9990 LIMESTONE: dark to medium gray to brown, light brown to tan in part, mottled in part, firm, microcrystalline to cryptocrystalline, earthy, slightly silty, argillaceous
- 9990-10500 LIMESTONE: dark to medium gray, light to medium brown, very trace tan in part, mottled in part, firm to hard, microcrystalline to cryptocrystalline, slightly to moderately silty, argillaceous, earthy texture
- 10500-10530 LIMESTONE: dark to medium gray, medium to dark brown gray, trace light gray, mottled in part, firm to hard, microcrystalline to cryptocrystalline, moderately silty, argillaceous, earthy texture, sub to blocky

FALSE BAKKEN: 10537' MD, 10483' TVD (-8543)

- 10530-10560 SHALE: black, moderately hard, sub blocky to sub platy, carbonaceous;
LIMESTONE: dark to medium gray, medium to dark brown gray, trace light gray, mottled in part, firm to hard, microcrystalline to cryptocrystalline, moderately silty, argillaceous, earthy texture, sub to blocky

UPPER BAKKEN SHALE: 10550' MD, 10490' TVD (-8550)

Formation tops
Sample Interval
(in feet)

SAMPLE DESCRIPTION

10560-10590 SHALE: black, moderately hard, sub blocky to sub platy, carbonaceous;
DOLOMITE: light to medium gray, cream to off white, tan in part, firm, slightly
hard, microcrystalline, silty, trace brown spotty oil stain, green to white
fluorescence, moderately blue to white diffuse cut

MIDDLE BAKKEN MEMBER: 10576'MD, 10505' TVD (-8565)

10590-10620 DOLOMITE: light to medium gray, cream to off white, tan in part, firm, slightly
hard, microcrystalline, silty, trace brown spotty oil stain, green to white
fluorescence, moderately blue to white diffuse cut

10620-10650 DOLOMITE: light to medium gray, cream to off white, tan in part, firm, slightly
hard, microcrystalline, silty, trace brown spotty oil stain, green to white
fluorescence, moderately blue to white diffuse cut; SHALE: dark brown to black,
firm, moderately hard, sub blocky to sub platy, carbonaceous

LOWER BAKKEN SHALE: 10641'MD, 10538' TVD (-8550)

10650-10710 SHALE: dark brown to black, firm, moderately hard, sub blocky to sub platy,
carbonaceous

10710-10740 SHALE: dark brown to black, firm, moderately hard, sub blocky to sub platy,
carbonaceous, DOLOMITE: light to medium brown, green to gray, firm,
microcrystalline, silty, slightly calcareous, trace brown spotty oil stain, scattered
yellow to green fluorescence, slow to moderately green diffuse cut

THREE FORKS: 10726'MD, 10568' TVD (-8628)

10740-10877 DOLOMITE: light to medium brown, green to gray, firm, microcrystalline, silty,
slightly calcareous, trace brown spotty oil stain, scattered yellow to green
fluorescence, slow to moderately green diffuse cut

10877-10900 DOLOMITE: light to medium brown, green to gray, firm, microcrystalline, silty,
slightly calcareous, trace brown spotty oil stain, sample cont with cement

10900-10950 DOLOMITE: light to medium brown, green to gray, firm, microcrystalline, silty,
slightly calcareous, trace brown spotty oil stain, scattered yellow to green
fluorescence, slow to moderately green diffuse cut

Formation tops
Sample Interval
(in feet)

SAMPLE DESCRIPTION

- 10950-11150 DOLOMITE: pink to off white, light brown in part, firm to moderately hard, microcrystalline to cryptocrystalline, calcareous; trace SHALE: light blue to green gray, calcareous, laminated in part, dull yellow fluorescence, dull yellow streaming cut
- 11150-11250 DOLOMITE: pink to off white, light brown in part, firm to moderately hard, microcrystalline to cryptocrystalline, calcareous; trace SHALE: light blue to green gray, calcareous, laminated in part, dull yellow fluorescence, slow to moderately dull yellow streaming cut
- 11250-11450 DOLOMITE: pink to off white, tan in part, firm to moderately hard, microcrystalline to cryptocrystalline, calcareous; trace SHALE: light blue to green gray, calcareous, laminated in part, green to yellow fluorescence, moderately to fast green streaming cut
- 11450-11700 DOLOMITE: pink to off white, tan in part, firm to moderately hard, microcrystalline to cryptocrystalline, calcareous; trace SHALE: light blue to green gray, calcareous, laminated in part, pyrite, green to yellow fluorescence, moderately to fast green streaming cut
- 11700-11750 DOLOMITE: pink to off white, tan in part, firm to moderately hard, microcrystalline to cryptocrystalline, calcareous; trace SHALE: light blue to green gray, calcareous, laminated in part, pyrite, dull yellow fluorescence, moderately to fast green cut
- 11750-11800 DOLOMITE: pink to off white, tan, light brown, firm to moderately hard, microcrystalline to cryptocrystalline, calcareous trace pyrite; trace SHALE: light blue to green gray, calcareous, laminated in part, trace pyrite, dull yellow fluorescence, slow green cut
- 11800-12000 DOLOMITE: pink to off white, tan, light brown, firm to moderately hard, microcrystalline to cryptocrystalline, calcareous trace pyrite; trace SHALE: light blue to green gray, calcareous, laminated in part, trace pyrite, dull yellow fluorescence, moderately to fast green cut
- 12000-12300 DOLOMITE: off white, tan to light brown, trace pink, firm to moderately hard, microcrystalline to cryptocrystalline, calcareous, trace pyrite; trace SHALE: light blue, green, gray, calcareous, laminated in part, dull yellow fluorescence, moderately to fast yellow green cut

Formation tops
Sample Interval
(in feet)

SAMPLE DESCRIPTION

- 12300-12500 DOLOMITE: off white to tan, light brown to pink, light gray, firm to moderately hard, microcrystalline to cryptocrystalline, calcareous, disseminated and nodules pyrite; SHALE: light blue to green, light gray, calcareous, laminated in part, scattered green yellow fluorescence, moderately to fast yellow green cut
- 12500-12700 DOLOMITE: off white to tan, light brown to pink, light gray, firm to moderately hard, microcrystalline to cryptocrystalline, calcareous, disseminated and nodules pyrite; SHALE: light blue to green, light gray, calcareous, laminated in part, scattered green yellow fluorescence, slow yellow green cut
- 12700-12800 DOLOMITE: pink to off white, tan, light brown, firm to moderately hard, microcrystalline to cryptocrystalline, calcareous trace pyrite; trace SHALE: light blue, green, gray, calcareous, laminated in part, trace pyrite, dull yellow fluorescence, moderately to fast green cut
- 12800-12950 DOLOMITE: pink to off white, tan in part, firm to moderately hard, microcrystalline to cryptocrystalline, calcareous; trace SHALE: light blue to green gray, calcareous, laminated in part, pyrite, dull yellow fluorescence, moderately to fast green streaming cut
- 12950-13050 DOLOMITE: pink to off white, light gray, tan in part, firm to moderately hard, microcrystalline to cryptocrystalline, calcareous; trace SHALE: light blue to green gray, calcareous, laminated in part, pyrite, dull yellow fluorescence, moderately to fast green streaming cut
- 13050-13150 DOLOMITE: pink to off white, tan in part, firm to moderately hard, microcrystalline to cryptocrystalline, calcareous; trace SHALE: light blue to green gray, calcareous, laminated in part, pyrite, dull yellow fluorescence, moderately to fast green streaming cut
- 13150-13450 DOLOMITE: pink to off white, tan in part, firm to moderately hard, microcrystalline to cryptocrystalline, calcareous; trace SHALE: light blue to green gray, calcareous, laminated in part, trace pyrite, dull yellow fluorescence, moderately to fast green streaming cut
- 13450-13850 DOLOMITE: pink to off white, tan in part, trace light gray, firm to moderately hard, microcrystalline to cryptocrystalline, calcareous; trace SHALE: light blue to green gray, calcareous, laminated in part, trace pyrite, dull yellow fluorescence, moderately yellow streaming cut

Formation tops
Sample Interval
(in feet)

SAMPLE DESCRIPTION

- 13850-14000 DOLOMITE: pink to cream, off white to tan, firm to moderately hard, microcrystalline to cryptocrystalline, calcareous; trace SHALE: light blue to green gray, calcareous, laminated in part, very trace nodules pyrite, scattered green fluorescence, moderately yellow diffuse cut
- 14000-14050 DOLOMITE: pink to cream, off white to tan, firm to moderately hard, microcrystalline to cryptocrystalline, calcareous; trace SHALE: light blue to green gray, calcareous, laminated in part, very trace nodules pyrite, scattered green fluorescence, slow yellow diffuse cut
- 14050-14100 DOLOMITE: cream, off white, tan, pink, firm to moderately hard, microcrystalline to cryptocrystalline, calcareous; trace SHALE: light blue to green gray, calcareous, laminated in part, very trace nodules pyrite, scattered green fluorescence, slow yellow diffuse cut
- 14100-14200 DOLOMITE: cream, off white, tan, pink, firm to moderately hard, microcrystalline to cryptocrystalline, calcareous; trace SHALE: light blue to green gray, calcareous, laminated in part, very trace nodules pyrite, scattered yellow green fluorescence, slow yellow diffuse cut
- 14200-14300 DOLOMITE: cream, off white, tan, pink, firm to moderately hard, microcrystalline to cryptocrystalline, calcareous; trace SHALE: light blue to green gray, calcareous, laminated in part, very trace nodules pyrite, scattered yellow green fluorescence, slow yellow
- 14300-14400 DOLOMITE: cream, off white, tan, pink, firm to moderately hard, microcrystalline to cryptocrystalline, calcareous; trace SHALE: light blue to green gray, calcareous, laminated in part, very trace nodules pyrite, scattered yellow fluorescence, slow yellow
- 14400-14550 DOLOMITE: off white, tan, pink, firm to moderately hard, microcrystalline to cryptocrystalline, calcareous; trace SHALE: light blue to green gray, calcareous, laminated in part, very trace nodules pyrite, scattered yellow fluorescence, slow yellow
- 14550-14850 DOLOMITE: off white, tan, pink, firm to moderately hard, microcrystalline to cryptocrystalline, calcareous; trace SHALE: light blue to green gray, calcareous, laminated in part, scattered yellow fluorescence, slow yellow
- 14850-15250 DOLOMITE: pink to off white, tan in part, trace light gray, firm to moderately hard, microcrystalline to cryptocrystalline, calcareous; trace SHALE: light blue to green gray, calcareous, laminated in part, trace pyrite, dull yellow fluorescence, moderately yellow streaming cut

Formation tops
Sample Interval
(in feet)

SAMPLE DESCRIPTION

- 15250-15550 DOLOMITE: pink to cream, off white to tan, firm to moderately hard, microcrystalline to cryptocrystalline, calcareous; trace SHALE: light blue to green gray, calcareous, laminated in part, scattered green fluorescence, moderately to slow yellow diffuse cut
- 15550-15650 DOLOMITE: pink to off white, tan in part, trace light gray, firm to moderately hard, microcrystalline to cryptocrystalline, calcareous; trace SHALE: light blue to green gray, calcareous, laminated in part, trace pyrite, dull yellow fluorescence, moderately yellow streaming cut
- 15650-15850 DOLOMITE: tan, off white, pink, firm to moderately hard, microcrystalline to cryptocrystalline, calcareous; trace SHALE: light blue to green gray, calcareous, laminated in part, scattered yellow fluorescence, slow yellow
- 15850-16050 DOLOMITE: tan, off white, pink, firm to moderately hard, microcrystalline to cryptocrystalline, calcareous; trace SHALE: light blue to green gray, calcareous, laminated in part, scattered dull yellow fluorescence, slow yellow, cut cont by lube
- 16050-16150 DOLOMITE: tan, light brown, off white, pink, firm to moderately hard, microcrystalline to cryptocrystalline, slightly calcareous; trace SHALE: light blue to green gray, calcareous, laminated in part, scattered dull yellow fluorescence, slow yellow, cut cont by lube
- 16150-16350 DOLOMITE: cream, off white, tan, pink, firm to moderately hard, microcrystalline to cryptocrystalline, calcareous; trace SHALE: light blue to green gray, calcareous, laminated in part, very trace nodules pyrite, scattered yellow green fluorescence, slow yellow diffuse cut
- 16350-16450 DOLOMITE: off white, pink to tan, light gray firm to moderately hard, microcrystalline to cryptocrystalline, calcareous; trace SHALE: dull light blue gray, white to light gray, calcareous, laminated in part, scattered dull yellow fluorescence, very slow dull yellow streaming cut; sample contaminated with lube
- 16450-16650 DOLOMITE: off white, cream to pink, light brown in part, firm to moderately hard, microcrystalline to cryptocrystalline, calcareous; SHALE: dull light blue gray, white to light gray, calcareous, laminated in part, scattered very dull yellow fluorescence, very slow dull yellow streaming cut
- 16650-16950 DOLOMITE: pink to red, white to cream, light brown in part, firm to moderately hard, microcrystalline to cryptocrystalline, calcareous; SHALE: dull light blue gray, white to light gray, calcareous, laminated in part, scattered very dull yellow fluorescence, moderately to slow yellow green streaming cut

Formation tops
Sample Interval
(in feet)

SAMPLE DESCRIPTION

- 16950-17000 DOLOMITE: pink to red, cream, light brown in part, firm to moderately hard, microcrystalline to cryptocrystalline, calcareous; SHALE: dull light blue gray, white to light gray, calcareous, laminated in part, scattered very dull yellow fluorescence, moderately to slow yellow green streaming cut
- 17000-17250 DOLOMITE: pink, cream, light brown in part, firm to moderately hard, microcrystalline to cryptocrystalline, calcareous; SHALE: dull light blue gray, white to light gray, calcareous, laminated in part, scattered very dull yellow fluorescence, moderately to slow yellow green streaming cut
- 17250-17350 DOLOMITE: tan, off white, pink, firm to moderately hard, microcrystalline to cryptocrystalline, calcareous; SHALE: light blue to green gray, calcareous, laminated in part, scattered yellow fluorescence, slow dull yellow blue white cut
- 17350-17450 DOLOMITE: tan, off white, light gray, pink, firm to moderately hard, microcrystalline to cryptocrystalline, calcareous; trace SHALE: light blue to green gray, calcareous, laminated in part, scattered yellow fluorescence, slow dull yellow blue white cut
- 17450-17500 DOLOMITE: tan, off white, light brown, cream, firm to moderately hard, microcrystalline to cryptocrystalline, calcareous; trace SHALE: light blue to green gray, light gray, calcareous, laminated in part, scattered yellow fluorescence, slow dull yellow blue white cut
- 17500-17550 DOLOMITE: tan, off white, light gray, light brown, cream, firm to moderately hard, microcrystalline to cryptocrystalline, calcareous; trace SHALE: light blue to green gray, light gray, calcareous, laminated in part, scattered yellow fluorescence, slow dull yellow blue white cut
- 17550-17650 DOLOMITE: tan, off white, light gray, light brown, cream, firm to moderately hard, microcrystalline to cryptocrystalline, calcareous; trace SHALE: light blue to green gray, calcareous, laminated in part, scattered yellow fluorescence, slow dull yellow blue white cut
- 17650-17750 DOLOMITE: off white, cream, tan, light brown, firm to moderately hard, microcrystalline to cryptocrystalline, calcareous; trace SHALE: light blue to green gray, calcareous, laminated in part, scattered yellow fluorescence, slow dull yellow blue white cut
- 17750-17950 DOLOMITE: off white, cream, tan, light brown, firm to moderately hard, microcrystalline to cryptocrystalline, calcareous; trace SHALE: light blue to green gray, light gray, calcareous, laminated in part, scattered yellow fluorescence, slow dull yellow blue white cut

Formation tops
Sample Interval
(in feet)

SAMPLE DESCRIPTION

- 17950-18150 DOLOMITE: cream, tan to light pink, off white, light brown, firm to moderately hard, microcrystalline to cryptocrystalline, calcareous; trace SHALE: light blue to green gray, light gray, calcareous, laminated in part, scattered yellow fluorescence, slow dull yellow blue white cut
- 18150-18350 DOLOMITE: cream, tan to light pink, off white, light brown, firm to moderately hard, microcrystalline to cryptocrystalline, calcareous; trace SHALE: light blue to green gray, light gray, calcareous, laminated in part, scattered yellow fluorescence, slow dull yellow blue white cut; sample contaminated with lube
- 18350-18550 DOLOMITE: off white, cream, tan, light brown, firm to moderately hard, microcrystalline to cryptocrystalline, calcareous; trace SHALE: light blue to green gray, calcareous, laminated in part, scattered yellow fluorescence, slow dull yellow blue white cut
- 18550-18650 DOLOMITE: cream, tan to light pink, off white, light brown, firm to moderately hard, microcrystalline to cryptocrystalline, calcareous; trace SHALE: light blue to green gray, light gray, calcareous, laminated in part, scattered yellow fluorescence, slow blue white streaming cut
- 18650-19100 DOLOMITE: cream, tan to light pink, off white, light brown, firm to moderately hard, microcrystalline to cryptocrystalline, calcareous; trace SHALE: light blue to green gray, light gray, calcareous, laminated in part, scattered yellow fluorescence, slow blue white cut
- 19100-19550 DOLOMITE: off white, light pink, cream, tan, light brown, firm to moderately hard, microcrystalline to cryptocrystalline, calcareous; SHALE: light blue to green gray, light gray, calcareous, laminated in part, scattered yellow fluorescence, slow blue white cut
- 19550-19650 DOLOMITE: off white, pink to tan, light gray firm to moderately hard, microcrystalline to cryptocrystalline, calcareous; trace SHALE: light blue gray, white to light gray, calcareous, laminated in part, scattered dull yellow fluorescence, slow dull yellow streaming cut
- 19650-20100 DOLOMITE: off white, pink to tan, light gray firm to moderately hard, microcrystalline to cryptocrystalline, calcareous; trace SHALE: light blue green gray, white to light gray, calcareous, laminated in part, trace nodules pyrite, scattered dull yellow fluorescence, slow dull yellow streaming cut

Formation tops
Sample Interval
(in feet)

SAMPLE DESCRIPTION

- 20100-20200 DOLOMITE: off white, pink to tan, light gray firm to moderately hard, microcrystalline to cryptocrystalline, calcareous; trace SHALE: light blue green gray, white to light gray, calcareous, laminated in part, scattered dull yellow fluorescence, slow dull yellow streaming cut
- 20200-20350 DOLOMITE: off white, pink to tan, light gray firm to moderately hard, microcrystalline to cryptocrystalline, calcareous; trace SHALE: light blue green gray, white to light gray, calcareous, laminated in part, scattered dull yellow fluorescence, moderately to slow green to white streaming cut
- 20350-20550 DOLOMITE: off white, tan, light brown, light gray, firm to moderately hard, microcrystalline to cryptocrystalline, calcareous; trace SHALE: light blue green gray, white to light gray, calcareous, laminated in part, scattered dull yellow fluorescence, moderately to slow green to white streaming cut
- 20550-20650 DOLOMITE: off white, tan, light gray, firm to moderately hard, microcrystalline to cryptocrystalline, calcareous; trace SHALE: light blue green gray, white to light gray, calcareous, laminated in part, scattered dull yellow fluorescence, moderately to slow green to white streaming cut
- 20650-20950 DOLOMITE: off white, tan, firm to moderately hard, microcrystalline to cryptocrystalline, calcareous; trace SHALE: light blue green gray, white to light gray, calcareous, laminated in part, scattered dull yellow fluorescence, moderately to slow green to white streaming cut
- 20950-21050 DOLOMITE: off white, pink to tan, light gray firm to moderately hard, microcrystalline to cryptocrystalline, calcareous; trace SHALE: light blue green gray, white to light gray, calcareous, laminated in part, scattered dull yellow fluorescence, moderately to slow green to white streaming cut
- 21050-21100 DOLOMITE: off white, pink to tan, light gray firm to moderately hard, microcrystalline to cryptocrystalline, calcareous; trace SHALE: light blue green gray, white to light gray, calcareous, laminated in part, scattered dull yellow fluorescence, moderately to slow green to white streaming cut; sample contaminated with lube
- 21100-21900 DOLOMITE: off white, pink to tan, light gray firm to moderately hard, microcrystalline to cryptocrystalline, calcareous; trace SHALE: light blue green gray, white to light gray, calcareous, laminated in part, scattered dull yellow fluorescence, slow green streaming cut; sample contaminated with lube

Formation tops
Sample Interval
(in feet)

SAMPLE DESCRIPTION

21900-22000 DOLOMITE: pink to tan, light gray firm to moderately hard, microcrystalline to cryptocrystalline, calcareous; trace SHALE: light blue green gray, white to light gray, calcareous, laminated in part, scattered dull yellow fluorescence, slow green streaming cut; sample contaminated with lube

22000-22100 DOLOMITE: off white, pink to tan, light gray firm to moderately hard, microcrystalline to cryptocrystalline, calcareous; trace SHALE: light blue green gray, white to light gray, calcareous, laminated in part, scattered dull yellow fluorescence, slow green streaming cut; sample contaminated with lube

22100-22750 DOLOMITE: off white, pink to tan, light brown, light gray firm to moderately hard, microcrystalline to cryptocrystalline, calcareous; trace SHALE: light blue green gray, white to light gray, calcareous, laminated in part, dull yellow fluorescence, slow green streaming cut; sample contaminated with lube

22750-22900 DOLOMITE: off white, pink to tan, light brown, light gray firm to moderately hard, microcrystalline to cryptocrystalline, calcareous, pyrite; trace SHALE: light blue green gray, white to light gray, calcareous, laminated in part, dull yellow fluorescence, slow green streaming cut; sample contaminated with lube

22900-23475 DOLOMITE: off white, tan, light brown, light gray firm to moderately hard, microcrystalline to cryptocrystalline, calcareous, pyrite; trace SHALE: light blue green gray, white to light gray, calcareous, laminated in part, dull yellow fluorescence, slow green streaming cut; sample contaminated with lube

GEOLOGICAL SUMMARY & CONCLUSIONS

Continental Resources, Inc drilled the Columbus Federal 2-16H in Sections 16, 9, and 4 of Township 153N, Range 101W into the First Bench of the Three Forks. The Well was spud using Cyclone 4 on October 25, 2013 at a surface location of 2508' FNL, and 222' FEL.

The surface was pre-drilled with surface casing set at 1965'MD. From there the 8 3/4" vertical portion was drilled to Kick Off Point (KOP) at 10135'MD. Leam provided Directional Drilling, and Measurement While Drilling (MWD) services. NorAm Wellsite Services provided both Mud Logging and Geosteering Services. No trips were necessary in the vertical.

The curve was drilled in 23.5 hours October 28-29, 2013. The wellbore entered the Three Forks top at 10726'MD, 10568'TVD, and landed 18' into the First Bench at 10877'MD, 10586'TVD.

The Lateral began December 2, 2013 at 18:45 hours. Drilling north under the river, formation took a dive down, as the wellbore struck the Lower Bakken Silt/Pronghorn calcareous shale. Cyclone 4 pulled back to 12560'MD to begin Sidetrack 1 on December 4th, 2013 at 10:00 hours. The first and only trip occurred at 16299'MD for a new Bit and Motor on December 7th, 2013 at 20:30 hours, and resumed drilling December 8th, 2013 at 17:52 hours.

The Columbus Federal 2-16 reached TD of 23,475'MD at 13:13 hours on December 13, 2013. The Columbus Federal 2-16H bottom hole location is 236.1' FNL & 1348.94' FEL Section 4, T153N, R101W.

NEWSCO

International Energy Services Inc.

Continental Resources

Company

33326

Job Number

8/13/2013

Date

Cyclone 4

Rig

Columbus Federal 2-16H

Well Name

McKenzie Co., ND

County & State

Surveyed from depth of: Surface to 1917'

GL to KB: 20'

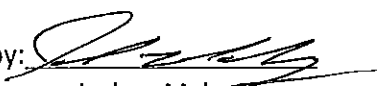
Type of Survey: Nvader/MWD

True North

Directional Supervisor/Surveyor: David Hopper

The data and calculations for this survey have been checked by me and conform to the standards and procedures set forth by Newsco International Energy Services Inc. This report represents a true and correct directional survey of this well based on the original data obtained at the well site. Wellbore coordinates are calculated using minimum curvature .

Certified by:


Joshua Mahoney



Directional Services U.S.A.

CLIENT: Continental					NEWSCO JOB # 33326							
DATE: 8/13/2013					WELL NAME: Columbus Federal 2-16H							
STATE: North Dakota					RIG: Cyclone 4							
FIELD: Baker					County: Mckenzie							
TIE-ON DATA					TARGET DATA							
C/L (10,30,100): 100.00 Feet					SENSOR TO BIT: 47.00							
MEASURED DEPTH: 0.00 Feet					KB TO GL : 6.00							
TVD: 0.00 Feet												
INCLINATION: 0.00 Deg.					Cyclone 4: 20.00							
AZIMUTH: 0.00 Deg.												
N(+) S(-): 0.00 Feet												
E(+) W(-): 0.00 Feet					All depths are measured from survey tool to Big Rigs KB							
V/SECTION PLANE: 0.00 Deg.												
V/SECTION : 0.00 Feet												
#	DEPTH	INC	AZM	CRSL	TVD	VS	+N/S-	+E/W-	DLS	/100 FEET		TF
1	142.00	0.20	326.80	142.00	142.00	0.21	0.21	-0.14	0.14	0.14	-23.38	MAG
2	220.00	0.20	353.20	78.00	220.00	0.46	0.46	-0.23	0.12	0.00	33.85	MAG
3	301.00	0.20	353.90	81.00	301.00	0.74	0.74	-0.26	0.00	0.00	0.86	MAG
4	381.00	0.10	327.50	80.00	381.00	0.94	0.94	-0.31	0.15	-0.13	-33.00	MAG
5	471.00	0.20	291.70	90.00	471.00	1.06	1.06	-0.50	0.15	0.11	-39.78	MAG
6	563.00	0.10	310.60	92.00	563.00	1.17	1.17	-0.71	0.12	-0.11	20.54	MAG
7	655.00	0.20	294.10	92.00	655.00	1.29	1.29	-0.92	0.12	0.11	-17.93	MAG
8	748.00	0.20	310.30	93.00	748.00	1.46	1.46	-1.19	0.06	0.00	17.42	MAG
9	843.00	0.20	295.90	95.00	843.00	1.64	1.64	-1.46	0.05	0.00	-15.16	MAG
10	934.00	0.10	272.70	91.00	934.00	1.71	1.71	-1.69	0.13	-0.11	-25.49	MAG
11	1026.00	0.20	237.90	92.00	1026.00	1.63	1.63	-1.90	0.14	0.11	-37.83	MAG
12	1120.00	0.40	192.50	94.00	1119.99	1.22	1.22	-2.11	0.31	0.21	-48.30	MAG
13	1213.00	0.40	241.40	93.00	1212.99	0.75	0.75	-2.47	0.36	0.00	52.58	MAG
14	1306.00	0.50	244.20	93.00	1305.99	0.42	0.42	-3.12	0.11	0.11	3.01	MAG
15	1396.00	0.40	259.70	90.00	1395.99	0.19	0.19	-3.78	0.17	-0.11	17.22	MAG
16	1490.00	0.30	281.10	94.00	1489.99	0.18	0.18	-4.34	0.17	-0.11	22.77	MAG
17	1583.00	0.40	271.30	93.00	1582.98	0.24	0.24	-4.91	0.12	0.11	-10.54	MAG
18	1677.00	0.20	323.00	94.00	1676.98	0.37	0.37	-5.34	0.34	-0.21	55.00	MAG
19	1769.00	0.20	327.20	92.00	1768.98	0.64	0.64	-5.52	0.02	0.00	4.57	MAG
20	1864.00	0.10	269.90	95.00	1863.98	0.78	0.78	-5.69	0.18	-0.11	-60.32	MAG
21	1917.00	0.30	262.10	53.00	1916.98	0.76	0.76	-5.88	0.38	0.38	-14.72	MAG
22												
23												
24												
25												
26												
27												
28												
#				CRSL	TVD	VS	+N/S-	+E/W-	DLS	/100 FEET		TF
28												
29												
30												
31												
32												
33												
34												
35												
36												
37												
38												
39												
40												
41												



Survey Certification Sheet

Continental Resources
Company

20131225D-ND
Job Number

01/16/2014
Date

Section 16, T153N, R101W
LOCATION

Columbus Federal 2-16H
Well Name

McKenzie County, ND
County & State

Enclosed, please find the survey performed on the referenced well by Leam Drilling Systems, LLC.
Other information required by your office is as follows:

Name & Title Of Surveyor	Drainhole Number	Surveyed Depths	Dates Performed	Type of Survey
Randy W. Rakowitz	OH	1,983'- 13,133'	10/29/2013-12/04/2013	MWD
Randy W. Rakowitz	ST	12,583'- 23,475'	12/04/2013-12/13/2013	MWD

The data and calculations for this survey have been checked and conform to the standards and procedures set forth by LEAM Drilling Systems, LLC. This report represents a true and correct Survey of this well, to the best of our knowledge, based on the original data obtained at the well site.


Nora Tucker
Well Planner

Physical & Mailing: 2010 East Davis, Conroe, TX 77301

(936) 756-7577 • (936) 756-7595 Fax • 1 (800) 756-7504



LEAM Drilling Systems, Inc.
2027A Airport Rd.
Conroe, TX 77301
(936) 569-1315

Survey Certification Form

State of North Dakota
McKenzie County

I, Randy W. Rakowitz, an employee of LEAM Drilling Systems, Inc., hereby certify that on the dates beginning on 10/29/2013 and ending on 12/4/2013, I conducted or supervised the taking of an MWD Survey from a beginning depth of 1,983' ft. MD to an ending depth of 13,133' ft. MD; that the depth is true, correct, complete, and within the limitations of the tools as set forth by LEAM Drilling Systems, Inc., that I am authorized and qualified to make this report; that this survey was conducted in reference to True North with a declination of 8.52° with respect to the well Columbus Federal 2-16H (OH) located in McKenzie County, North Dakota as requested by Continental Resources.

A handwritten signature in black ink, appearing to read 'Randy W. Rakowitz', is written over a solid black horizontal line.

Randy W. Rakowitz
Sr. MWD Operations Coordinator



LEAM Drilling Systems, Inc.
2027A Airport Rd.
Conroe, TX 77301
(936) 569-1315

Survey Certification Form

State of North Dakota
McKenzie County

I, Randy W. Rakowitz, an employee of LEAM Drilling Systems, Inc., hereby certify that on the dates beginning on 12/4/2013 and ending on 12/13/2013, I conducted or supervised the taking of an MWD Survey from a beginning depth of 12,583' ft. MD to an ending depth of 23,475' ft. MD; that the depth is true, correct, complete, and within the limitations of the tools as set forth by LEAM Drilling Systems, Inc., that I am authorized and qualified to make this report; that this survey was conducted in reference to True North with a declination of 8.39° with respect to the well Columbus Federal 2-16H (ST) located in McKenzie County, North Dakota as requested by Continental Resources.

A handwritten signature in black ink, appearing to read 'Randy W. Rakowitz', is written over a solid black horizontal line.

Randy W. Rakowitz
Sr. MWD Operations Coordinator

LEAM Drilling Systems LLC

Survey Report

OH

Company:	Continental Resources	Local Co-ordinate Reference:	Well 2-16H
Project:	McKenzie County, ND	TVD Reference:	GL 1920+KB 20 @ 1940.00usft (Cyclone 4)
Site:	Columbus Federal 2	MD Reference:	GL 1920+KB 20 @ 1940.00usft (Cyclone 4)
Well:	2-16H	North Reference:	True
Wellbore:	OH	Survey Calculation Method:	Minimum Curvature
Design:	OH	Database:	EDM 5000.1 Multi User DB

Project	McKenzie County, ND		
Map System:	US State Plane 1983	System Datum:	Mean Sea Level
Geo Datum:	North American Datum 1983		
Map Zone:	North Dakota Northern Zone		

Site	Columbus Federal 2, Sec. 16 - T153N - R101W		
Site Position:		Northing:	408,198.98 usft
From:	Lat/Long	Easting:	1,193,822.32 usft
Position Uncertainty:	0.00 usft	Slot Radius:	0 "
		Latitude:	48° 4' 31.258 N
		Longitude:	103° 40' 11.343 W
		Grid Convergence:	-2.36 °

Well	2-16H		
Well Position	+N/-S	0.00 usft	Northing:
	+E/-W	0.00 usft	Easting:
Position Uncertainty	0.00 usft	Wellhead Elevation:	usft
		Latitude:	48° 4' 31.258 N
		Longitude:	103° 40' 11.343 W
		Ground Level:	1,920.00 usft

Wellbore	OH				
Magnetics	Model Name	Sample Date	Declination (°)	Dip Angle (°)	Field Strength (nT)
	BGGM2013	10/11/13	8.52	72.97	56,411

Design	OH				
Audit Notes:					
Version:	1.0	Phase:	ACTUAL	Tie On Depth:	0.00
Vertical Section:	Depth From (TVD) (usft)	+N/-S (usft)	+E/-W (usft)	Direction (°)	
	0.00	0.00	0.00	355.08	

Survey Program	Date	01/16/14			
From (usft)	To (usft)	Survey (Wellbore)	Tool Name	Description	
142.00	1,917.00	Survey #1-NEUSCO (OH)	MWD-ISCWSA	MWD - Standard	
1,983.00	13,057.00	Survey #2 (OH)	LEAM MWD-ADJ	MWD - Standard	
13,133.00	13,133.00	Survey #3 (OH)	Project	Projection	

Survey										
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)	
1,917.00	0.30	262.10	1,916.98	0.75	-5.87	1.25	0.38	0.38	-14.72	
1,983.00	0.70	256.10	1,982.98	0.63	-6.44	1.18	0.61	0.61	-9.09	
2,077.00	1.30	241.20	2,076.96	-0.02	-7.93	0.66	0.69	0.64	-15.85	
2,170.00	1.30	238.70	2,169.94	-1.08	-9.76	-0.24	0.06	0.00	-2.69	
2,263.00	1.60	243.70	2,262.91	-2.20	-11.82	-1.18	0.35	0.32	5.38	
2,357.00	1.70	244.90	2,356.87	-3.37	-14.26	-2.14	0.11	0.11	1.28	
2,450.00	2.10	226.30	2,449.82	-5.13	-16.74	-3.68	0.78	0.43	-20.00	
2,543.00	2.30	211.90	2,542.75	-7.90	-18.96	-6.24	0.63	0.22	-15.48	
2,638.00	2.20	188.10	2,637.68	-11.32	-20.22	-9.54	0.98	-0.11	-25.05	

LEAM Drilling Systems LLC

Survey Report

Company:	Continental Resources	Local Co-ordinate Reference:	Well 2-16H
Project:	McKenzie County, ND	TVD Reference:	GL 1920+KB 20 @ 1940.00usft (Cyclone 4)
Site:	Columbus Federal 2	MD Reference:	GL 1920+KB 20 @ 1940.00usft (Cyclone 4)
Well:	2-16H	North Reference:	True
Wellbore:	OH	Survey Calculation Method:	Minimum Curvature
Design:	OH	Database:	EDM 5000.1 Multi User DB

Survey									
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
2,732.00	2.20	173.20	2,731.61	-14.90	-20.26	-13.10	0.61	0.00	-15.85
2,825.00	1.80	154.00	2,824.56	-17.98	-19.41	-16.25	0.83	-0.43	-20.65
2,918.00	1.10	117.80	2,917.53	-19.71	-17.98	-18.10	1.20	-0.75	-38.92
3,011.00	1.40	63.50	3,010.51	-19.62	-16.18	-18.16	1.26	0.32	-58.39
3,105.00	1.70	61.20	3,104.48	-18.44	-13.93	-17.17	0.33	0.32	-2.45
3,198.00	1.60	64.70	3,197.44	-17.22	-11.54	-16.16	0.15	-0.11	3.76
3,291.00	1.70	68.40	3,290.40	-16.15	-9.09	-15.32	0.16	0.11	3.98
3,384.00	1.60	72.10	3,383.36	-15.25	-6.57	-14.63	0.16	-0.11	3.98
3,477.00	1.30	68.40	3,476.33	-14.46	-4.35	-14.03	0.34	-0.32	-3.98
3,571.00	1.10	65.10	3,570.31	-13.69	-2.54	-13.42	0.23	-0.21	-3.51
3,664.00	0.80	76.70	3,663.30	-13.16	-1.10	-13.02	0.38	-0.32	12.47
3,758.00	0.60	77.60	3,757.29	-12.91	0.02	-12.86	0.21	-0.21	0.96
3,851.00	0.50	74.90	3,850.28	-12.70	0.89	-12.73	0.11	-0.11	-2.90
3,944.00	0.60	78.40	3,943.28	-12.49	1.75	-12.60	0.11	0.11	3.76
4,038.00	0.40	74.40	4,037.28	-12.31	2.55	-12.48	0.22	-0.21	-4.26
4,132.00	0.40	89.50	4,131.27	-12.21	3.20	-12.44	0.11	0.00	16.06
4,225.00	0.40	91.40	4,224.27	-12.22	3.85	-12.50	0.01	0.00	2.04
4,319.00	0.40	103.70	4,318.27	-12.31	4.49	-12.64	0.09	0.00	13.09
4,412.00	0.30	102.50	4,411.27	-12.43	5.05	-12.82	0.11	-0.11	-1.29
4,505.00	0.30	147.30	4,504.27	-12.69	5.42	-13.11	0.25	0.00	48.17
4,599.00	0.40	191.60	4,598.27	-13.22	5.48	-13.64	0.30	0.11	47.13
4,692.00	0.40	214.70	4,691.26	-13.81	5.23	-14.20	0.17	0.00	24.84
4,785.00	0.50	181.80	4,784.26	-14.48	5.03	-14.86	0.29	0.11	-35.38
4,879.00	0.50	145.60	4,878.26	-15.23	5.25	-15.62	0.33	0.00	-38.51
4,972.00	0.30	118.90	4,971.26	-15.68	5.70	-16.11	0.29	-0.22	-28.71
5,066.00	0.60	36.80	5,065.25	-15.40	6.21	-15.88	0.67	0.32	-87.34
5,159.00	0.90	61.40	5,158.24	-14.66	7.14	-15.22	0.47	0.32	26.45
5,253.00	0.90	67.50	5,252.23	-14.03	8.47	-14.70	0.10	0.00	6.49
5,347.00	1.00	83.00	5,346.22	-13.65	9.97	-14.45	0.29	0.11	16.49
5,440.00	1.20	76.90	5,439.20	-13.33	11.72	-14.28	0.25	0.22	-6.56
5,533.00	1.40	65.40	5,532.18	-12.63	13.70	-13.76	0.35	0.22	-12.37
5,627.00	1.80	55.40	5,626.14	-11.32	15.96	-12.64	0.52	0.43	-10.64
5,720.00	1.80	46.60	5,719.10	-9.48	18.22	-11.01	0.30	0.00	-9.46
5,813.00	2.00	56.10	5,812.05	-7.57	20.63	-9.32	0.40	0.22	10.22
5,907.00	1.80	57.50	5,905.99	-5.87	23.24	-7.84	0.22	-0.21	1.49
6,000.00	1.00	48.00	5,998.97	-4.54	25.07	-6.67	0.89	-0.86	-10.22
6,094.00	0.90	44.00	6,092.95	-3.46	26.20	-5.69	0.13	-0.11	-4.26
6,187.00	0.10	18.10	6,185.95	-2.86	26.73	-5.14	0.87	-0.86	-27.85
6,281.00	0.40	343.70	6,279.95	-2.46	26.66	-4.74	0.34	0.32	-36.60
6,375.00	0.30	36.10	6,373.95	-1.95	26.71	-4.23	0.34	-0.11	55.74
6,469.00	0.30	336.70	6,467.95	-1.52	26.76	-3.81	0.32	0.00	-63.19
6,562.00	0.50	318.70	6,560.94	-1.00	26.40	-3.26	0.25	0.22	-19.35
6,655.00	0.40	312.80	6,653.94	-0.47	25.89	-2.69	0.12	-0.11	-6.34

LEAM Drilling Systems LLC

Survey Report

Company:	Continental Resources	Local Co-ordinate Reference:	Well 2-16H
Project:	McKenzie County, ND	TVD Reference:	GL 1920+KB 20 @ 1940.00usft (Cyclone 4)
Site:	Columbus Federal 2	MD Reference:	GL 1920+KB 20 @ 1940.00usft (Cyclone 4)
Well:	2-16H	North Reference:	True
Wellbore:	OH	Survey Calculation Method:	Minimum Curvature
Design:	OH	Database:	EDM 5000.1 Multi User DB

Survey										
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)	
6,749.00	0.10	343.50	6,747.94	-0.17	25.63	-2.37	0.34	-0.32	32.66	
6,843.00	0.30	69.10	6,841.94	0.00	25.83	-2.22	0.33	0.21	91.06	
6,936.00	0.40	68.90	6,934.94	0.20	26.36	-2.06	0.11	0.11	-0.22	
7,030.00	0.50	325.90	7,028.94	0.66	26.44	-1.61	0.75	0.11	-109.57	
7,123.00	1.00	319.60	7,121.93	1.61	25.69	-0.60	0.54	0.54	-6.77	
7,216.00	1.00	319.80	7,214.91	2.85	24.64	0.73	0.00	0.00	0.22	
7,309.00	0.90	312.80	7,307.90	3.97	23.58	1.93	0.16	-0.11	-7.53	
7,403.00	0.80	301.90	7,401.89	4.82	22.48	2.87	0.20	-0.11	-11.60	
7,497.00	0.90	287.30	7,495.88	5.38	21.22	3.54	0.25	0.11	-15.53	
7,590.00	0.60	274.80	7,588.87	5.64	20.03	3.90	0.37	-0.32	-13.44	
7,683.00	0.60	270.70	7,681.87	5.69	19.06	4.03	0.05	0.00	-4.41	
7,777.00	0.50	267.80	7,775.86	5.68	18.16	4.10	0.11	-0.11	-3.09	
7,870.00	0.70	252.30	7,868.86	5.49	17.21	3.99	0.28	0.22	-16.67	
7,963.00	0.80	262.70	7,961.85	5.23	16.03	3.84	0.18	0.11	11.18	
8,056.00	0.40	255.60	8,054.84	5.07	15.07	3.76	0.44	-0.43	-7.63	
8,149.00	0.40	252.50	8,147.84	4.89	14.45	3.63	0.02	0.00	-3.33	
8,243.00	0.30	237.90	8,241.84	4.66	13.92	3.45	0.14	-0.11	-15.53	
8,336.00	0.40	248.60	8,334.84	4.41	13.42	3.25	0.13	0.11	11.51	
8,429.00	0.60	252.30	8,427.83	4.15	12.65	3.05	0.22	0.22	3.98	
8,522.00	0.60	273.40	8,520.83	4.03	11.70	3.01	0.24	0.00	22.69	
8,616.00	0.70	285.00	8,614.82	4.21	10.65	3.28	0.18	0.11	12.34	
8,709.00	0.70	297.60	8,707.82	4.62	9.60	3.78	0.17	0.00	13.55	
8,803.00	0.80	289.40	8,801.81	5.10	8.47	4.35	0.16	0.11	-8.72	
8,896.00	1.00	280.10	8,894.80	5.46	7.06	4.83	0.27	0.22	-10.00	
8,990.00	0.60	286.20	8,988.79	5.74	5.78	5.22	0.43	-0.43	6.49	
9,083.00	0.40	301.10	9,081.78	6.04	5.04	5.59	0.25	-0.22	16.02	
9,176.00	0.40	291.30	9,174.78	6.33	4.46	5.92	0.07	0.00	-10.54	
9,270.00	0.40	288.70	9,268.78	6.55	3.84	6.20	0.02	0.00	-2.77	
9,363.00	0.40	277.10	9,361.78	6.70	3.21	6.40	0.09	0.00	-12.47	
9,457.00	0.40	260.40	9,455.78	6.68	2.56	6.44	0.12	0.00	-17.77	
9,550.00	0.40	253.20	9,548.77	6.54	1.93	6.35	0.05	0.00	-7.74	
9,643.00	0.50	253.50	9,641.77	6.33	1.23	6.20	0.11	0.11	0.32	
9,736.00	0.30	222.60	9,734.77	6.03	0.68	5.95	0.31	-0.22	-33.23	
9,830.00	0.30	244.20	9,828.77	5.74	0.29	5.70	0.12	0.00	22.98	
9,923.00	0.30	256.70	9,921.77	5.58	-0.17	5.58	0.07	0.00	13.44	
10,016.00	0.40	274.30	10,014.76	5.55	-0.73	5.59	0.16	0.11	18.92	
10,081.00	0.40	256.90	10,079.76	5.52	-1.18	5.60	0.19	0.00	-26.77	
10,112.00	0.40	260.70	10,110.76	5.47	-1.39	5.57	0.09	0.00	12.26	
10,143.00	2.00	317.50	10,141.75	5.85	-1.86	5.99	5.85	5.16	183.23	
10,174.00	7.10	328.00	10,172.65	7.88	-3.24	8.13	16.60	16.45	33.87	
10,205.00	12.10	327.90	10,203.20	12.26	-5.98	12.73	16.13	16.13	-0.32	
10,236.00	16.70	325.40	10,233.22	18.68	-10.24	19.49	14.97	14.84	-8.06	
10,267.00	21.20	321.20	10,262.53	26.72	-16.29	28.02	15.16	14.52	-13.55	

LEAM Drilling Systems LLC

Survey Report

Company:	Continental Resources	Local Co-ordinate Reference:	Well 2-16H
Project:	McKenzie County, ND	TVD Reference:	GL 1920+KB 20 @ 1940.00usft (Cyclone 4)
Site:	Columbus Federal 2	MD Reference:	GL 1920+KB 20 @ 1940.00usft (Cyclone 4)
Well:	2-16H	North Reference:	True
Wellbore:	OH	Survey Calculation Method:	Minimum Curvature
Design:	OH	Database:	EDM 5000.1 Multi User DB

Survey									
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
10,298.00	24.40	318.70	10,291.11	35.90	-24.03	37.83	10.78	10.32	-8.06
10,329.00	26.90	315.20	10,319.05	45.69	-33.20	48.37	9.43	8.06	-11.29
10,361.00	29.80	312.00	10,347.21	56.15	-44.21	59.74	10.23	9.06	-10.00
10,392.00	31.70	314.30	10,373.86	66.99	-55.77	71.53	7.21	6.13	7.42
10,423.00	36.60	313.80	10,399.50	79.09	-68.27	84.65	15.83	15.81	-1.61
10,454.00	40.30	312.40	10,423.78	92.25	-82.35	98.97	12.26	11.94	-4.52
10,485.00	44.10	309.80	10,446.74	105.92	-98.05	113.94	13.49	12.26	-8.39
10,516.00	47.80	307.80	10,468.29	119.87	-115.42	129.32	12.80	11.94	-6.45
10,547.00	51.60	307.30	10,488.34	134.27	-134.16	145.28	12.32	12.26	-1.61
10,578.00	56.30	308.40	10,506.58	149.65	-153.94	162.30	15.43	15.16	3.55
10,609.00	59.90	310.10	10,522.96	166.31	-174.31	180.64	12.51	11.61	5.48
10,641.00	62.80	311.90	10,538.30	184.73	-195.50	200.82	10.32	9.06	5.63
10,672.00	66.50	313.10	10,551.57	203.66	-216.14	221.44	12.44	11.94	3.87
10,703.00	70.60	313.80	10,562.90	223.50	-237.08	243.01	13.39	13.23	2.26
10,734.00	75.30	313.50	10,571.99	243.95	-258.52	265.22	15.19	15.16	-0.97
10,765.00	79.40	312.40	10,578.78	264.55	-280.66	287.65	13.67	13.23	-3.55
10,796.00	84.20	311.00	10,583.20	284.95	-303.56	309.94	16.12	15.48	-4.52
10,821.00	87.50	310.80	10,585.01	301.28	-322.41	327.82	13.22	13.20	-0.80
10,873.00	89.00	310.80	10,586.59	335.24	-361.75	365.03	2.88	2.88	0.00
10,968.00	88.00	311.90	10,589.08	397.98	-433.04	433.65	1.56	-1.05	1.16
11,063.00	89.70	314.70	10,590.99	463.11	-502.16	504.47	3.45	1.79	2.95
11,158.00	91.40	317.30	10,590.08	531.43	-568.14	578.20	3.27	1.79	2.74
11,253.00	92.70	321.40	10,586.68	603.45	-629.97	655.25	4.52	1.37	4.32
11,348.00	91.20	324.40	10,583.44	679.16	-687.23	735.60	3.53	-1.58	3.16
11,443.00	90.90	325.90	10,581.70	757.11	-741.51	817.91	1.61	-0.32	1.58
11,538.00	90.90	329.50	10,580.21	837.39	-792.26	902.25	3.79	0.00	3.79
11,633.00	90.20	332.30	10,579.30	920.38	-838.45	988.90	3.04	-0.74	2.95
11,728.00	88.80	334.90	10,580.13	1,005.46	-880.69	1,077.29	3.11	-1.47	2.74
11,823.00	88.50	338.40	10,582.37	1,092.64	-918.32	1,167.38	3.70	-0.32	3.68
11,918.00	90.20	340.50	10,583.44	1,181.58	-951.66	1,258.85	2.84	1.79	2.21
12,013.00	90.30	343.00	10,583.03	1,271.80	-981.41	1,351.28	2.63	0.11	2.63
12,108.00	90.80	345.60	10,582.12	1,363.24	-1,007.12	1,444.59	2.79	0.53	2.74
12,203.00	90.90	348.30	10,580.71	1,455.77	-1,028.56	1,538.62	2.84	0.11	2.84
12,298.00	90.60	350.50	10,579.46	1,549.13	-1,046.04	1,633.14	2.34	-0.32	2.32
12,393.00	88.10	353.40	10,580.54	1,643.17	-1,059.34	1,727.97	4.03	-2.63	3.05
12,488.00	89.50	355.60	10,582.53	1,737.71	-1,068.44	1,822.94	2.74	1.47	2.32
12,583.00	91.20	358.30	10,581.95	1,832.56	-1,073.49	1,917.87	3.36	1.79	2.84
12,678.00	89.20	0.70	10,581.62	1,927.54	-1,074.32	2,012.58	3.29	-2.11	2.53
12,772.00	90.20	0.60	10,582.11	2,021.53	-1,073.25	2,106.13	1.07	1.06	-0.11
12,868.00	89.70	0.60	10,582.19	2,117.53	-1,072.25	2,201.69	0.52	-0.52	0.00
12,962.00	90.20	0.90	10,582.28	2,211.52	-1,071.02	2,295.23	0.62	0.53	0.32
13,057.00	87.80	0.20	10,583.93	2,306.49	-1,070.11	2,389.77	2.63	-2.53	-0.74
13,133.00	87.80	0.20	10,586.85	2,382.44	-1,069.84	2,465.41	0.00	0.00	0.00
Projection to Bit									

LEAM Drilling Systems LLC

Survey Report

ST

Company:	Continental Resources	Local Co-ordinate Reference:	Well 2-16H
Project:	McKenzie County, ND	TVD Reference:	GL 1920+KB 20 @ 1940.00usft (Cyclone 4)
Site:	Columbus Federal 2	MD Reference:	GL 1920+KB 20 @ 1940.00usft (Cyclone 4)
Well:	2-16H	North Reference:	True
Wellbore:	ST01	Survey Calculation Method:	Minimum Curvature
Design:	ST01	Database:	EDM 5000.1 Multi User DB

Project	McKenzie County, ND		
Map System:	US State Plane 1983	System Datum:	Mean Sea Level
Geo Datum:	North American Datum 1983		
Map Zone:	North Dakota Northern Zone		

Site	Columbus Federal 2, Sec. 16 - T153N - R101W		
Site Position:		Northing:	408,198.98 usft
From:	Lat/Long	Easting:	1,193,822.32 usft
Position Uncertainty:	0.00 usft	Slot Radius:	0 "
		Latitude:	48° 4' 31.258 N
		Longitude:	103° 40' 11.343 W
		Grid Convergence:	-2.36 °

Well	2-16H		
Well Position	+N/-S	0.00 usft	Northing:
	+E/-W	0.00 usft	Easting:
Position Uncertainty	0.00 usft	Wellhead Elevation:	usft
		Latitude:	48° 4' 31.258 N
		Longitude:	103° 40' 11.343 W
		Ground Level:	1,920.00 usft

Wellbore	ST01				
Magnetics	Model Name	Sample Date	Declination (°)	Dip Angle (°)	Field Strength (nT)
	IGRF2010	12/06/13	8.39	73.01	56,485

Design	ST01				
Audit Notes:					
Version:	1.0	Phase:	ACTUAL	Tie On Depth:	12,488.00
Vertical Section:	Depth From (TVD) (usft)	+N/-S (usft)	+E/-W (usft)	Direction (°)	
	0.00	0.00	0.00	354.94	

Survey Program	Date	01/16/14			
From (usft)	To (usft)	Survey (Wellbore)	Tool Name	Description	
142.00	1,917.00	Survey #1-NEUSCO (OH)	MWD-ISCWSA	MWD - Standard	
1,983.00	12,488.00	Survey #2 (OH)	LEAM MWD-ADJ	MWD - Standard	
12,583.00	23,427.00	Survey #1 (ST01)	LEAM MWD-ADJ	MWD - Standard	
23,475.00	23,475.00	Survey #2 (ST01)	Project	Projection	

Survey										
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)	
12,488.00	89.50	355.60	10,582.53	1,737.71	-1,068.44	1,825.22	2.74	1.47	2.32	
12,583.00	85.30	355.60	10,586.84	1,832.31	-1,075.72	1,920.09	4.42	-4.42	0.00	
12,678.00	88.20	354.60	10,592.22	1,926.80	-1,083.82	2,014.93	3.23	3.05	-1.05	
12,772.00	90.50	355.80	10,593.29	2,020.45	-1,091.68	2,108.91	2.76	2.45	1.28	
12,868.00	90.80	359.00	10,592.20	2,116.34	-1,096.04	2,204.81	3.35	0.31	3.33	
12,962.00	90.00	0.20	10,591.55	2,210.33	-1,096.69	2,298.49	1.53	-0.85	1.28	
13,057.00	88.30	1.10	10,592.95	2,305.31	-1,095.62	2,393.00	2.02	-1.79	0.95	
13,152.00	88.80	0.70	10,595.36	2,400.27	-1,094.12	2,487.46	0.67	0.53	-0.42	

LEAM Drilling Systems LLC

Survey Report

Company:	Continental Resources	Local Co-ordinate Reference:	Well 2-16H
Project:	McKenzie County, ND	TVD Reference:	GL 1920+KB 20 @ 1940.00usft (Cyclone 4)
Site:	Columbus Federal 2	MD Reference:	GL 1920+KB 20 @ 1940.00usft (Cyclone 4)
Well:	2-16H	North Reference:	True
Wellbore:	ST01	Survey Calculation Method:	Minimum Curvature
Design:	ST01	Database:	EDM 5000.1 Multi User DB

Survey									
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
13,247.00	89.50	0.70	10,596.77	2,495.25	-1,092.96	2,581.97	0.74	0.74	0.00
13,342.00	88.30	0.70	10,598.59	2,590.22	-1,091.80	2,676.47	1.26	-1.26	0.00
13,434.00	89.50	1.30	10,600.36	2,682.19	-1,090.20	2,767.93	1.46	1.30	0.65
13,526.00	89.30	1.40	10,601.32	2,774.16	-1,088.03	2,859.35	0.24	-0.22	0.11
13,618.00	88.40	1.60	10,603.17	2,866.11	-1,085.62	2,950.73	1.00	-0.98	0.22
13,710.00	88.40	1.60	10,605.74	2,958.04	-1,083.06	3,042.07	0.00	0.00	0.00
13,802.00	88.10	1.60	10,608.55	3,049.96	-1,080.49	3,133.41	0.33	-0.33	0.00
13,894.00	89.40	2.30	10,610.55	3,141.88	-1,077.36	3,224.70	1.60	1.41	0.76
13,986.00	90.10	1.10	10,610.95	3,233.84	-1,074.63	3,316.05	1.51	0.76	-1.30
14,078.00	90.50	1.30	10,610.47	3,325.81	-1,072.70	3,407.50	0.49	0.43	0.22
14,169.00	88.80	1.10	10,611.03	3,416.79	-1,070.80	3,497.96	1.88	-1.87	-0.22
14,261.00	90.20	359.90	10,611.83	3,508.78	-1,069.99	3,589.51	2.00	1.52	-1.30
14,354.00	88.40	359.30	10,612.97	3,601.76	-1,070.64	3,682.20	2.04	-1.94	-0.65
14,446.00	91.00	359.50	10,613.45	3,693.75	-1,071.61	3,773.91	2.83	2.83	0.22
14,538.00	89.40	358.80	10,613.13	3,785.74	-1,072.97	3,865.66	1.90	-1.74	-0.76
14,631.00	88.70	357.80	10,614.67	3,878.68	-1,075.73	3,958.48	1.31	-0.75	-1.08
14,723.00	91.50	359.20	10,614.51	3,970.64	-1,078.14	4,050.29	3.40	3.04	1.52
14,816.00	91.30	358.60	10,612.24	4,063.59	-1,079.92	4,143.04	0.68	-0.22	-0.65
14,908.00	91.30	358.50	10,610.15	4,155.54	-1,082.25	4,234.83	0.11	0.00	-0.11
15,001.00	90.40	0.20	10,608.77	4,248.52	-1,083.30	4,327.54	2.07	-0.97	1.83
15,093.00	91.40	0.20	10,607.33	4,340.50	-1,082.98	4,419.14	1.09	1.09	0.00
15,186.00	89.30	0.70	10,606.76	4,433.49	-1,082.25	4,511.71	2.32	-2.26	0.54
15,279.00	89.60	0.70	10,607.65	4,526.48	-1,081.12	4,604.23	0.32	0.32	0.00
15,372.00	91.10	359.00	10,607.08	4,619.48	-1,081.36	4,696.88	2.44	1.61	-1.83
15,465.00	91.70	358.80	10,604.81	4,712.43	-1,083.14	4,789.63	0.68	0.65	-0.22
15,557.00	90.00	359.50	10,603.44	4,804.41	-1,084.51	4,881.37	2.00	-1.85	0.76
15,649.00	90.80	359.70	10,602.80	4,896.40	-1,085.15	4,973.06	0.90	0.87	0.22
15,741.00	91.30	359.50	10,601.12	4,988.38	-1,085.79	5,064.74	0.59	0.54	-0.22
15,835.00	90.70	359.30	10,599.48	5,082.36	-1,086.78	5,158.44	0.67	-0.64	-0.21
15,927.00	89.90	359.90	10,598.99	5,174.36	-1,087.42	5,250.13	1.09	-0.87	0.65
16,020.00	90.30	359.90	10,598.83	5,267.36	-1,087.58	5,342.79	0.43	0.43	0.00
16,115.00	89.40	359.70	10,599.08	5,362.36	-1,087.91	5,437.44	0.97	-0.95	-0.21
16,210.00	89.00	359.30	10,600.41	5,457.34	-1,088.74	5,532.13	0.60	-0.42	-0.42
16,306.00	89.20	358.60	10,601.92	5,553.31	-1,090.50	5,627.88	0.76	0.21	-0.73
16,401.00	90.50	0.00	10,602.16	5,648.30	-1,091.66	5,722.60	2.01	1.37	1.47
16,496.00	91.30	359.90	10,600.67	5,743.29	-1,091.75	5,817.23	0.85	0.84	-0.11
16,590.00	90.50	359.20	10,599.20	5,837.27	-1,092.48	5,910.91	1.13	-0.85	-0.74
16,685.00	91.30	0.00	10,597.70	5,932.26	-1,093.15	6,005.58	1.19	0.84	0.84
16,780.00	91.20	359.00	10,595.63	6,027.23	-1,093.98	6,100.26	1.06	-0.11	-1.05
16,875.00	92.10	358.50	10,592.90	6,122.17	-1,096.05	6,195.01	1.08	0.95	-0.53
16,970.00	92.00	357.60	10,589.50	6,217.05	-1,099.28	6,289.81	0.95	-0.11	-0.95
17,065.00	92.10	358.10	10,586.10	6,311.92	-1,102.84	6,384.62	0.54	0.11	0.53
17,160.00	92.10	356.90	10,582.62	6,406.77	-1,106.98	6,479.46	1.26	0.00	-1.26

LEAM Drilling Systems LLC

Survey Report

Company:	Continental Resources	Local Co-ordinate Reference:	Well 2-16H
Project:	McKenzie County, ND	TVD Reference:	GL 1920+KB 20 @ 1940.00usft (Cyclone 4)
Site:	Columbus Federal 2	MD Reference:	GL 1920+KB 20 @ 1940.00usft (Cyclone 4)
Well:	2-16H	North Reference:	True
Wellbore:	ST01	Survey Calculation Method:	Minimum Curvature
Design:	ST01	Database:	EDM 5000.1 Multi User DB

Survey									
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
17,255.00	91.50	359.30	10,579.63	6,501.66	-1,110.13	6,574.26	2.60	-0.63	2.53
17,350.00	92.90	0.90	10,575.99	6,596.58	-1,109.96	6,668.80	2.24	1.47	1.68
17,445.00	90.70	0.90	10,573.00	6,691.52	-1,108.47	6,763.23	2.32	-2.32	0.00
17,540.00	90.80	0.40	10,571.76	6,786.51	-1,107.39	6,857.75	0.54	0.11	-0.53
17,635.00	90.00	359.50	10,571.10	6,881.50	-1,107.48	6,952.39	1.27	-0.84	-0.95
17,730.00	90.00	359.70	10,571.10	6,976.50	-1,108.14	7,047.07	0.21	0.00	0.21
17,824.00	90.00	359.00	10,571.10	7,070.49	-1,109.21	7,140.79	0.74	0.00	-0.74
17,919.00	89.90	358.10	10,571.18	7,165.46	-1,111.61	7,235.60	0.95	-0.11	-0.95
18,014.00	90.00	358.30	10,571.26	7,260.41	-1,114.59	7,330.45	0.24	0.11	0.21
18,109.00	91.40	0.00	10,570.10	7,355.39	-1,116.00	7,425.18	2.32	1.47	1.79
18,204.00	90.00	359.30	10,568.94	7,450.38	-1,116.58	7,519.85	1.65	-1.47	-0.74
18,299.00	92.20	0.20	10,567.12	7,545.35	-1,117.00	7,614.49	2.50	2.32	0.95
18,394.00	90.70	0.60	10,564.71	7,640.32	-1,116.34	7,709.02	1.63	-1.58	0.42
18,489.00	89.20	0.60	10,564.79	7,735.31	-1,115.34	7,803.56	1.58	-1.58	0.00
18,584.00	89.30	0.70	10,566.04	7,830.30	-1,114.26	7,898.08	0.15	0.11	0.11
18,679.00	90.20	0.70	10,566.45	7,925.29	-1,113.10	7,992.60	0.95	0.95	0.00
18,774.00	90.00	0.90	10,566.29	8,020.28	-1,111.78	8,087.10	0.30	-0.21	0.21
18,869.00	88.90	0.60	10,567.20	8,115.26	-1,110.53	8,181.60	1.20	-1.16	-0.32
18,964.00	89.80	1.30	10,568.28	8,210.24	-1,108.96	8,276.07	1.20	0.95	0.74
19,059.00	90.60	0.60	10,567.95	8,305.23	-1,107.38	8,370.55	1.12	0.84	-0.74
19,154.00	90.70	1.10	10,566.87	8,400.21	-1,105.97	8,465.04	0.54	0.11	0.53
19,249.00	89.50	0.20	10,566.70	8,495.20	-1,104.90	8,559.56	1.58	-1.26	-0.95
19,344.00	90.60	1.80	10,566.62	8,590.18	-1,103.24	8,654.03	2.04	1.16	1.68
19,439.00	90.20	0.70	10,565.96	8,685.16	-1,101.17	8,748.45	1.23	-0.42	-1.16
19,534.00	92.20	0.20	10,563.97	8,780.13	-1,100.42	8,842.98	2.17	2.11	-0.53
19,629.00	92.70	359.90	10,559.91	8,875.04	-1,100.34	8,937.52	0.61	0.53	-0.32
19,724.00	90.70	0.40	10,557.09	8,969.99	-1,100.09	9,032.08	2.17	-2.11	0.53
19,819.00	89.40	0.90	10,557.00	9,064.98	-1,099.01	9,126.60	1.47	-1.37	0.53
19,914.00	90.00	0.70	10,557.50	9,159.97	-1,097.68	9,221.10	0.67	0.63	-0.21
20,009.00	90.90	0.40	10,556.76	9,254.96	-1,096.77	9,315.65	1.00	0.95	-0.32
20,104.00	89.30	0.70	10,556.59	9,349.96	-1,095.86	9,410.19	1.71	-1.68	0.32
20,199.00	88.50	0.60	10,558.41	9,444.93	-1,094.78	9,504.70	0.85	-0.84	-0.11
20,293.00	88.40	0.60	10,560.96	9,538.89	-1,093.80	9,598.20	0.11	-0.11	0.00
20,388.00	88.90	359.70	10,563.19	9,633.86	-1,093.55	9,692.78	1.08	0.53	-0.95
20,483.00	88.50	1.40	10,565.35	9,728.83	-1,092.64	9,787.30	1.84	-0.42	1.79
20,578.00	89.70	0.90	10,566.84	9,823.80	-1,090.73	9,881.73	1.37	1.26	-0.53
20,673.00	90.90	0.90	10,566.34	9,918.78	-1,089.24	9,976.21	1.26	1.26	0.00
20,768.00	90.70	0.20	10,565.02	10,013.77	-1,088.33	10,070.75	0.77	-0.21	-0.74
20,863.00	90.20	0.60	10,564.27	10,108.76	-1,087.66	10,165.31	0.67	-0.53	0.42
20,958.00	89.90	359.90	10,564.19	10,203.76	-1,087.25	10,259.90	0.80	-0.32	-0.74
21,053.00	89.30	0.00	10,564.85	10,298.76	-1,087.33	10,354.54	0.64	-0.63	0.11
21,148.00	90.90	0.00	10,564.69	10,393.76	-1,087.33	10,449.16	1.68	1.68	0.00
21,243.00	90.90	359.50	10,563.19	10,488.74	-1,087.75	10,543.81	0.53	0.00	-0.53

LEAM Drilling Systems LLC

Survey Report

Company:	Continental Resources	Local Co-ordinate Reference:	Well 2-16H
Project:	McKenzie County, ND	TVD Reference:	GL 1920+KB 20 @ 1940.00usft (Cyclone 4)
Site:	Columbus Federal 2	MD Reference:	GL 1920+KB 20 @ 1940.00usft (Cyclone 4)
Well:	2-16H	North Reference:	True
Wellbore:	ST01	Survey Calculation Method:	Minimum Curvature
Design:	ST01	Database:	EDM 5000.1 Multi User DB

Survey										
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)	
21,338.00	90.50	359.30	10,562.03	10,583.73	-1,088.74	10,638.52	0.47	-0.42	-0.21	
21,433.00	89.20	359.20	10,562.28	10,678.72	-1,089.99	10,733.25	1.37	-1.37	-0.11	
21,528.00	90.00	359.20	10,562.95	10,773.71	-1,091.31	10,827.98	0.84	0.84	0.00	
21,623.00	89.60	358.50	10,563.28	10,868.69	-1,093.22	10,922.76	0.85	-0.42	-0.74	
21,718.00	90.00	0.20	10,563.61	10,963.68	-1,094.30	11,017.47	1.84	0.42	1.79	
21,813.00	89.30	0.40	10,564.19	11,058.67	-1,093.80	11,112.05	0.77	-0.74	0.21	
21,908.00	90.70	0.20	10,564.19	11,153.67	-1,093.30	11,206.64	1.49	1.47	-0.21	
22,003.00	90.70	359.50	10,563.03	11,248.66	-1,093.55	11,301.28	0.74	0.00	-0.74	
22,098.00	89.70	359.50	10,562.70	11,343.66	-1,094.38	11,395.98	1.05	-1.05	0.00	
22,193.00	89.50	359.50	10,563.36	11,438.65	-1,095.21	11,490.67	0.21	-0.21	0.00	
22,288.00	89.40	359.50	10,564.27	11,533.64	-1,096.04	11,585.37	0.11	-0.11	0.00	
22,383.00	90.70	359.50	10,564.19	11,628.64	-1,096.87	11,680.07	1.37	1.37	0.00	
22,478.00	91.10	359.30	10,562.70	11,723.62	-1,097.86	11,774.77	0.47	0.42	-0.21	
22,573.00	91.50	359.00	10,560.54	11,818.58	-1,099.27	11,869.48	0.53	0.42	-0.32	
22,668.00	91.50	357.60	10,558.05	11,913.51	-1,102.09	11,964.29	1.47	0.00	-1.47	
22,763.00	91.50	357.20	10,555.57	12,008.38	-1,106.40	12,059.16	0.42	0.00	-0.42	
22,858.00	89.80	357.90	10,554.49	12,103.28	-1,110.46	12,154.06	1.94	-1.79	0.74	
22,952.00	89.50	359.20	10,555.06	12,197.25	-1,112.83	12,247.87	1.42	-0.32	1.38	
23,047.00	89.60	359.30	10,555.81	12,292.24	-1,114.08	12,342.59	0.15	0.11	0.11	
23,142.00	90.00	358.60	10,556.14	12,387.22	-1,115.82	12,437.36	0.85	0.42	-0.74	
23,237.00	90.60	358.50	10,555.64	12,482.19	-1,118.22	12,532.17	0.64	0.63	-0.11	
23,332.00	89.60	358.10	10,555.48	12,577.14	-1,121.04	12,627.00	1.13	-1.05	-0.42	
23,427.00	90.00	357.40	10,555.81	12,672.07	-1,124.77	12,721.89	0.85	0.42	-0.74	
23,475.00	90.00	357.40	10,555.81	12,720.02	-1,126.95	12,769.84	0.00	0.00	0.00	
Projection to Bit										



SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
800 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)



Well File No. **25159**

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☒ Notice of Intent	Approximate Start Date August 16, 2013	☐ Drilling Prognosis	☐ Spill Report
☐ Report of Work Done	Date Work Completed	☐ Redrilling or Repair	☐ Shooting
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date	☐ Casing or Liner	☐ Acidizing
		☐ Plug Well	☐ Fracture Treatment
		☐ Supplemental History	☐ Change Production Method
		☐ Temporarily Abandon	☐ Reclamation
		☒ Other Cement to surface	

Well Name and Number Columbus Federal 2-16H							
Footages		Qtr-Qtr	Section	Township	Range		
2508 F N L 222 F E L		SENE	16	153 N	101 W		
Field		Pool	County				
Bakker		Bakken	McKenzie				

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

During the cement job on the Columbus Federal 2-16H, returns were lost after dropping the plug. Returns resumed temporarily but then became very low. Cement was not returned to surface. A CBL was scheduled the morning of Thursday, August 15th, and indicated the top of cement to be at approximately 500'. Cement will be brought to surface on the intermediate casing string.

"A 1" top job was performed. We got 1" pipe 180' deep. We got cement to surface and continued pumping and then staging to make sure the cement would not drop."

***The Operator is to maintain a pressure gauge on the 9-5/8" X 7" annulus for continuous monitoring and to report any pressure development or pressure management events.**

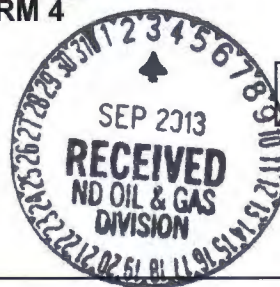
Company Continental Resources, Inc.		Telephone Number (405) 234-9000	
Address P.O. Box 268870			
City Oklahoma City		State OK	Zip Code 73126
Signature <i>Becky Barnes</i>		Printed Name Becky Barnes	
Title Regulatory Compliance Specialist		Date August 19, 2013	
Email Address becky.barnes@clr.com			

FOR STATE USE ONLY	
☒ Received	☐ Approved
Date	9-16-2013
By	<i>Richard A. Suggs</i>
Title	Richard A. Suggs Geologist



SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (03-2004)



Well File No.
25159

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☒ Notice of Intent	Approximate Start Date October 14, 2013
☐ Report of Work Done	Date Work Completed
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☒ Other	Flow back exemption

Well Name and Number Columbus Federal 2-16H					
Footages 2508 F N L	222 F E L	Qtr-Qtr SENE	Section 16	Township 153 N	Range 101 W
Field Baker		Pool Bakken		County McKenzie	

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Continental Resources, Inc. requests a waiver from the tubing/pkr requirement included in NDIC 43-02-03-21: Casing, Tubing, and Cementing Requirements during the completion period immediately following the upcoming fracture stimulation. The following assurances apply:

- 1) The well is equipped with 26#/ft P-110 7" casing at surface with an API burst rating of 9960 psig for the 26 #/ft casing.
- 2) The frac design will use a safety factor of 0.85 * API burst rating to determine the max pressure.
- 3) Damage to the casing during the frac would be detected immediately by monitoring equipment.
- 4) The casing is exposed to significantly lower rates and pressures during flow back than during the frac job.
- 5) The frac fluid and formation fluids have very low corrosion and erosion rates
- 6) Production equipment will be installed as soon as possible after the well ceases flowing.
- 7) A 300# gauge will be installed on surface casing during flowback period.

Company Continental Resources, Inc.	Telephone Number 405-234-9000
Address P.O. Box 269000	
City Oklahoma City	State OK
Signature 	Zip Code 73126
Printed Name Jim Landrigan	
Title Completion Engineer	Date August 29, 2013

FOR STATE USE ONLY	
☐ Received	☒ Approved
Date September 4, 2013	
By 	
Title PETROLEUM ENGINEER	

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2008)



Well File No.
25159

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☒ Notice of Intent	Approximate Start Date August 14, 2013
☐ Report of Work Done	Date Work Completed
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☒ Other	Spud with Small Rig

Well Name and Number Columbus Federal 2-16H							
Footages 2508 F N L		222 F E L	Qtr-Qtr SENE	Section 16	Township 153 N	Range 101 W	
Field Bakker		Pool Bakken		County McKenzie			

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s) Advanced Energy Services, LLC			
Address P.O. Box 85	City South Boardman	State Michigan	Zip Code 49680-0085

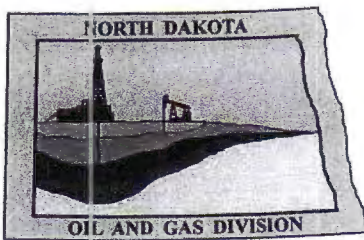
DETAILS OF WORK

Continental Resources, Inc. requests permission for suspension of drilling for up to 90 days for the referenced well under NDAC 43-02-03-55. Continental Resources, Inc. intends to drill the surface hole with freshwater based drilling mud and set surface casing with a small drilling rig and move off within 3 to 5 days. The casing will be set at a depth pre-approved by the NDIC per the Application for Permit to Drill NDAC 43-02-03-21. No saltwater will be used in the drilling and cementing operations of the surface casing. Once the surface casing is cemented, a plug or mechanical seal will be placed at the top of the casing to prevent any foreign matter from getting into the well. A rig capable of drilling to TD will move onto the location within the 90 days previously outlined to complete the drilling and casing plan as per the APD. The undersigned states that this request for suspension of drilling operations in accordance with the Subsection 4 of Section 43-02-03-55 of the NDAC, is being requested to take advantage of the cost savings and time savings of using an initial rig that is smaller than the rig necessary to drill a well to total depth but is not intended to alter or extend the terms and conditions of, or suspend any obligation under, any oil and gas lease with acreage in or under the spacing or drilling unit for the above-referenced well. Continental Resources, Inc. understands NDAC 43-02-03-31 requirements regarding confidentiality pertaining to this permit. The drilling pit will be fenced immediately after construction if the well pad is located in a pasture (NDAC 43-02-03-19 & 19.1). Continental Resources, Inc. will plug and abandon the well and reclaim the well site if the well is not drilled by the larger rotary rig within 90 days after spudding the well with the smaller drilling rig.

Company Continental Resources, Inc.		Telephone Number (405) 234-9000	
Address P.O. Box 268870			
City Oklahoma City		State OK	Zip Code 73126
Signature <i>Becky Barnes</i>	Printed Name Becky Barnes		
Title Regulatory Compliance Specialist	Date July 30, 2013		
Email Address becky.barnes@clr.com			

FOR STATE USE ONLY	
☐ Received	☒ Approved
Date 8-02-2013	
By <i>David Burns</i>	
Title David Burns Engineering Tech.	

CR7 notice NPLS Inspector Richard Burns 701-770-3554 with spud 8/10



Oil and Gas Division

Lynn D. Helms - Director

Bruce E. Hicks - Assistant Director

Department of Mineral Resources

Lynn D. Helms - Director

North Dakota Industrial Commission

www.dmr.nd.gov/oilgas

25159
JA

BECKY BARNES
CONTINENTAL RESOURCES, INC.
PO BOX 1032
ENID, OK 73702-1032 USA

Date: 3/18/2013

RE: CORES AND SAMPLES

Well Name: **COLUMBUS FEDERAL 2-16H** Well File No.: **25159**
Location: **SENE 16-153-101** County: MCKENZIE
Permit Type: **Development - HORIZONTAL**
Field: **BAKER** Target Horizon: THREE FORKS

Dear BECKY BARNES:

North Dakota Century Code (NDCC) Section 38-08-04 provides for the preservation of cores and samples and their shipment to the State Geologist when requested. The following is required on the above referenced well:

- 1) All cores, core chips and samples must be submitted to the State Geologist as provided for the NDCC Section 38-08-04 and North Dakota Administrative Code 43-02-03-38.1.
- 2) Samples shall include all cuttings from:

Base of the Last Charles Salt

Samples of cuttings shall be taken at 30' maximum intervals through all vertical, build and horizontal sections. Samples must be washed, dried, packed in sample envelopes in correct order with labels showing operator, well name, location and depth, and forwarded in standard boxes to the State Geologist within 30 days of the completion of drilling operations.

- 3) Cores: ALL CORES cut shall be preserved in correct order, properly boxed, and forwarded to the State Geologist within 90 days of completion of drilling operations. Any extension of time must have written approval from the State Geologist.
- 4) All cores, core chips, and samples must be shipped, prepaid, to the State Geologist at the following address:
**ND Geological Survey Core Library
Campus Road and Cornell
Grand Forks, ND 58202**
- 5) NDCC Section 38-08-16 allows for a civil penalty for any violation of Chapter 38 08 not to exceed \$12,500 for each offense, and each day's violation is a separate offense.

Sincerely

Richard A. Suggs
Geologist



SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)



Well File No.

25159

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☐ Notice of Intent	Approximate Start Date
☐ Report of Work Done	Date Work Completed
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Progress	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☒ Other	<u>Open Hole Log Waiver</u>

Well Name and Number Columbus Federal 2-16H					
Footages	Qtr-Qtr	Section	Township	Range	
2508 F N L	222 F E L	SENE	16	153 N	101 W
Field	Pool	County			
	Bakken	McKenzie			

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Requested variance to not run openhole logs. GR/CBL/CCL will be run from deepest point obtainable to base of surface casing.

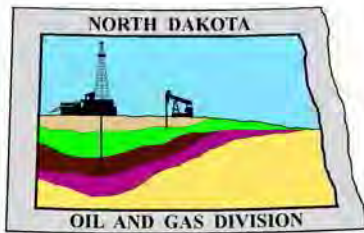
Offset logs used will be the Nance Petroleum, Corps of Engineers 31-10, Sec 10-153N-101W, McKenzie County, ND.

The Gamma Ray Log will be run all the way to surface and all mud logs will be submitted as one digital tiff formatted file and one digital LAS formatted file.

** Approval per loss run on # 10710 - Novak 1-9-3A*

Company Continental Resources, Inc.		Telephone Number 580-233-8955	
Address P.O. Box 1032			
City Enid	State OK	Zip Code 73702	
Signature <i>Terry L. Olson</i>	Printed Name Terry L. Olson		
Title Regulatory Compliance Specialist	Date June 12, 2012		
Email Address Terry.Olson@clr.com			

FOR STATE USE ONLY	
☐ Received	☒ Approved
Date	<i>3/11/2013</i>
By	<i>[Signature]</i>
Title	Richard A. Suggs Geologist



Oil and Gas Division

Lynn D. Helms - Director

Bruce E. Hicks - Assistant Director

Department of Mineral Resources

Lynn D. Helms - Director

North Dakota Industrial Commission

www.oilgas.nd.gov

March 11, 2013

Terry L. Olson
Regulatory Compliance Specialist
CONTINENTAL RESOURCES, INC.
P.O. Box 1032
Enid, OK 73702

RE: **HORIZONTAL WELL
COLUMBUS FEDERAL 2-16H
SENE Section 16-153N-101W
McKenzie County
Well File # 25159**

Dear Terry:

Pursuant to Commission Order No. 21551, approval to drill the above captioned well is hereby given. The approval is granted on the condition that all portions of the well bore not isolated by cement, be no closer than the **200' setback** from the north & south boundaries and **500' setback** from the east & west boundaries within the 2560 acre spacing unit consisting of Sections 4, 9, 16, & 21-T153N-R101W. **Tool error is not required pursuant to order.**

PERMIT STIPULATIONS: Due to surficial water adjacent to the well site, a dike is required surrounding the entire location. A LINER MUST BE PLACED UNDER THE LOCATION. The MOUSE AND RAT HOLE MUST BE CEMENTED. A spill contingency plan must be in place prior to spudding the wells. No Drilling Pit will be allowed. Continental must drill the Columbus Federal 2-16H, 3-16H and Tallahassee 2-16H, 3-16H back to back. One horizontal well shall be drilled and completed in the standup 1280-acre spacing unit described as Sections 4 and 9, T153N-R101W, McKenzie County, north Dakota, prior to completing any horizontal well in the 2560-acre spacing unit described as Sections 4, 9, 16, and 21-T153N-R101W McKenzie County, North Dakota. CONTINENTAL RESOURCES must contact NDIC Field Inspector Richard Dunn at 701-770-3554 prior to location construction.

Drilling pit

NDAC 43-02-03-19.4 states that "a pit may be utilized to bury drill cuttings and solids generated during well drilling and completion operations, providing the pit can be constructed, used and reclaimed in a manner that will prevent pollution of the land surface and freshwaters. Reserve and circulation of mud system through earthen pits are prohibited. All pits shall be inspected by an authorized representative of the director prior to lining and use. Drill cuttings and solids must be stabilized in a manner approved by the director prior to placement in a cuttings pit."

Form 1 Changes & Hard Lines

Any changes, shortening of casing point or lengthening at Total Depth must have prior approval by the NDIC. The proposed directional plan is at a legal location. The minimum legal coordinate from the well head at casing point is: 278' W. Also, based on the azimuth of the proposed lateral the maximum legal coordinate from the well head is: 12755' N and a minimum legal coordinate from the well head of 278' W.

Location Construction Commencement (Three Day Waiting Period)

Operators shall not commence operations on a drill site until the 3rd business day following publication of the approved drilling permit on the NDIC - OGD Daily Activity Report. If circumstances require operations to commence before the 3rd business day following publication on the Daily Activity Report, the waiting period may be waived by the Director. Application for a waiver must be by sworn affidavit providing the information necessary to evaluate the extenuating circumstances, the factors of NDAC 43-02-03-16.2 (1), (a)-(f), and any other information that would allow the Director to conclude that in the event another owner seeks revocation of the drilling permit, the applicant should retain the permit.

Permit Fee & Notification

Payment was received in the amount of \$100 via credit card. It is requested that notification be given immediately upon the spudding of the well. This information should be relayed to the Oil & Gas Division, Bismarck, via telephone. The following information must be included: Well name, legal location, permit number, drilling contractor, company representative, date and time of spudding. Office hours are 8:00 a.m. to 12:00 p.m. and 1:00 p.m. to 5:00 p.m. Central Time. Our telephone number is (701) 328-8020, leave a message if after hours or on the weekend.

Survey Requirements for Horizontal, Horizontal Re-entry, and Directional Wells

NDAC Section 43-02-03-25 (Deviation Tests and Directional Surveys) states in part (that) the survey contractor shall file a certified copy of all surveys with the director free of charge within thirty days of completion. Surveys must be submitted as one electronic copy, or in a form approved by the director. However, the director may require the directional survey to be filed immediately after completion if the survey is needed to conduct the operation of the director's office in a timely manner. Certified surveys must be submitted via email in one adobe document, with a certification cover page to certsurvey@nd.gov.

Survey points shall be of such frequency to accurately determine the entire location of the well bore.

Specifically, the Horizontal and Directional well survey frequency is 100 feet in the vertical, 30 feet in the curve (or when sliding) and 90 feet in the lateral.

Confidential status

Your request for confidential status of all information furnished to the Director, or his representatives, is hereby granted. Such information, except production runs, shall remain confidential for six months commencing on the date the well is spud.

Confidential status notwithstanding, the Director and his representatives shall have access to all well records wherever located. Your company personnel, or any person performing work for your company shall permit the Director and his representatives to come upon any lease, property, well, or drilling rig operated or controlled by them, complying with all safety rules, and to inspect the records and operation of such wells and to have access at all times to any and all records of wells. The Commission's field personnel periodically inspect producing and drilling wells. Any information regarding such wells shall be made available to them at any time upon request. The information so obtained by the field personnel shall be maintained in strict confidence and shall be available only to the Commission and its staff.

Surface casing cement

Tail cement utilized on surface casing must have a minimum compressive strength of 500 psi within 12 hours, and tail cement utilized on production casing must have a minimum compressive strength of 500 psi before drilling the plug or initiating tests.

Logs

NDAC Section 43-02-03-31 requires the running of (1) a suite of open hole logs from which formation tops and porosity zones can be determined, (2) a Gamma Ray Log run from total depth to ground level elevation of the well bore, and (3) a log from which the presence and quality of cement can be determined (Standard CBL or Ultrasonic cement evaluation log) in every well in which production or intermediate casing has been set, this log must be run prior to completing the well. All logs run must be submitted free of charge, as one digital TIFF (tagged image file format) copy and one digital LAS (log ASCII) formatted copy. Digital logs may be submitted on a standard CD, DVD, or attached to an email sent to digitallogs@nd.gov. Thank you for your cooperation.

Sincerely,

David Tabor
Engineering Technician IV



APPLICATION FOR PERMIT TO DRILL HORIZONTAL WELL - FORM 1H

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 54269 (08-2005)

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

Type of Work New Location	Type of Well Oil & Gas	Approximate Date Work Will Start 6 / 13 / 2012	Confidential Status Yes
Operator CONTINENTAL RESOURCES, INC.			Telephone Number 580-233-8955
Address P.O. Box 1032		City Enid	State OK Zip Code 73702

☒ Notice has been provided to the owner of any permanently occupied dwelling within 1,320 feet.

☒ This well is not located within five hundred feet of an occupied dwelling.

WELL INFORMATION (If more than one lateral proposed, enter data for additional laterals on page 2)

Well Name COLUMBUS FEDERAL				Well Number 2-16H			
Surface Footages 2508 F N L 222 F E L		Qtr-Qtr SENE	Section 16	Township 153 N	Range 101 W	County McKenzie	
Longstring Casing Point Footages 202 F S L 1176 F E L		Qtr-Qtr SESE	Section 9	Township 153 N	Range 101 W	County McKenzie	
Longstring Casing Point Coordinates From Well Head 2709 N From WH 954 W From WH		Azimuth 341 °	Longstring Total Depth 13290 Feet MD 10607 Feet TVD				
Bottom Hole Footages From Nearest Section Line 200 F N L 1320 F E L		Qtr-Qtr NWNE	Section 4	Township 153 N	Range 101 W	County McKenzie	
Bottom Hole Coordinates From Well Head 12755 N From WH 1098 W From WH		KOP Lateral 1 10166 Feet MD	Azimuth Lateral 1 0 °		Estimated Total Depth Lateral 1 23354 Feet MD 10607 Feet TVD		
Latitude of Well Head 48 ° 04 ' 31.26 "		Longitude of Well Head -103 ° 40 ' 11.34 "		NAD Reference NAD83	Description of (Subject to NDIC Approval) Spacing Unit: Sec 4, 9, 16, & 21 153 101		
Ground Elevation 1921 Feet Above S.L.	Acres in Spacing/Drilling Unit 2560	Spacing/Drilling Unit Setback Requirement 200 Feet N/S 500 Feet E/W			Industrial Commission Order 21551		
North Line of Spacing/Drilling Unit 5279 Feet	South Line of Spacing/Drilling Unit 5300 Feet	East Line of Spacing/Drilling Unit 20988 Feet			West Line of Spacing/Drilling Unit 20969 Feet		
Objective Horizons Three Forks					Pierre Shale Top 1841		
Proposed Surface Casing	Size 9 - 5/8 "	Weight 36 Lb./Ft.	Depth 1940 Feet	Cement Volume 732 Sacks	NOTE: Surface hole must be drilled with fresh water and surface casing must be cemented back to surface.		
Proposed Longstring Casing	Size 7 - "	Weight(s) 26-32 Lb./Ft.	Longstring Total Depth 13290 Feet MD 10607 Feet TVD		Cement Volume 1170 Sacks	Cement Top 0 Feet	Top Dakota Sand 4826 Feet
Base Last Charles Salt (If Applicable) 9031 Feet		NOTE: Intermediate or longstring casing string must be cemented above the top Dakota Group Sand.					
Proposed Logs CBL/GR from deepest depth obtainable to ground surface/mud							
Drilling Mud Type (Vertical Hole - Below Surface Casing) Invert				Drilling Mud Type (Lateral) Brine			
Survey Type in Vertical Portion of Well MWD Every 100 Feet		Survey Frequency: Build Section 30 Feet		Survey Frequency: Lateral 90 Feet		Survey Contractor LEAM Drilling Services	

NOTE: A Gamma Ray log must be run to ground surface and a CBL must be run on intermediate or longstring casing string if set.

Surveys are required at least every 30 feet in the build section and every 90 feet in the lateral section of a horizontal well. Measurement inaccuracies are not considered when determining compliance with the spacing/drilling unit boundary setback requirement except in the following scenarios: 1) When the angle between the well bore and the respective boundary is 10 degrees or less; or 2) If Industry standard methods and equipment are not utilized. Consult the applicable field order for exceptions.

If measurement inaccuracies are required to be considered, a 2° MWD measurement inaccuracy will be applied to the horizontal portion of the well bore. This measurement inaccuracy is applied to the well bore from KOP to TD.

REQUIRED ATTACHMENTS: Certified surveyor's plat, horizontal section plat, estimated geological tops, proposed mud/cementing plan, directional plot/plan, \$100 fee.
See Page 2 for Comments section and signature block.

COMMENTS, ADDITIONAL INFORMATION, AND/OR LIST OF ATTACHMENTS

--

Lateral 2

KOP Lateral 2 Feet MD	Azimuth Lateral 2 °	Estimated Total Depth Lateral 2 Feet MD Feet TVD	KOP Coordinates From Well Head From WH From WH		
Formation Entry Point Coordinates From Well Head From WH From WH		Bottom Hole Coordinates From Well Head From WH From WH			
KOP Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County
Bottom Hole Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County

Lateral 3

KOP Lateral 3 Feet MD	Azimuth Lateral 3 °	Estimated Total Depth Lateral 3 Feet MD Feet TVD	KOP Coordinates From Well Head From WH From WH		
Formation Entry Point Coordinates From Well Head From WH From WH		Bottom Hole Coordinates From Well Head From WH From WH			
KOP Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County
Bottom Hole Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County

Lateral 4

KOP Lateral 4 Feet MD	Azimuth Lateral 4 °	Estimated Total Depth Lateral 4 Feet MD Feet TVD	KOP Coordinates From Well Head From WH From WH		
Formation Entry Point Coordinates From Well Head From WH From WH		Bottom Hole Coordinates From Well Head From WH From WH			
KOP Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County
Bottom Hole Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County

Lateral 5

KOP Lateral 5 Feet MD	Azimuth Lateral 5 °	Estimated Total Depth Lateral 5 Feet MD Feet TVD	KOP Coordinates From Well Head From WH From WH		
Formation Entry Point Coordinates From Well Head From WH From WH		Bottom Hole Coordinates From Well Head From WH From WH			
KOP Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County
Bottom Hole Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County

I hereby swear or affirm the information provided is true, complete and correct as determined from all available records.		Date 6 / 12 / 2012
ePermit	Printed Name Terry L. Olson	Title Regulatory Compliance Specialist

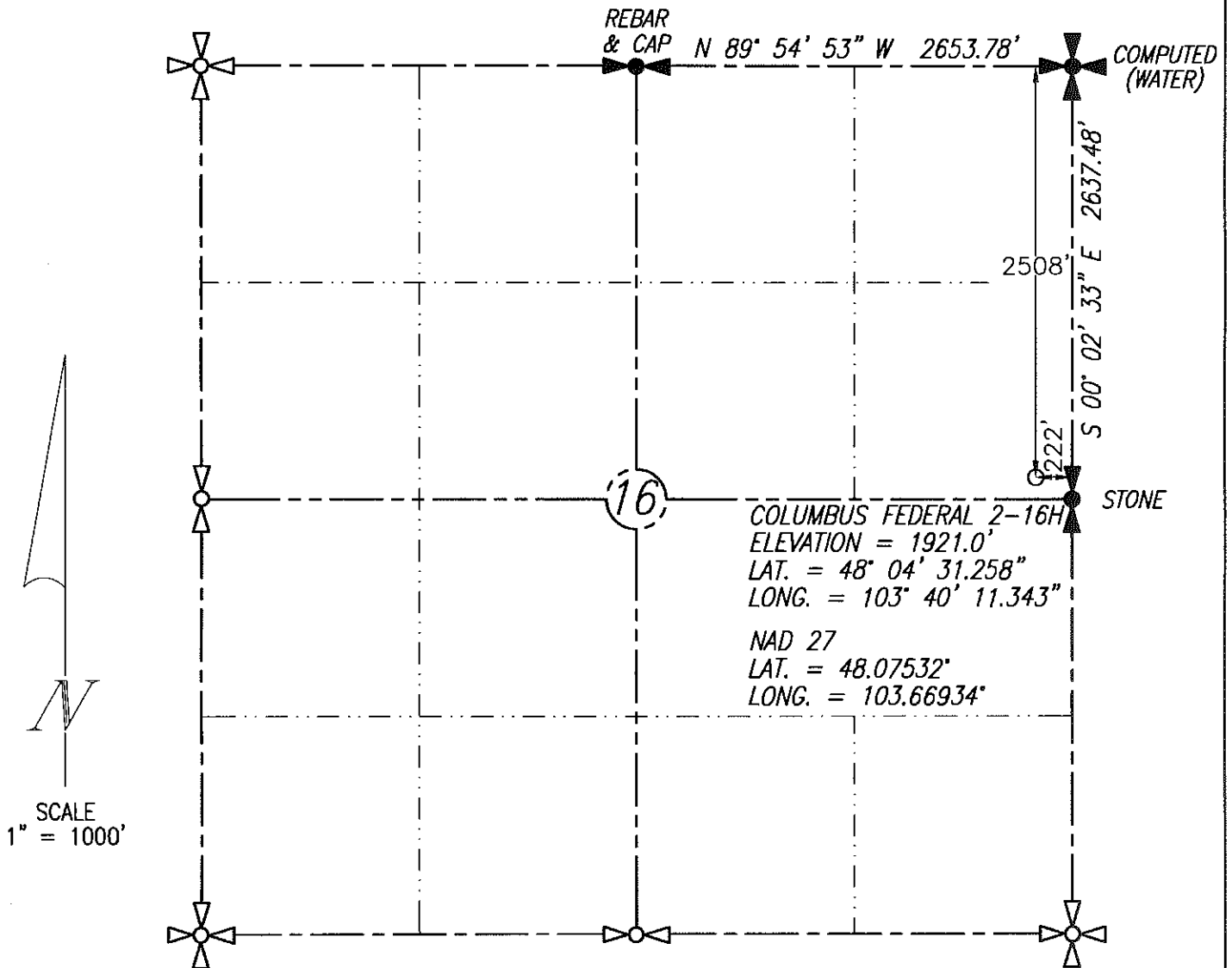
FOR STATE USE ONLY

Permit and File Number 25159	API Number 33 - 053 - 04855
Field BAKER	
Pool BAKKEN	Permit Type DEVELOPMENT

FOR STATE USE ONLY

Date Approved 3 / 11 / 2013
By David Tabor
Title Engineering Technician IV

WELL LOCATION PLAT
CONTINENTAL RESOURCES INC.
COLUMBUS FEDERAL 2-16H
SECTION 16, T153N, R101W
MCKENZIE COUNTY, NORTH DAKOTA
2508' FNL & 222' FEL



I CERTIFY THAT THIS PLAT CORRECTLY REPRESENTS
WORK PERFORMED BY ME OR UNDER MY RESPONSIBLE
CHARGE, AND IS TRUE AND CORRECT TO THE BEST OF
MY KNOWLEDGE AND BELIEF

John P. Johnson
JOHN P. JOHNSON L.S. 3366
LAND
SURVEYOR
L.S. 3366
NORTH DAKOTA
DATE STAKED: 1-4-2012
BASIS OF VERTICAL DATUM:
NAVD 1988 GEOID 09

PERSON AUTHORIZING SURVEY;
CHAD NEWBY

EXPLANATION AREA: NAD83(CORS96)

BASIS OF BEARING: TRUE NORTH

BROSZ ENGINEERING INC.

BOX 357
BOWMAN, N.D. 58623
PHONE: 701-523-3340
FAX: 701-523-5243
PROJECT NO. 12-10

GEOLOGIC PROGNOSIS

Well Name: Columbus Fed. 2-16H

Rig: Cyclone 20

Prospect: Williston

Target: Three Forks

Spacing: 1280

☐ Pre-Staked

☒ Staked

Rig Grade Elevation: 1920'

KB: 21'

RKB: 1941'

SHL: 2508' FNL & 222' FEL

Sec. 16 - 153N - 101W

McKenzie , ND

BHL: 200' FNL & 1320' FEL

Sec. 4 - 153N - 101W

McKenzie , ND

==>

FORMATION	SUBSEA	TVD
Pierre Shale	100	1,841
Greenhorn	-2,484	4,425
Dakota Group (fka Mowry)	-2,885	4,826
Base of Dakota Sand	-3,764	5,705
Dunham Salt Top	-4,792	6,733
Dunham Salt Base	-4,852	6,793
Pine Salt Top	-5,162	7,103
Pine Salt Base	-5,218	7,159
Minnekahta	-5,244	7,185
Opeche Salt Top	NA	
Opeche Salt Base	NA	
Minnelusa Group	-5,530	7,471
Tyler	-5,686	7,627
Kibbey	-6,238	8,179
Top Charles	-6,392	8,333
Base Last Charles Salt	-7,090	9,031
Mission Canyon	-7,314	9,255
Lodgepole	-7,880	9,821
Upper Bakken Shale	-8,570	10,511
Middle Bakken Member	-8,587	10,528
Lower Bakken Shale	-8,629	10,570
Three Forks	-8,646	10,587
Three Forks Target	-8,666	10,607
End of Lateral	-8,666	10,607

20' into 3forks
up 30' and down 30'

DRILLING PROGRAM

06/08/12

Lease and Well No.

Columbus Fed. 2-16H

MUD PROGRAM

Depth	Type	Weight	Remarks
0 ' - 1940 '	Fresh water	8.4-8.8	Add Soap Sticks for Mud Rings
1940 ' - 6500 '	Invert	9.3-9.5	35-50 sec, 10-30 cc's
6500 ' - 13290 '	Invert	9.6-10.0	40-55 sec, 10-15 cc's O/W 70/30 to 80/20
13290 ' - 23362 '	Brine	8.7-10.0	Cuttings pit

TUBULAR PROGRAM

String Type	Hole Size	Depth	Feet	Casing Diameter	Weight, Grade, Connection	ERW/ Seamless	Critical Inspection
Surf	13 1/2 "	1940 '	1940 '	9 5/8 "	9-5/8", 36 #, J-55, STC	ERW	BCI & Drift
Float shoe, shoe joint & float collar. Centralize bottom 3 jts and every 4th jt to surface.							
Int	8 3/4 "	80 '	80 '	7 "	7", 32#, P-110 IC, LTC	ERW	BCI & Drift
		4000 '	3920 '	7 "	7", 26#, P-110 IC, LTC	ERW	BCI & Drift
		8130 '	4130 '	7 "	7", 29#, P-110 IC, LTC	ERW	BCI & Drift
		9230 '	1100 '	7 "	7", 32#, P-110 IC, LTC	ERW	BCI & Drift
		13290 '	4060 '	7 "	7", 29#, P-110 IC, LTC	ERW	BCI & Drift
Float shoe, shoe joint & float collar. Centralize bottom 3 joints. Centralize thru curve and across all salts.							
Liner	6 "	23362 '	13196 '	4 1/2 "	4-1/2", 11.6 #, P-110, BTC		
Tubing		10170 '	10170 '	2 7/8 "	2-7/8", 6.5 #, L-80, EUE		

Notes: Pipe to end up in hole from top to bottom as shown.

CEMENT PROGRAM

String Type	SHOE/DV Depth	Stage Lead/Tail	Cement Bottom	Cement Top	No Sacks	Cement System	Cement Yield	Cement Weight
		Lead	1360 '	0 '	431	35/65 Poz/Class "C", 3% CaCl, 12% gel	2.39	12
Surf	1940	Tail	1940 '	1360 '	301	Class "C", 2% CaCl	1.46	14.3
(Basis: Gauge hole + 55% excess, tail 30% of length, lead to surface.)								
Int	13290	Lead	7830 '	0 '	473	35/65 Poz/Class "C", 3% KCl, 5 #/sk Silica	3.21	11.3
		Tail	13290 '	7830 '	697	Class "G", 3% KCl, 35% Silica	1.59	15.6
(Basis: Gauge hole + 35% excess, Tail to 500 ft above top of Charles Salt, Lead to Surface)								

BOP PROGRAM

Hole Size	Configuration	Pressure Rating	BOP Test Press	Casing Test Press
13 1/2 "	Circulate Conductor			
8 3/4 "	RRA	5M	5000	
6 "	RRA	5M	5000	
Utilize commercial testers every 30 days.				

LOGGING PROGRAM

FROM	TO	TOOLS
BLS	TD	2 man mud logger trailer, 30 ft samples
Surf csg	TD	MWD Surveys, 90 ft & 30 ft thru curve
KOP	TD	MWD Gamma Ray
Surface	DDO	CBL, CCL, Gr

Remarks: 4 1/2 inch Liner to be run. Liner Top @ KOP (10166 ft TVD) and 29 swell packers to be run with liner.
Liner top to be tested successfully to 4500 psi prior to rigging down.
Casing caliper to be run to determine if a frac string is needed.

Prepared By:

Pat McCollom

Date:

04/17/12

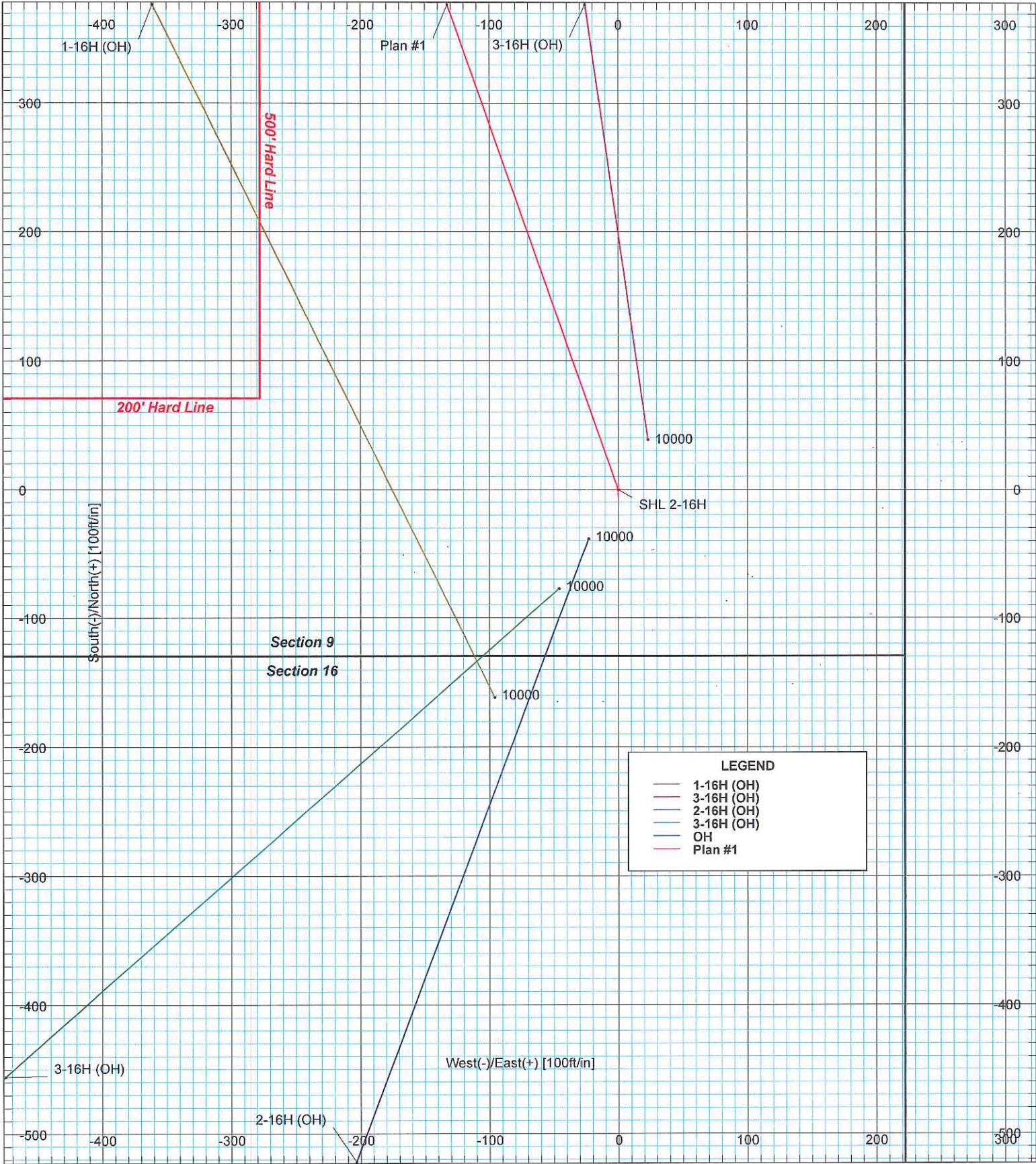
Continental Resources

Field: McKenzie County, ND
Site: Columbus Federal 2
Well: 2-16H
Wellpath: OH
Plan: Plan #1



Azimuths to True North
Magnetic North: 8.65°

Magnetic Field
Strength: 56676nT
Dip Angle: 73.09°
Date: 04/18/2012
Model: IGRF2010



LEAM DRILLING SYSTEMS, INC.
2010 East Davis Conroe, Texas 77301
Phone: 936-756-7577 Fax: 936-756-7595

Plan: Plan #1 (2-16H/OH)
Created By: Justin Andoe Date: 04/23/2012
Checked: _____ Date: _____
Reviewed: _____ Date: _____



To: Todd Holweger, NDIC

From: Shawn Svob

Date: 4/5/2012

Re: Continental Resources standard CCL, CBL, 4-1/2" liner running and testing procedures

Continental Resources' standard practice for running the cement bond log and casing caliper log is to run both logs immediately after coming out of the hole after TD, prior to running the 4-1/2" liner, to the deepest depth obtainable; however, if there are well control concerns that require us to run the liner sooner, only the CBL will be run and the CCL will be run after setting the liner.

Based on the CCL results, we determine the actual API minimum burst allowance for the 7" casing. If the downgraded API burst pressure is below our minimum required frac pressures, we will run a 4-1/2" frac string; if severe wear or holes are found in the casing, we will run a 5" cemented, to surface, tie back string.

The CBL log is run in order to determine the top of cement, as required by the NDIC. Our current 4-1/2" liner program for a 1280 unit is 30, evenly spaced, stages with 29 swellable packers. The liner shoe is set approximately 180 feet off bottom. The shoe stage below the last packer has 2 joints, a double valved float, one joint, and a ported guide shoe – appx 130 ft. The liner is run using a running tool on the end of 4" DP. The 7" packer/hanger is set about 40 ft above KOP between two casing collars but conditions occasionally occur that require setting higher, either through unexpected failure or in order to isolate casing wear close to KOP. Recently we have tried 40 stage liners and the trend to explore the optimum stage count will continue.. Once the liner is at depth, a ball is dropped through the DP, the ball is pressured up against the setting tool to approximately 2500 psi, and the 7" packer/hanger is set.

A push pull test is done to confirm the hanger has set. Then, a 4500 psi pressure test is completed on the back side of the 4" DP to confirm the packer has set. The setting tool is then backed off and the 4" DP/running tool is laid down.

Immediately after the rotary rig has been moved off the well location, the 7" csg and liner packer/ hanger are tested to the frac pressure. The testers will rig up and test the tubing head to 5000 psi. Next a test plug will be run and set, using wire line, in the top of the 7" packer/hanger. Testers will pressure up to our frac pressure, typically 8500 psi, to confirm the 7" is ready for completion.

Shawn Svob
Drilling Operations Coordinator

Continental Resources

SITE DETAILS

Field: McKenzie County, ND
Site: Columbus Federal 2
Well: 2-16H
Wellpath: OH
Plan: Plan #2

Columbus 2
Sec. 16 - T153N - R101W
SHL 2508' FNL & 222' FEL
PBHL 200' FNL & 1320' FEL

Latitude: 48°04'31.258N
Longitude: 103°40'11.343W

Ground Level: 1920.00
KB: 21.00

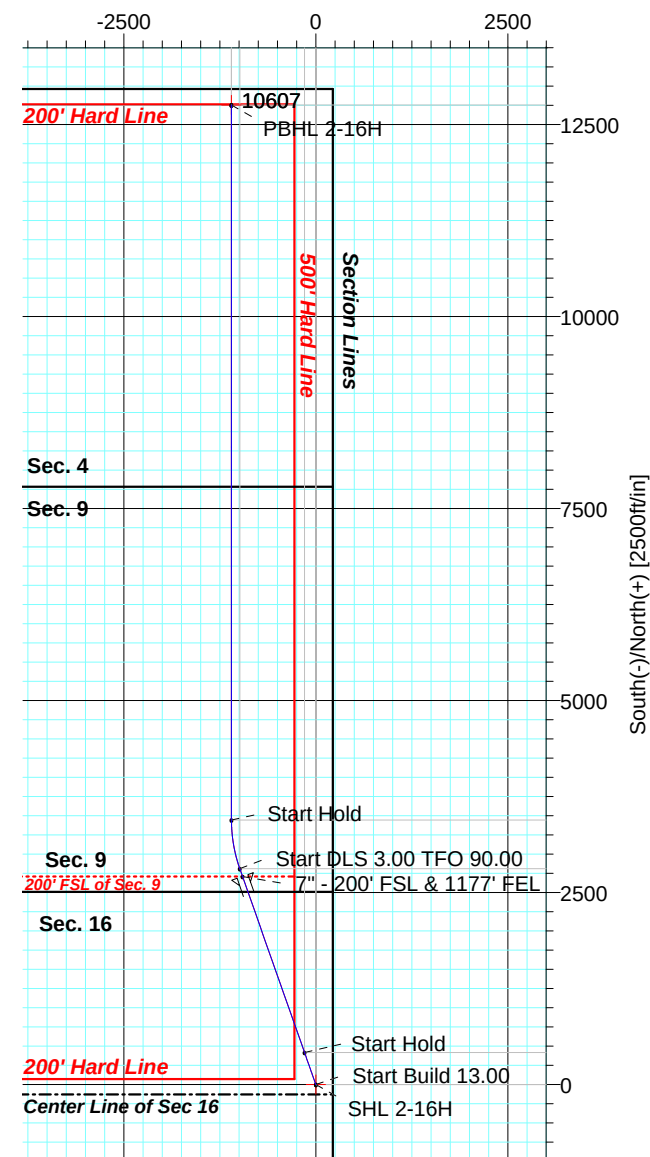


Azimuths to True North
Magnetic North: 8.65°

Magnetic Field
Strength: 56676nT
Dip Angle: 73.09°
Date: 04/18/2012
Model: IGRF2010



West(-)/East(+) [2500ft/in]



TARGET DETAILS

Name	TVD	+N/-S	+E/-W	Northing	Easting	Latitude	Longitude	Shape
SHL 2-16H	0.00	0.00	0.00	408199.81	1193824.71	48°04'31.258N	103°40'11.343W	Point
PBHL 2-16H10607.00	12755.00	-1098.00		420989.19	1193252.59	48°06'37.136N	103°40'27.523W	Point

SECTION DETAILS

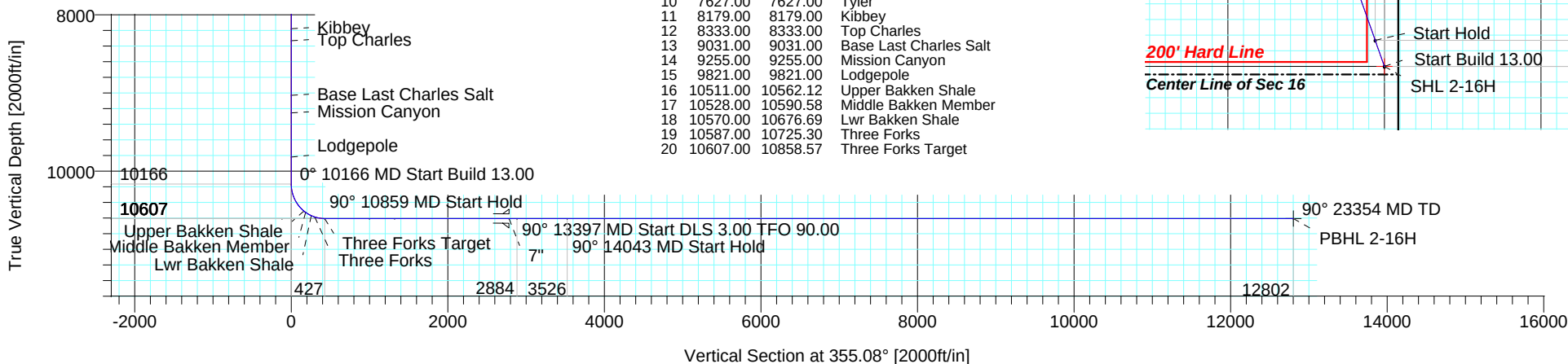
Sec	MD	Inc	Azi	TVD	+N/-S	+E/-W	DLeg	TFace	VSec	Target
1	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
2	10166.26	0.00	0.00	10166.26	0.00	0.00	0.00	0.00	0.00	
3	10858.57	90.00	340.60	10607.00	415.71	-146.41	13.00	340.60	426.73	
4	13396.57	90.00	340.60	10607.00	2809.57	-989.54	0.00	0.00	2884.09	
5	14043.31	90.00	0.00	10607.00	3444.03	-1098.00	3.00	90.00	3525.51	
6	23354.29	90.00	0.00	10607.00	12755.00	-1098.00	0.00	0.00	12802.17	PBHL 2-16H

CASING DETAILS

No.	TVD	MD	Name	Size
1	10607.00	13290.00	7"	7.00

FORMATION TOP DETAILS

No.	TVDPath	MDPath	Formation
1	1841.00	1841.00	Pierre Shale
2	4425.00	4425.00	Greenhorn
3	4826.00	4826.00	Dakota Group
4	5705.00	5705.00	Dunham Salt Top
5	6793.00	6793.00	Dunham Salt Base
6	7103.00	7103.00	Pine Salt Top
7	7159.00	7159.00	Pine Salt Base
8	7185.00	7185.00	Minnekahta
9	7471.00	7471.00	Minnelusa Group
10	7627.00	7627.00	Tyler
11	8179.00	8179.00	Kibbey
12	8333.00	8333.00	Top Charles
13	9031.00	9031.00	Base Last Charles Salt
14	9255.00	9255.00	Mission Canyon
15	9821.00	9821.00	Lodgepole
16	10511.00	10562.12	Upper Bakken Shale
17	10528.00	10590.58	Middle Bakken Member
18	10570.00	10676.69	Lwr Bakken Shale
19	10587.00	10725.30	Three Forks
20	10607.00	10858.57	Three Forks Target



LEAM DRILLING SYSTEMS LLC
2010 East Davis, Conroe, Texas 77301
Phone: 936-756-7577 Fax: 936-756-7595

Plan: Plan #2 (2-16H/OH)

Created By: David Vogler Date: 02/13/2013
Checked: _____ Date: _____
Reviewed: _____ Date: _____
Approved: _____ Date: _____

LEAM Drilling Systems LLC

Planning Report

Company: Continental Resources Field: McKenzie County, ND Site: Columbus Federal 2 Well: 2-16H Wellpath: OH	Date: 02/13/2013 Co-ordinate(NE) Reference: Vertical (TVD) Reference: Section (VS) Reference: Survey Calculation Method:	Time: 16:18:50 Well: 2-16H, True North GL 1920+KB 21 1941.0 Well (0.00N,0.00E,355.08Azi) Minimum Curvature	Page: 1 Db: Adapti
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Field: McKenzie County, ND	
Map System: US State Plane Coordinate System 1983 Geo Datum: GRS 1980 Sys Datum: Mean Sea Level	Map Zone: North Dakota, Northern Zone Coordinate System: Well Centre Geomagnetic Model: IGRF2010

Site: Columbus Federal 2 Sec. 16 - T153N - R101W 2508' FNL & 222' FEL Site Position: From: Geographic Position Uncertainty: 0.00 ft Ground Level: 1920.00 ft	Northing: 408199.81 ft Easting: 1193824.71 ft	Latitude: 48 4 31.258 N Longitude: 103 40 11.343 W North Reference: True Grid Convergence: -2.36 deg
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Well: 2-16H	Slot Name:
Well Position: +N/-S 0.00 ft +E/-W 0.00 ft Position Uncertainty: 0.00 ft	Latitude: 48 4 31.258 N Longitude: 103 40 11.343 W

Wellpath: OH Current Datum: GL 1920+KB 21 Magnetic Data: 04/18/2012 Field Strength: 56676 nT Vertical Section: Depth From (TVD) ft	Height 1941.00 ft +N/-S ft	Drilled From: Surface Tie-on Depth: 0.00 ft Above System Datum: Mean Sea Level Declination: 8.65 deg Mag Dip Angle: 73.09 deg +E/-W ft Direction deg
0.00	0.00	0.00 355.08

Plan: Plan #2 Principal: Yes	Date Composed: 04/18/2012 Version: 1 Tied-to: User Defined
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Plan Section Information										
MD ft	Incl deg	Azim deg	TVD ft	+N/-S ft	+E/-W ft	DLS deg/100ft	Build deg/100ft	Turn deg/100ft	TFO deg	Target
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
10166.26	0.00	0.00	10166.26	0.00	0.00	0.00	0.00	0.00	0.00	
10858.57	90.00	340.60	10607.00	415.71	-146.41	13.00	13.00	0.00	340.60	
13396.57	90.00	340.60	10607.00	2809.57	-989.54	0.00	0.00	0.00	0.00	
14043.31	90.00	0.00	10607.00	3444.03	-1098.00	3.00	0.00	3.00	90.00	
23354.29	90.00	0.00	10607.00	12755.00	-1098.00	0.00	0.00	0.00	0.00	PBHL 2-16H

Survey										
MD ft	Incl deg	Azim deg	TVD ft	N/S ft	E/W ft	VS ft	DLS deg/100ft	Build deg/100ft	Turn deg/100ft	Tool/Comment
10166.26	0.00	0.00	10166.26	0.00	0.00	0.00	0.00	0.00	0.00	
10175.00	1.14	340.60	10175.00	0.08	-0.03	0.08	13.00	13.00	0.00	
10200.00	4.39	340.60	10199.97	1.22	-0.43	1.25	13.00	13.00	0.00	
10225.00	7.64	340.60	10224.83	3.69	-1.30	3.78	13.00	13.00	0.00	
10250.00	10.89	340.60	10249.50	7.48	-2.63	7.68	13.00	13.00	0.00	
10275.00	14.14	340.60	10273.90	12.59	-4.43	12.92	13.00	13.00	0.00	
10300.00	17.39	340.60	10297.96	18.99	-6.69	19.50	13.00	13.00	0.00	
10325.00	20.64	340.60	10321.59	26.67	-9.39	27.38	13.00	13.00	0.00	
10350.00	23.89	340.60	10344.72	35.60	-12.54	36.55	13.00	13.00	0.00	
10375.00	27.14	340.60	10367.28	45.76	-16.12	46.97	13.00	13.00	0.00	
10400.00	30.39	340.60	10389.20	57.10	-20.11	58.62	13.00	13.00	0.00	
10425.00	33.64	340.60	10410.39	69.60	-24.51	71.45	13.00	13.00	0.00	
10450.00	36.89	340.60	10430.80	83.21	-29.31	85.42	13.00	13.00	0.00	
10475.00	40.14	340.60	10450.36	97.89	-34.48	100.49	13.00	13.00	0.00	
10500.00	43.39	340.60	10469.01	113.60	-40.01	116.61	13.00	13.00	0.00	

LEAM Drilling Systems LLC

Planning Report

Company: Continental Resources	Date: 02/13/2013	Time: 16:18:50	Page: 2
Field: McKenzie County, ND	Co-ordinate(NE) Reference: Well: 2-16H, True North		
Site: Columbus Federal 2	Vertical (TVD) Reference: GL 1920+KB 21 1941.0		
Well: 2-16H	Section (VS) Reference: Well (0.00N,0.00E,355.08Azi)		
Wellpath: OH	Survey Calculation Method: Minimum Curvature	Db: Adapti	

Survey

MD ft	Incl deg	Azim deg	TVD ft	N/S ft	E/W ft	VS ft	DLS deg/100ft	Build deg/100ft	Turn deg/100ft	Tool/Comment
10525.00	46.64	340.60	10486.68	130.27	-45.88	133.73	13.00	13.00	0.00	
10550.00	49.89	340.60	10503.32	147.86	-52.08	151.79	13.00	13.00	0.00	
10562.12	51.46	340.60	10511.00	156.70	-55.19	160.86	13.00	13.00	0.00	Upper Bakken Shale
10575.00	53.14	340.60	10518.88	166.32	-58.58	170.73	13.00	13.00	0.00	
10590.58	55.16	340.60	10528.00	178.23	-62.77	182.95	13.00	13.00	0.00	Middle Bakken Member
10600.00	56.39	340.60	10533.30	185.57	-65.36	190.50	13.00	13.00	0.00	
10625.00	59.64	340.60	10546.54	205.57	-72.40	211.02	13.00	13.00	0.00	
10650.00	62.89	340.60	10558.56	226.24	-79.68	232.24	13.00	13.00	0.00	
10675.00	66.14	340.60	10569.32	247.53	-87.18	254.09	13.00	13.00	0.00	
10676.69	66.36	340.60	10570.00	248.99	-87.69	255.59	13.00	13.00	0.00	Lwr Bakken Shale
10700.00	69.39	340.60	10578.78	269.35	-94.87	276.49	13.00	13.00	0.00	
10725.00	72.64	340.60	10586.91	291.64	-102.72	299.38	13.00	13.00	0.00	
10725.30	72.67	340.60	10587.00	291.91	-102.81	299.65	13.00	13.00	0.00	Three Forks
10750.00	75.89	340.60	10593.69	314.34	-110.71	322.67	13.00	13.00	0.00	
10775.00	79.14	340.60	10599.10	337.36	-118.82	346.30	13.00	13.00	0.00	
10800.00	82.39	340.60	10603.11	360.63	-127.01	370.19	13.00	13.00	0.00	
10825.00	85.64	340.60	10605.72	384.08	-135.27	394.26	13.00	13.00	0.00	
10850.00	88.89	340.60	10606.91	407.63	-143.57	418.44	13.00	13.00	0.00	
10858.57	90.00	340.60	10607.00	415.71	-146.41	426.73	13.00	13.00	0.00	Three Forks Target
10900.00	90.00	340.60	10607.00	454.79	-160.18	466.85	0.00	0.00	0.00	
11000.00	90.00	340.60	10607.00	549.11	-193.40	563.67	0.00	0.00	0.00	
11100.00	90.00	340.60	10607.00	643.43	-226.62	660.49	0.00	0.00	0.00	
11200.00	90.00	340.60	10607.00	737.75	-259.84	757.32	0.00	0.00	0.00	
11300.00	90.00	340.60	10607.00	832.07	-293.06	854.14	0.00	0.00	0.00	
11400.00	90.00	340.60	10607.00	926.39	-326.28	950.96	0.00	0.00	0.00	
11500.00	90.00	340.60	10607.00	1020.71	-359.50	1047.78	0.00	0.00	0.00	
11600.00	90.00	340.60	10607.00	1115.03	-392.72	1144.61	0.00	0.00	0.00	
11700.00	90.00	340.60	10607.00	1209.35	-425.94	1241.43	0.00	0.00	0.00	
11800.00	90.00	340.60	10607.00	1303.67	-459.16	1338.25	0.00	0.00	0.00	
11900.00	90.00	340.60	10607.00	1397.99	-492.38	1435.07	0.00	0.00	0.00	
12000.00	90.00	340.60	10607.00	1492.32	-525.60	1531.89	0.00	0.00	0.00	
12100.00	90.00	340.60	10607.00	1586.64	-558.82	1628.72	0.00	0.00	0.00	
12200.00	90.00	340.60	10607.00	1680.96	-592.04	1725.54	0.00	0.00	0.00	
12300.00	90.00	340.60	10607.00	1775.28	-625.26	1822.36	0.00	0.00	0.00	
12400.00	90.00	340.60	10607.00	1869.60	-658.48	1919.18	0.00	0.00	0.00	
12500.00	90.00	340.60	10607.00	1963.92	-691.70	2016.01	0.00	0.00	0.00	
12600.00	90.00	340.60	10607.00	2058.24	-724.92	2112.83	0.00	0.00	0.00	
12700.00	90.00	340.60	10607.00	2152.56	-758.14	2209.65	0.00	0.00	0.00	
12800.00	90.00	340.60	10607.00	2246.88	-791.36	2306.47	0.00	0.00	0.00	
12900.00	90.00	340.60	10607.00	2341.20	-824.58	2403.30	0.00	0.00	0.00	
13000.00	90.00	340.60	10607.00	2435.52	-857.80	2500.12	0.00	0.00	0.00	
13100.00	90.00	340.60	10607.00	2529.85	-891.02	2596.94	0.00	0.00	0.00	
13200.00	90.00	340.60	10607.00	2624.17	-924.24	2693.76	0.00	0.00	0.00	
13290.00	90.00	340.60	10607.00	2709.05	-954.14	2780.90	0.00	0.00	0.00	7"
13300.00	90.00	340.60	10607.00	2718.49	-957.46	2790.59	0.00	0.00	0.00	
13396.57	90.00	340.60	10607.00	2809.57	-989.54	2884.09	0.00	0.00	0.00	
13400.00	90.00	340.70	10607.00	2812.81	-990.68	2887.41	3.00	0.00	3.00	
13500.00	90.00	343.70	10607.00	2908.01	-1021.24	2984.88	3.00	0.00	3.00	
13600.00	90.00	346.70	10607.00	3004.68	-1046.78	3083.39	3.00	0.00	3.00	
13700.00	90.00	349.70	10607.00	3102.56	-1067.23	3182.66	3.00	0.00	3.00	
13800.00	90.00	352.70	10607.00	3201.37	-1082.52	3282.42	3.00	0.00	3.00	
13900.00	90.00	355.70	10607.00	3300.85	-1092.63	3382.39	3.00	0.00	3.00	
14000.00	90.00	358.70	10607.00	3400.72	-1097.51	3482.31	3.00	0.00	3.00	

LEAM Drilling Systems LLC

Planning Report

Company: Continental Resources	Date: 02/13/2013	Time: 16:18:50	Page: 3
Field: McKenzie County, ND	Co-ordinate(NE) Reference: Well: 2-16H, True North		
Site: Columbus Federal 2	Vertical (TVD) Reference: GL 1920+KB 21 1941.0		
Well: 2-16H	Section (VS) Reference: Well (0.00N,0.00E,355.08Azi)		
Wellpath: OH	Survey Calculation Method: Minimum Curvature	Db: Adapti	

Survey

MD ft	Incl deg	Azim deg	TVD ft	N/S ft	E/W ft	VS ft	DLS deg/100ft	Build deg/100ft	Turn deg/100ft	Tool/Comment
14043.31	90.00	0.00	10607.00	3444.03	-1098.00	3525.51	3.00	0.00	3.00	
14100.00	90.00	0.00	10607.00	3500.71	-1098.00	3581.98	0.00	0.00	0.00	
14200.00	90.00	0.00	10607.00	3600.71	-1098.00	3681.62	0.00	0.00	0.00	
14300.00	90.00	0.00	10607.00	3700.71	-1098.00	3781.25	0.00	0.00	0.00	
14400.00	90.00	0.00	10607.00	3800.71	-1098.00	3880.88	0.00	0.00	0.00	
14500.00	90.00	0.00	10607.00	3900.71	-1098.00	3980.51	0.00	0.00	0.00	
14600.00	90.00	0.00	10607.00	4000.71	-1098.00	4080.14	0.00	0.00	0.00	
14700.00	90.00	0.00	10607.00	4100.71	-1098.00	4179.77	0.00	0.00	0.00	
14800.00	90.00	0.00	10607.00	4200.71	-1098.00	4279.40	0.00	0.00	0.00	
14900.00	90.00	0.00	10607.00	4300.71	-1098.00	4379.04	0.00	0.00	0.00	
15000.00	90.00	0.00	10607.00	4400.71	-1098.00	4478.67	0.00	0.00	0.00	
15100.00	90.00	0.00	10607.00	4500.71	-1098.00	4578.30	0.00	0.00	0.00	
15200.00	90.00	0.00	10607.00	4600.71	-1098.00	4677.93	0.00	0.00	0.00	
15300.00	90.00	0.00	10607.00	4700.71	-1098.00	4777.56	0.00	0.00	0.00	
15400.00	90.00	0.00	10607.00	4800.71	-1098.00	4877.19	0.00	0.00	0.00	
15500.00	90.00	0.00	10607.00	4900.71	-1098.00	4976.83	0.00	0.00	0.00	
15600.00	90.00	0.00	10607.00	5000.71	-1098.00	5076.46	0.00	0.00	0.00	
15700.00	90.00	0.00	10607.00	5100.71	-1098.00	5176.09	0.00	0.00	0.00	
15800.00	90.00	0.00	10607.00	5200.71	-1098.00	5275.72	0.00	0.00	0.00	
15900.00	90.00	0.00	10607.00	5300.71	-1098.00	5375.35	0.00	0.00	0.00	
16000.00	90.00	0.00	10607.00	5400.71	-1098.00	5474.98	0.00	0.00	0.00	
16100.00	90.00	0.00	10607.00	5500.71	-1098.00	5574.61	0.00	0.00	0.00	
16200.00	90.00	0.00	10607.00	5600.71	-1098.00	5674.25	0.00	0.00	0.00	
16300.00	90.00	0.00	10607.00	5700.71	-1098.00	5773.88	0.00	0.00	0.00	
16400.00	90.00	0.00	10607.00	5800.71	-1098.00	5873.51	0.00	0.00	0.00	
16500.00	90.00	0.00	10607.00	5900.71	-1098.00	5973.14	0.00	0.00	0.00	
16600.00	90.00	0.00	10607.00	6000.71	-1098.00	6072.77	0.00	0.00	0.00	
16700.00	90.00	0.00	10607.00	6100.71	-1098.00	6172.40	0.00	0.00	0.00	
16800.00	90.00	0.00	10607.00	6200.71	-1098.00	6272.04	0.00	0.00	0.00	
16900.00	90.00	0.00	10607.00	6300.71	-1098.00	6371.67	0.00	0.00	0.00	
17000.00	90.00	0.00	10607.00	6400.71	-1098.00	6471.30	0.00	0.00	0.00	
17100.00	90.00	0.00	10607.00	6500.71	-1098.00	6570.93	0.00	0.00	0.00	
17200.00	90.00	0.00	10607.00	6600.71	-1098.00	6670.56	0.00	0.00	0.00	
17300.00	90.00	0.00	10607.00	6700.71	-1098.00	6770.19	0.00	0.00	0.00	
17400.00	90.00	0.00	10607.00	6800.71	-1098.00	6869.82	0.00	0.00	0.00	
17500.00	90.00	0.00	10607.00	6900.71	-1098.00	6969.46	0.00	0.00	0.00	
17600.00	90.00	0.00	10607.00	7000.71	-1098.00	7069.09	0.00	0.00	0.00	
17700.00	90.00	0.00	10607.00	7100.71	-1098.00	7168.72	0.00	0.00	0.00	
17800.00	90.00	0.00	10607.00	7200.71	-1098.00	7268.35	0.00	0.00	0.00	
17900.00	90.00	0.00	10607.00	7300.71	-1098.00	7367.98	0.00	0.00	0.00	
18000.00	90.00	0.00	10607.00	7400.71	-1098.00	7467.61	0.00	0.00	0.00	
18100.00	90.00	0.00	10607.00	7500.71	-1098.00	7567.25	0.00	0.00	0.00	
18200.00	90.00	0.00	10607.00	7600.71	-1098.00	7666.88	0.00	0.00	0.00	
18300.00	90.00	0.00	10607.00	7700.71	-1098.00	7766.51	0.00	0.00	0.00	
18400.00	90.00	0.00	10607.00	7800.71	-1098.00	7866.14	0.00	0.00	0.00	
18500.00	90.00	0.00	10607.00	7900.71	-1098.00	7965.77	0.00	0.00	0.00	
18600.00	90.00	0.00	10607.00	8000.71	-1098.00	8065.40	0.00	0.00	0.00	
18700.00	90.00	0.00	10607.00	8100.71	-1098.00	8165.03	0.00	0.00	0.00	
18800.00	90.00	0.00	10607.00	8200.71	-1098.00	8264.67	0.00	0.00	0.00	
18900.00	90.00	0.00	10607.00	8300.71	-1098.00	8364.30	0.00	0.00	0.00	
19000.00	90.00	0.00	10607.00	8400.71	-1098.00	8463.93	0.00	0.00	0.00	
19100.00	90.00	0.00	10607.00	8500.71	-1098.00	8563.56	0.00	0.00	0.00	

LEAM Drilling Systems LLC

Planning Report

Company: Continental Resources	Date: 02/13/2013	Time: 16:18:50	Page: 4
Field: McKenzie County, ND	Co-ordinate(NE) Reference: Well: 2-16H, True North		
Site: Columbus Federal 2	Vertical (TVD) Reference: GL 1920+KB 21 1941.0		
Well: 2-16H	Section (VS) Reference: Well (0.00N,0.00E,355.08Azi)		
Wellpath: OH	Survey Calculation Method: Minimum Curvature	Db: Adapti	

Survey

MD ft	Incl deg	Azim deg	TVD ft	N/S ft	E/W ft	VS ft	DLS deg/100ft	Build deg/100ft	Turn deg/100ft	Tool/Comment
19200.00	90.00	0.00	10607.00	8600.71	-1098.00	8663.19	0.00	0.00	0.00	
19300.00	90.00	0.00	10607.00	8700.71	-1098.00	8762.82	0.00	0.00	0.00	
19400.00	90.00	0.00	10607.00	8800.71	-1098.00	8862.46	0.00	0.00	0.00	
19500.00	90.00	0.00	10607.00	8900.71	-1098.00	8962.09	0.00	0.00	0.00	
19600.00	90.00	0.00	10607.00	9000.71	-1098.00	9061.72	0.00	0.00	0.00	
19700.00	90.00	0.00	10607.00	9100.71	-1098.00	9161.35	0.00	0.00	0.00	
19800.00	90.00	0.00	10607.00	9200.71	-1098.00	9260.98	0.00	0.00	0.00	
19900.00	90.00	0.00	10607.00	9300.71	-1098.00	9360.61	0.00	0.00	0.00	
20000.00	90.00	0.00	10607.00	9400.71	-1098.00	9460.24	0.00	0.00	0.00	
20100.00	90.00	0.00	10607.00	9500.71	-1098.00	9559.88	0.00	0.00	0.00	
20200.00	90.00	0.00	10607.00	9600.71	-1098.00	9659.51	0.00	0.00	0.00	
20300.00	90.00	0.00	10607.00	9700.71	-1098.00	9759.14	0.00	0.00	0.00	
20400.00	90.00	0.00	10607.00	9800.71	-1098.00	9858.77	0.00	0.00	0.00	
20500.00	90.00	0.00	10607.00	9900.71	-1098.00	9958.40	0.00	0.00	0.00	
20600.00	90.00	0.00	10607.00	10000.71	-1098.00	10058.03	0.00	0.00	0.00	
20700.00	90.00	0.00	10607.00	10100.71	-1098.00	10157.67	0.00	0.00	0.00	
20800.00	90.00	0.00	10607.00	10200.71	-1098.00	10257.30	0.00	0.00	0.00	
20900.00	90.00	0.00	10607.00	10300.71	-1098.00	10356.93	0.00	0.00	0.00	
21000.00	90.00	0.00	10607.00	10400.71	-1098.00	10456.56	0.00	0.00	0.00	
21100.00	90.00	0.00	10607.00	10500.71	-1098.00	10556.19	0.00	0.00	0.00	
21200.00	90.00	0.00	10607.00	10600.71	-1098.00	10655.82	0.00	0.00	0.00	
21300.00	90.00	0.00	10607.00	10700.71	-1098.00	10755.45	0.00	0.00	0.00	
21400.00	90.00	0.00	10607.00	10800.71	-1098.00	10855.09	0.00	0.00	0.00	
21500.00	90.00	0.00	10607.00	10900.71	-1098.00	10954.72	0.00	0.00	0.00	
21600.00	90.00	0.00	10607.00	11000.71	-1098.00	11054.35	0.00	0.00	0.00	
21700.00	90.00	0.00	10607.00	11100.71	-1098.00	11153.98	0.00	0.00	0.00	
21800.00	90.00	0.00	10607.00	11200.71	-1098.00	11253.61	0.00	0.00	0.00	
21900.00	90.00	0.00	10607.00	11300.71	-1098.00	11353.24	0.00	0.00	0.00	
22000.00	90.00	0.00	10607.00	11400.71	-1098.00	11452.88	0.00	0.00	0.00	
22100.00	90.00	0.00	10607.00	11500.71	-1098.00	11552.51	0.00	0.00	0.00	
22200.00	90.00	0.00	10607.00	11600.71	-1098.00	11652.14	0.00	0.00	0.00	
22300.00	90.00	0.00	10607.00	11700.71	-1098.00	11751.77	0.00	0.00	0.00	
22400.00	90.00	0.00	10607.00	11800.71	-1098.00	11851.40	0.00	0.00	0.00	
22500.00	90.00	0.00	10607.00	11900.71	-1098.00	11951.03	0.00	0.00	0.00	
22600.00	90.00	0.00	10607.00	12000.71	-1098.00	12050.66	0.00	0.00	0.00	
22700.00	90.00	0.00	10607.00	12100.71	-1098.00	12150.30	0.00	0.00	0.00	
22800.00	90.00	0.00	10607.00	12200.71	-1098.00	12249.93	0.00	0.00	0.00	
22900.00	90.00	0.00	10607.00	12300.71	-1098.00	12349.56	0.00	0.00	0.00	
23000.00	90.00	0.00	10607.00	12400.71	-1098.00	12449.19	0.00	0.00	0.00	
23100.00	90.00	0.00	10607.00	12500.71	-1098.00	12548.82	0.00	0.00	0.00	
23200.00	90.00	0.00	10607.00	12600.71	-1098.00	12648.45	0.00	0.00	0.00	
23300.00	90.00	0.00	10607.00	12700.71	-1098.00	12748.09	0.00	0.00	0.00	
23354.29	90.00	0.00	10607.00	12755.00	-1098.00	12802.17	0.00	0.00	0.00	PBHL 2-16H

Targets

Name	Description Dip.	Dir.	TVD ft	+N/-S ft	+E/-W ft	Map Northing ft	Map Easting ft	<--- Latitude ---> Deg Min Sec			<--- Longitude ---> Deg Min Sec		
SHL 2-16H			0.00	0.00	0.00	408199.81	1193824.71	48	4	31.258 N	103	40	11.343 W
PBHL 2-16H			10607.00	12755.00	-1098.00	420989.19	1193252.59	48	6	37.136 N	103	40	27.523 W

LEAM Drilling Systems LLC

Planning Report

Company: Continental Resources Field: McKenzie County, ND Site: Columbus Federal 2 Well: 2-16H Wellpath: OH	Date: 02/13/2013 Co-ordinate(NE) Reference: Well: 2-16H, True North Vertical (TVD) Reference: GL 1920+KB 21 1941.0 Section (VS) Reference: Well (0.00N,0.00E,355.08Azi) Survey Calculation Method: Minimum Curvature	Page: 5 Db: Adapti
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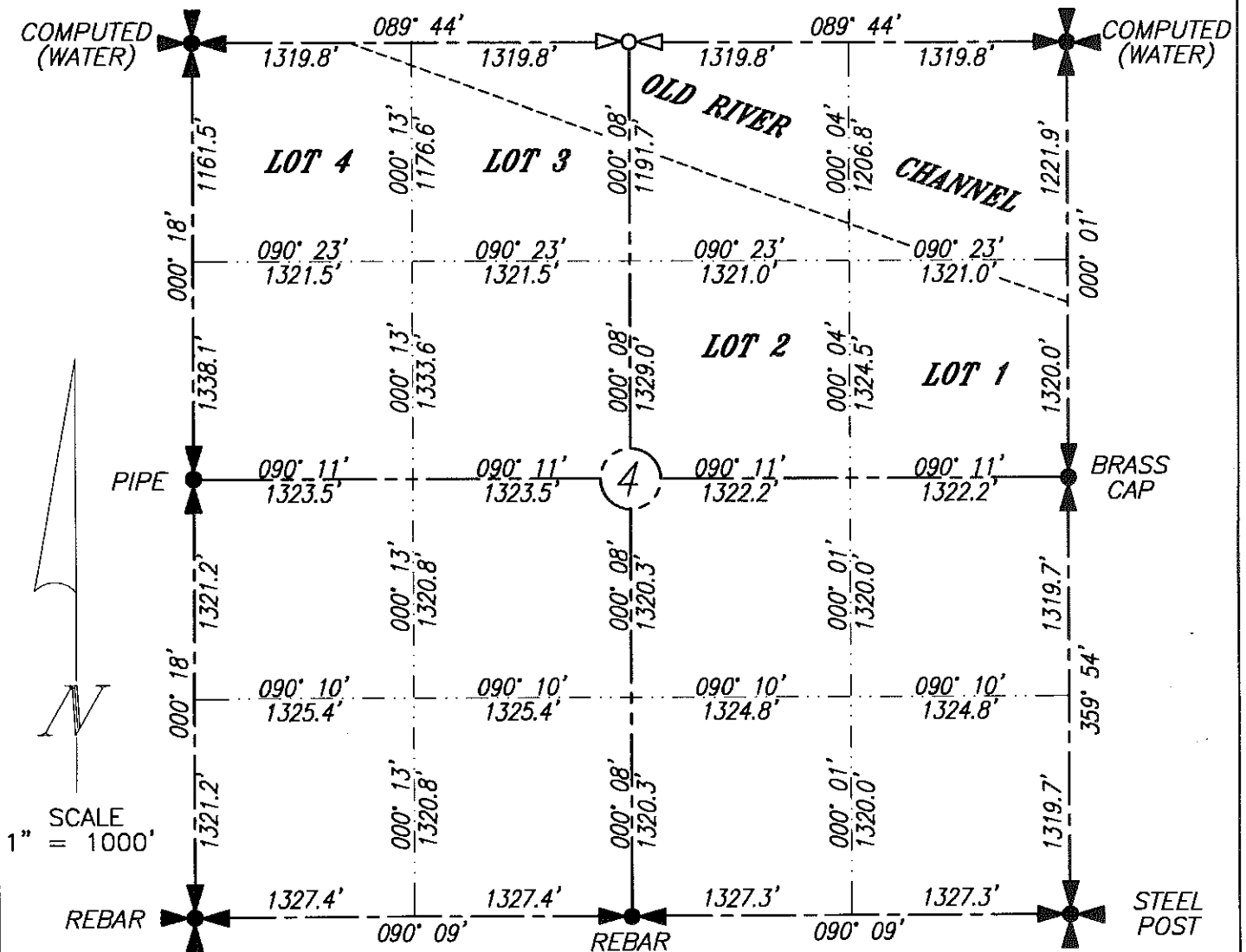
Casing Points

MD ft	TVD ft	Diameter in	Hole Size in	Name
13290.00	10607.00	7.00	8.50	7"

Formations

MD ft	TVD ft	Formations	Lithology	Dip Angle deg	Dip Direction deg
1841.00	1841.00	Pierre Shale		0.00	0.00
4425.00	4425.00	Greenhorn		0.00	0.00
4826.00	4826.00	Dakota Group		0.00	0.00
5705.00	5705.00	Dunham Salt Top		0.00	0.00
6793.00	6793.00	Dunham Salt Base		0.00	0.00
7103.00	7103.00	Pine Salt Top		0.00	0.00
7159.00	7159.00	Pine Salt Base		0.00	0.00
7185.00	7185.00	Minnekahta		0.00	0.00
7471.00	7471.00	Minnelusa Group		0.00	0.00
7627.00	7627.00	Tyler		0.00	0.00
8179.00	8179.00	Kibbey		0.00	0.00
8333.00	8333.00	Top Charles		0.00	0.00
9031.00	9031.00	Base Last Charles Salt		0.00	0.00
9255.00	9255.00	Mission Canyon		0.00	0.00
9821.00	9821.00	Lodgepole		0.00	0.00
10562.12	10511.00	Upper Bakken Shale		0.00	0.00
10590.58	10528.00	Middle Bakken Member		0.00	0.00
10676.69	10570.00	Lwr Bakken Shale		0.00	0.00
10725.30	10587.00	Three Forks		0.00	0.00
10858.57	10607.00	Three Forks Target		0.00	0.00

HORIZONTAL SECTION PLAT
CONTINENTAL RESOURCES INC.
COLUMBUS FEDERAL 2-16H
SECTION 4, T153N, R101W
MCKENZIE COUNTY, NORTH DAKOTA



ALL CORNERS SHOWN ON THIS PLAT WERE FOUND IN THE FIELD
DISTANCES TO ALL OTHERS ARE CALCULATED.
ALL BEARINGS SHOWN ARE ASSUMED.

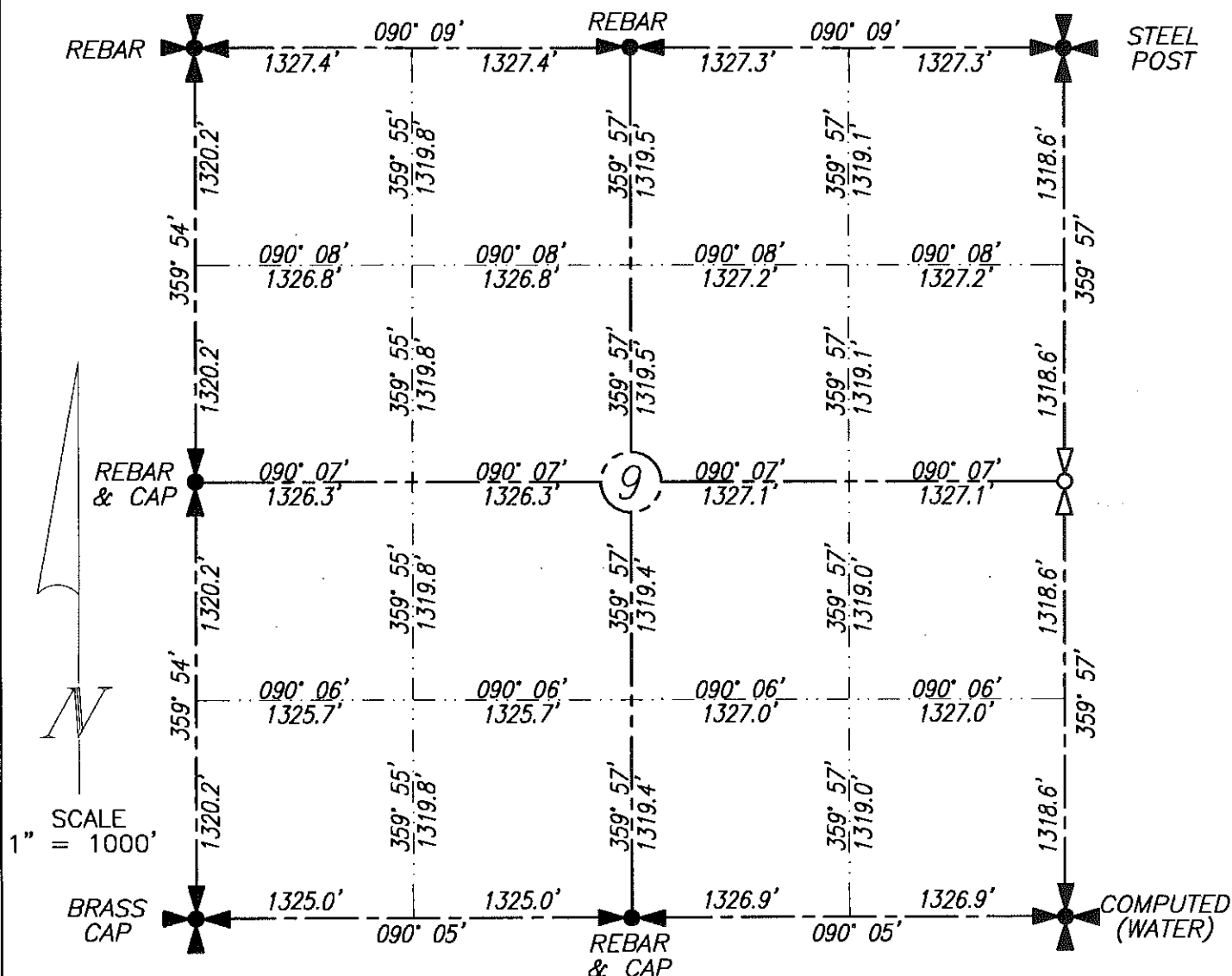
I CERTIFY THAT THIS PLAT CORRECTLY REPRESENTS
WORK PERFORMED BY ME OR UNDER MY RESPONSIBLE
CHARGE AND IS TRUE AND CORRECT TO THE BEST OF
MY KNOWLEDGE AND BELIEF

JOHN PAULSON
NORTH DAKOTA 4-11-12
JOHN PAULSON R.L.S. 3366

BROSZ ENGINEERING INC.

BOX 357
BOWMAN, N.D. 58623
PHONE: 701-523-3340
FAX: 701-523-5243
PROJECT NO. 12-10

HORIZONTAL SECTION PLAT
CONTINENTAL RESOURCES INC.
COLUMBUS FEDERAL 2-16H
SECTION 9, T153N, R101W
MCKENZIE COUNTY, NORTH DAKOTA



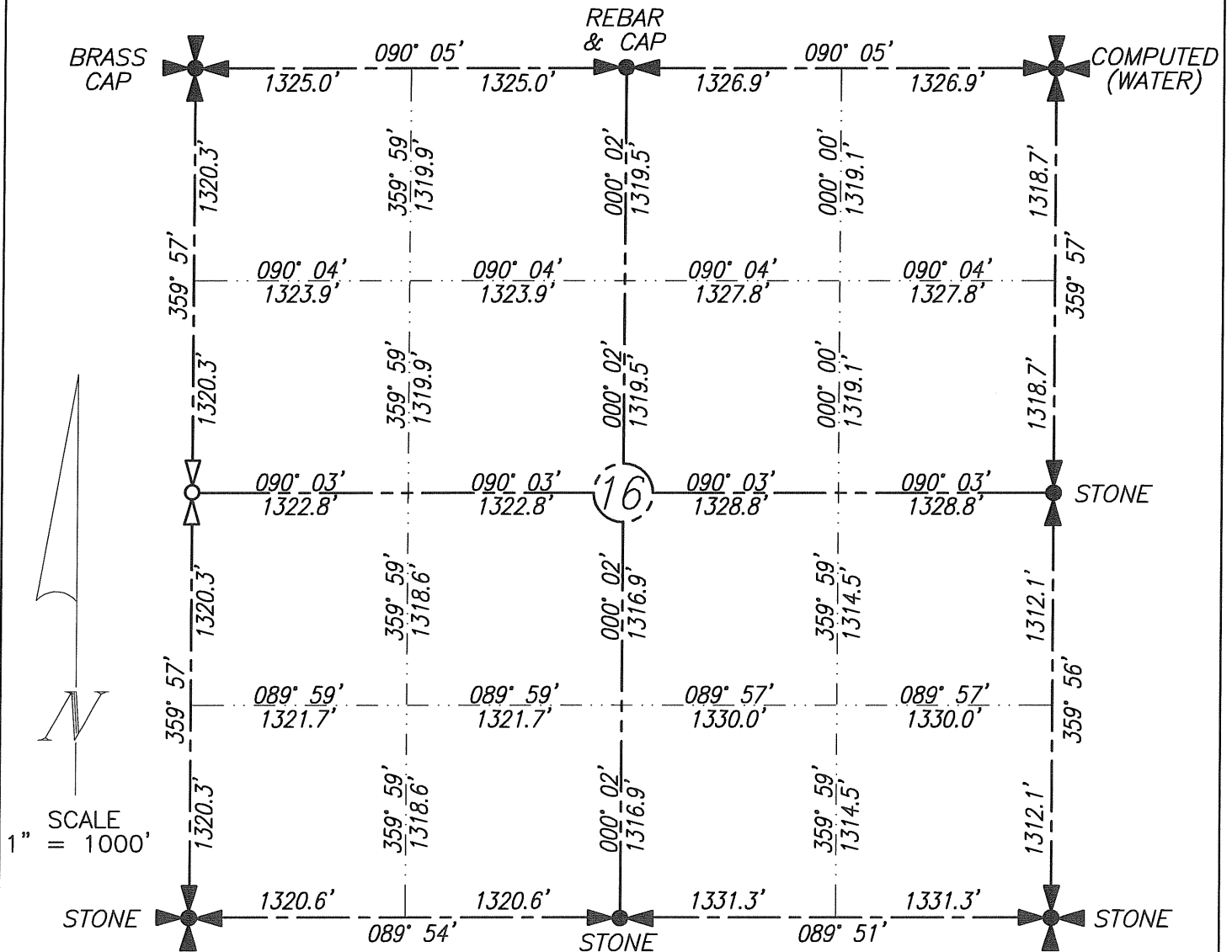
ALL CORNERS SHOWN ON THIS PLAT WERE FOUND IN THE FIELD
DISTANCES TO ALL OTHERS ARE CALCULATED.
ALL BEARINGS SHOWN ARE ASSUMED.

JOHN PAULSON
I CERTIFY THAT THIS PLAT CORRECTLY REPRESENTS
WORK PERFORMED BY ME OR UNDER MY RESPONSIBLE
CHARGE, AND IS TRUE AND CORRECT TO THE BEST OF
MY KNOWLEDGE AND BELIEF
JOHN PAULSON R.L.S. 3366

BROSZ ENGINEERING INC.

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BOWMAN, N.D. 58623
PHONE: 701-523-3340
FAX: 701-523-5243
PROJECT NO. 12-10

HORIZONTAL SECTION PLAT
CONTINENTAL RESOURCES INC.
COLUMBUS FEDERAL 2-16H
SECTION 16, T153N, R101W
MCKENZIE COUNTY, NORTH DAKOTA



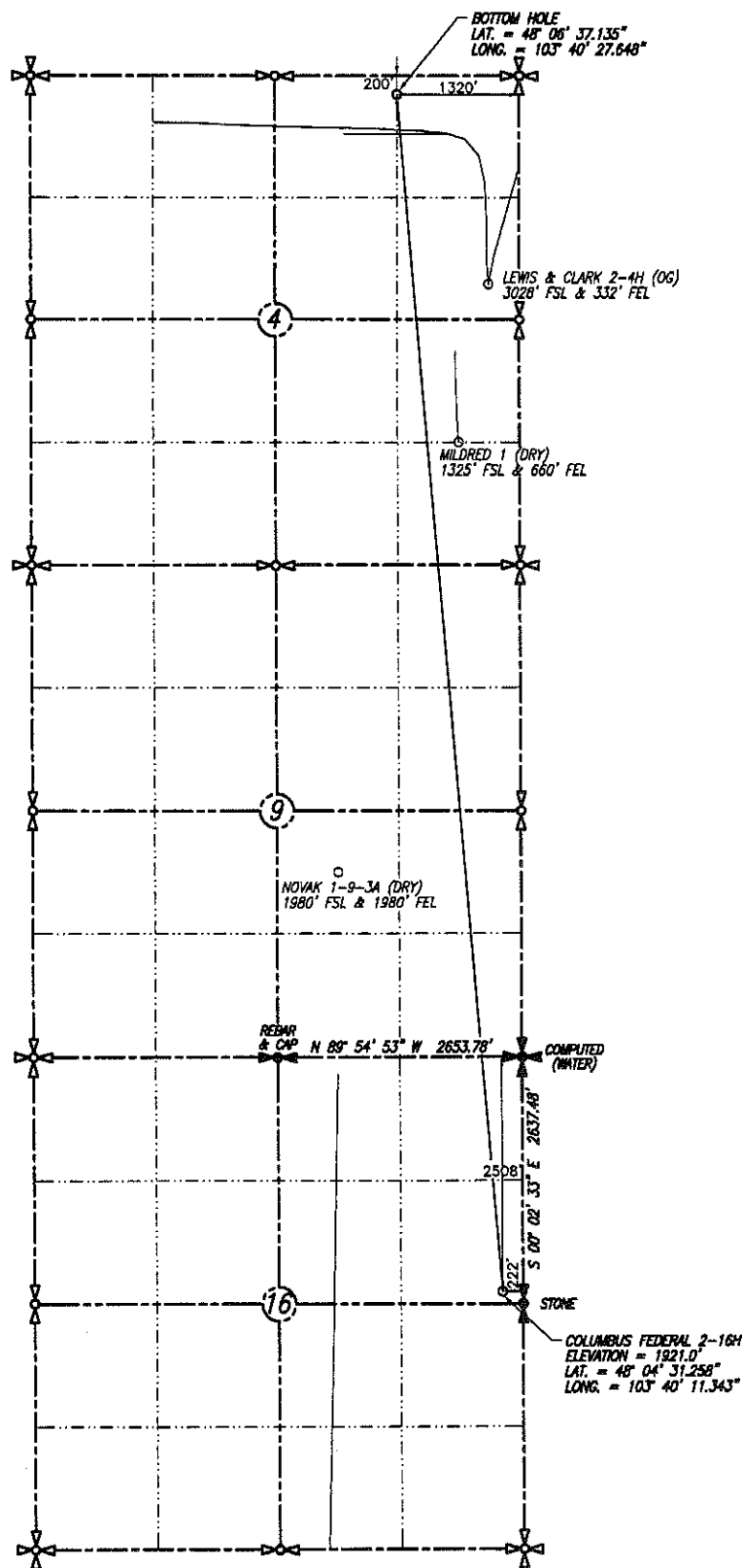
ALL CORNERS SHOWN ON THIS PLAT WERE FOUND IN THE FIELD
DISTANCES TO ALL OTHERS ARE CALCULATED.
ALL BEARINGS SHOWN ARE ASSUMED.

I CERTIFY THAT THIS PLAT CORRECTLY REPRESENTS
WORK PERFORMED BY ME OR UNDER MY RESPONSIBLE
CHARGE, AND IS TRUE AND CORRECT TO THE BEST OF
MY KNOWLEDGE AND BELIEF
JOHN PAULSON, R.L.S. 3366
4-11-12

BROSZ ENGINEERING INC.

BOX 357
BOWMAN, N.D. 58623
PHONE: 701-523-3340
FAX: 701-523-5243
PROJECT NO. 12-10

BOTTOM HOLE LOCATION PLAT
 CONTINENTAL RESOURCES INC.
 COLUMBUS FEDERAL 2-16H
 SECTION 16, T153N, R101W
 MCKENZIE COUNTY, NORTH DAKOTA
 2508' FNL & 222' FEL



I CERTIFY THAT THIS PLAT CORRECTLY REPRESENTS
 WORK PERFORMED BY ME OR UNDER MY RESPONSIBLE
 CHARGE AND IS TRUE AND CORRECT TO THE BEST OF
 MY KNOWLEDGE AND BELIEF

PERSON AUTHORIZING SURVEY;
 CHAD NEWBY

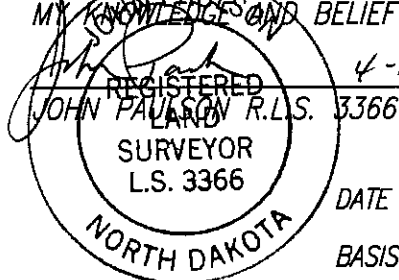
EXPLANATION AREA: NAD83(CORS96)

BASIS OF BEARING: TRUE NORTH

BROSZ ENGINEERING INC.

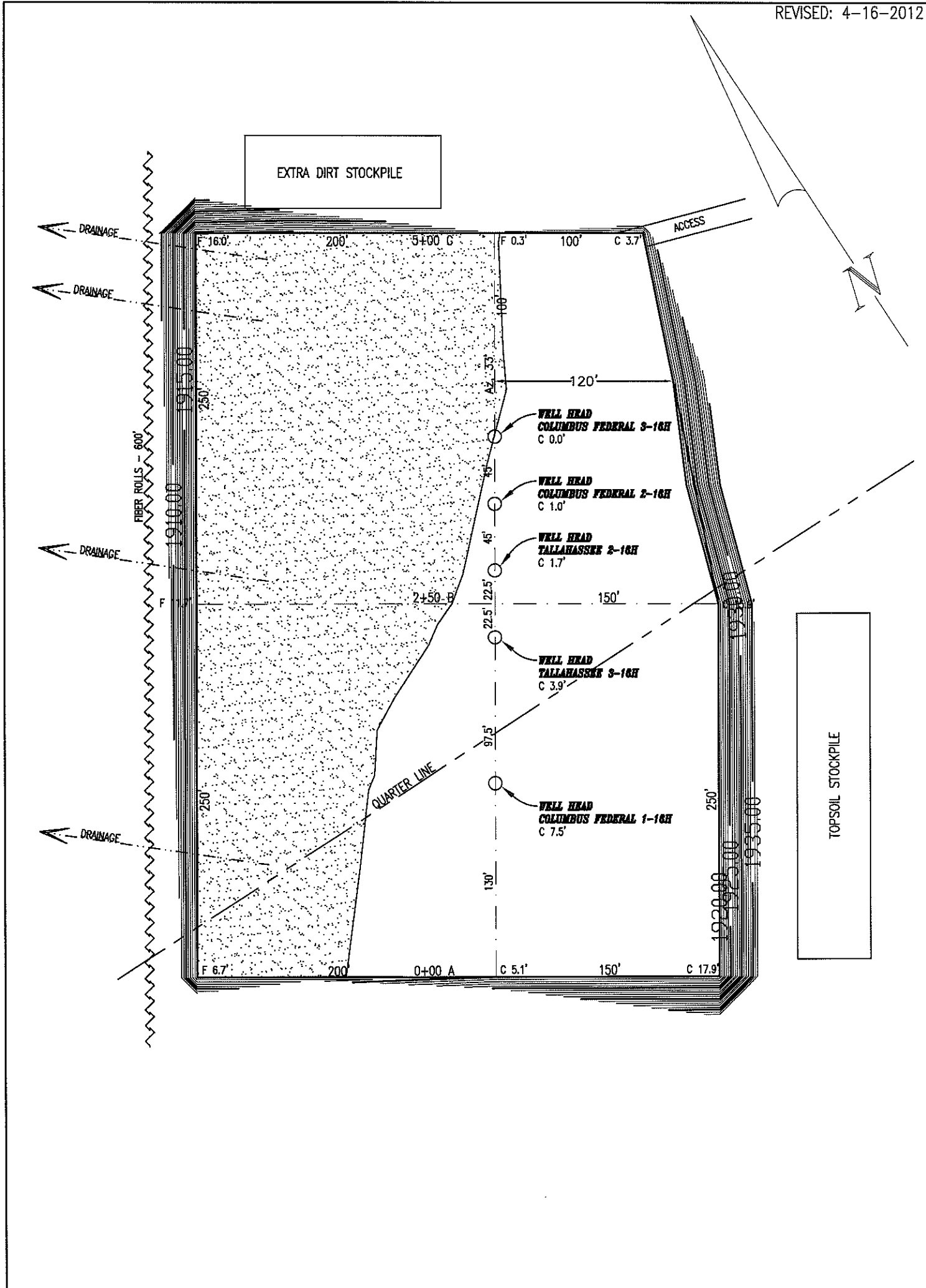
BOX 357
 BOWMAN, N.D. 58623
 PHONE: 701-523-3340
 FAX: 701-523-5243

PROJECT NO. 12-10

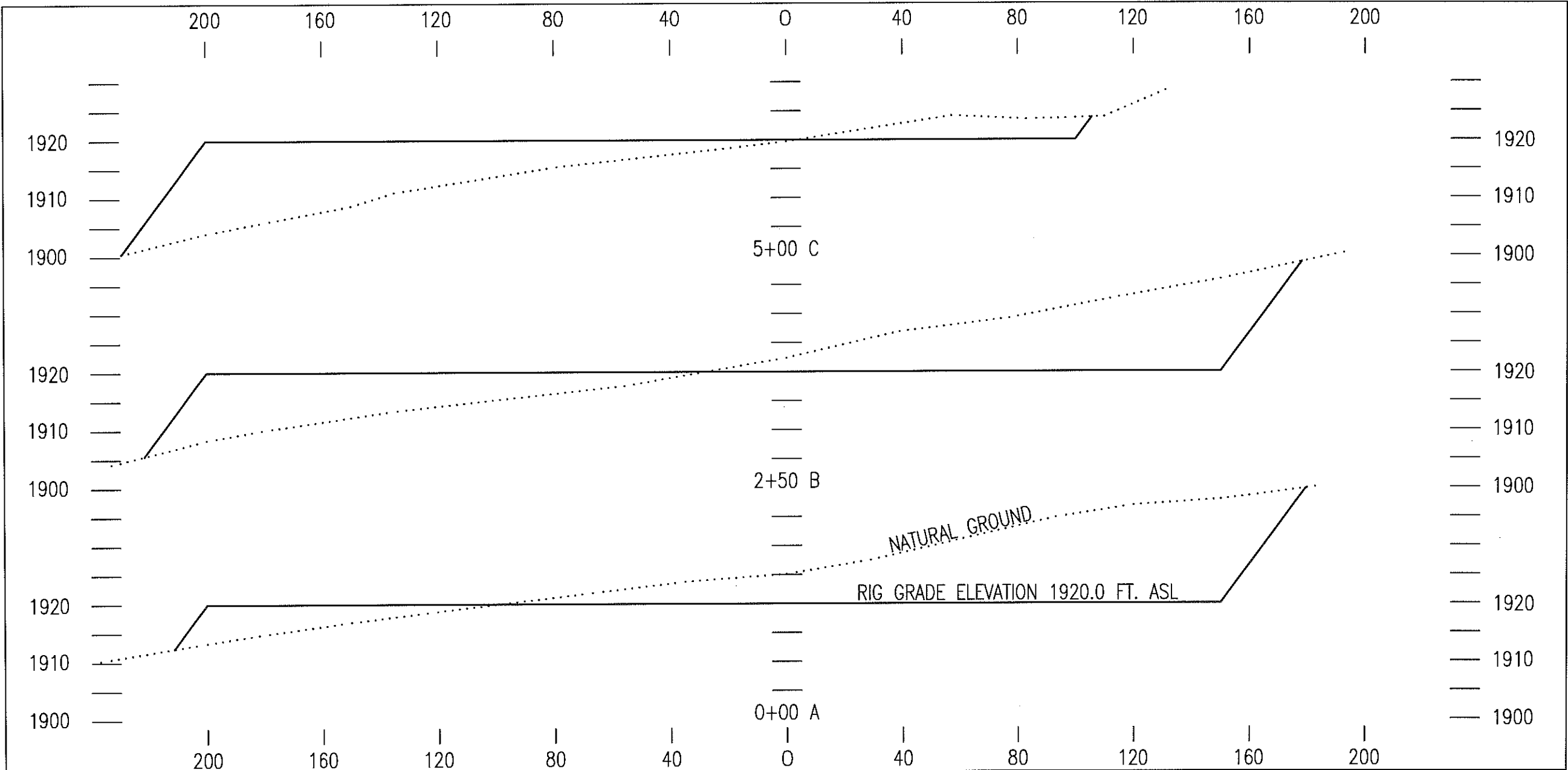


DATE STAKED: 1-4-2012

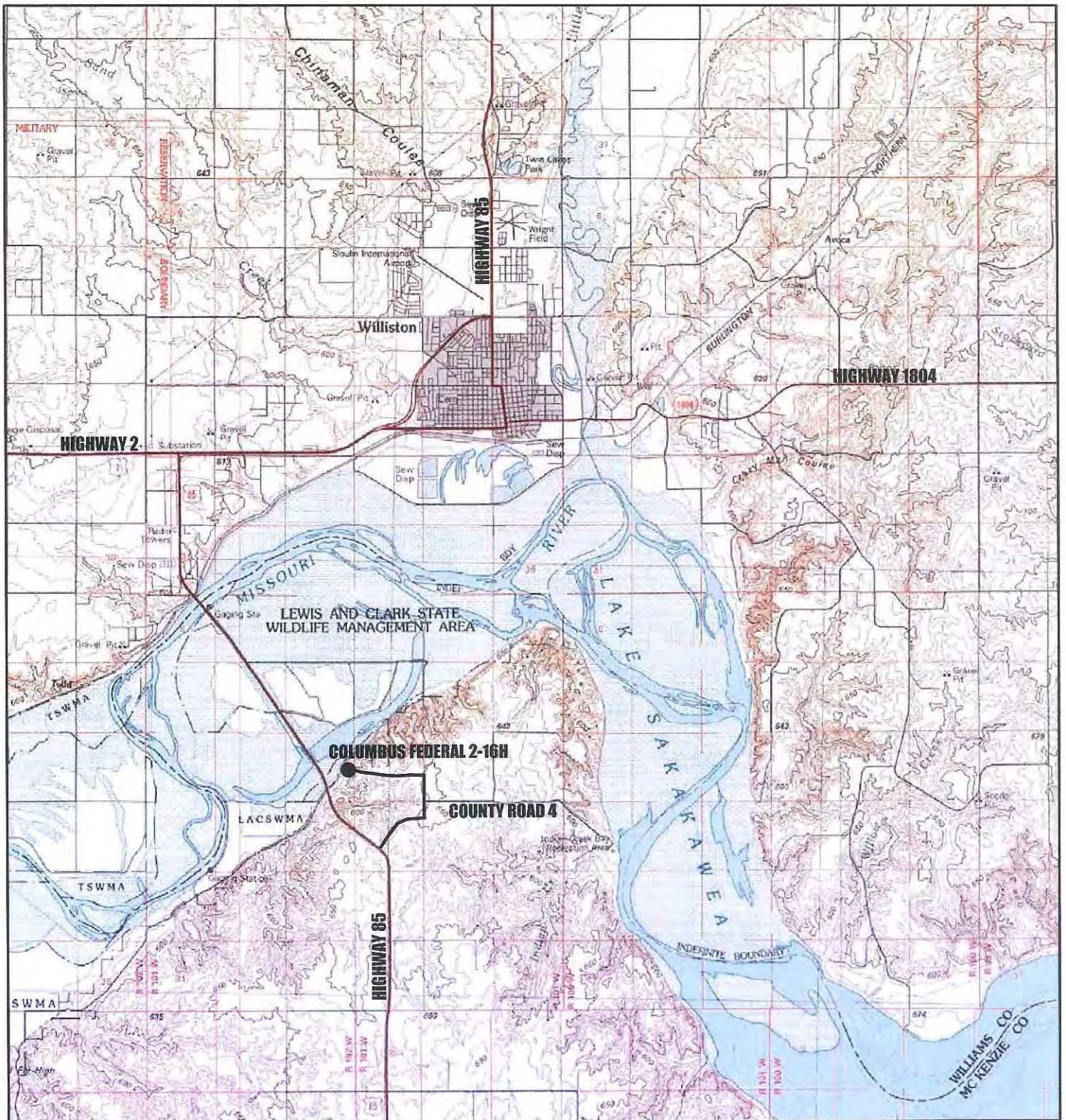
BASIS OF VERTICAL DATUM:
 NAVD 1988 GEOID 09



CONTINENTAL RESOURCES INC. PO BOX 1032 ENID, OKLAHOMA 73702	ESTIMATED EARTH QUANTITIES		ALL INDICATED CUTS & FILLS ARE STAKED GRADE ELEVATIONS.
	TOP-SOIL:	3,125 CUBIC YARDS	
	SUB-SOIL:	28,719 CUBIC YARDS	BACKSLOPES ASSUMED AT 1 1/2 : 1 %
CUT & FILL EXHIBIT 6 COLUMBUS FEDERAL 2-16H SECTION 16, T153N, R101W MCKENZIE COUNTY, NORTH DAKOTA	TOTAL CUT:	31,844 CUBIC YARDS	
	TOTAL FILL:	27,716 CUBIC YARDS	
	Use excess materials in access road fill		
Ground Elevation at Well Head:		1921.0 ft. ASL	
Finished Rig Grade Elevation:		1920.0 ft. ASL	
DRAFT: HEDGE	SCALE 1" = 80'	DATE: 1-4-2012	PROJECT NO. 12-10



CONTINENTAL RESOURCES INC. PO BOX 1032 ENID, OKLAHOMA 73702	DRILL PAD PROFILE EXHIBIT 7 COLUMBUS FEDERAL 2-16H SECTION 16, T153N, R101W MCKENZIE COUNTY, NORTH DAKOTA	DRAWN BY: JH	PROJ. NO. 12-10	ESTIMATED QUANTITIES		
		SCALE: Horz. 1" = 40' Vert: 1" =20'				
		BROSZ ENGINEERING INC. BOX 357 BOWMAN, N.D. 58623 PHONE: 701-523-3340		TOP-SOIL:	3,125	CUBIC YARDS
		SUB-SOIL:	28,719	CUBIC YARDS		
		TOTAL CUT:	31,844	CUBIC YARDS		
		TOTAL FILL:	27,716	CUBIC YARDS		



CONTINENTAL RESOURCES INC.
EXHIBIT 1
VICINITY MAP

COLUMBUS FEDERAL 2-16H
SECTION 16, T153N, R101W
MCKENZIE COUNTY, NORTH DAKOTA

Tabor, David

From: Tabor, David
Sent: Wednesday, March 06, 2013 2:25 PM
To: Tabor, David
Subject: FW: Columbus Federal 3-16H and pad.

From: Robert Sandbo [<mailto:Robert.Sandbo@clr.com>]
Sent: Friday, March 01, 2013 4:48 PM
To: Tabor, David
Subject: RE: Columbus Federal 3-16H and pad.

Thanks David. I got the email. Here is our schedule for the pad as of right now:

Columbus Federal 1-16H to spud on 3/24/2013 (may spud with the small rig around one week earlier if possible). We had planned on drilling the Columbus 1-16H first and then come back in a year or so and drill the other 4 but if we get the permits for all and the order on the 1280 allows us to drill them all back to back, we will most likely go ahead and drill them all at once. We will take a look at the 1280 order that covers the Columbus 1-16H and see if we can batch drill them.

You should have everything on the 3-16H and the affidavit sometime Monday.

Thanks for the help,

Bob Sandbo
Regulatory Compliance Supervisor

Continental Resources, Inc.
20 N. Broadway
OKC, OK 73102
P: 405-234-9020
F: 405-234-9562
C: 405-708-0691
robert.sandbo@clr.com
www.clr.com



March 7, 2013

Industrial Commission of North Dakota
Oil & Gas Division
600 East Boulevard, Dept 405
Bismarck, North Dakota 58505

Continental Resources, Inc. (CRI) respectfully submits the following information concerning the drilling of the Columbus Federal 1-16H, 2-16H & 3-16H1, and the Tallahassee 2-21H & 3-21H1

Township 153N, Range 101W

Section 16, NE/4 SE/4 & SE/4 NE/4 McKenzie County, North Dakota.

The Columbus Federal and Tallahassee well(s) are located in an environmentally sensitive area in close proximity to the Missouri River. Therefore, Continental Resources Inc. would like to propose the following automatic shut down equipment and level sensing monitoring equipment be deployed on the site to aid in the prevention of any accidental release or safety issue. A schematic diagram and pictures of an existing tank battery with emergency shutdown equipment has been included with this affidavit.

- 1) Tank Side (oil & water tanks)
 - a. ABB Levelmaster dual float
 - b. High level switch as backup
 - c. Battery box with solar backup
- 2) Treater & Separator
 - a. Temperature device in each vessel
 - b. Pressure transducer in each vessel
 - c. Level switch in each vessel
 - d. Battery box with solar backup
- 3) Wellhead
 - a. TotalFlow Controller
 - b. Emergency Shutdown Valve Package
 - c. Battery box with solar backup
 - d. Pressure transducer on casing & tubing
- 4) Flare
 - a. Scrubber pot high level switch
 - b. Monitoring flare for low temp output – alarm if flare goes out

- 5) System Automation through the proposed equipment will provide an independent control system on all equipment on site which will be able to shut the well(s) in should any of the other equipment be incapacitated or functioning improperly.

Don Kennedy
Don Kennedy, Sr. Production Engineer
Continental Resources, Inc.

[illegible]

On the 7th day of March, 2013, before me, a Notary Public in and for said County and State, personally appeared **Don Kennedy**, known to me to be the Sr. Production Engineer of Continental Resources, Inc., the Corporation that executed the within instrument, and acknowledged to me that such Corporation executed the same.

Betsy Barnes
Notary Public

Oklahoma County, Oklahoma

My Commission Expires: 7/5/2015

Commission No.: 11006023

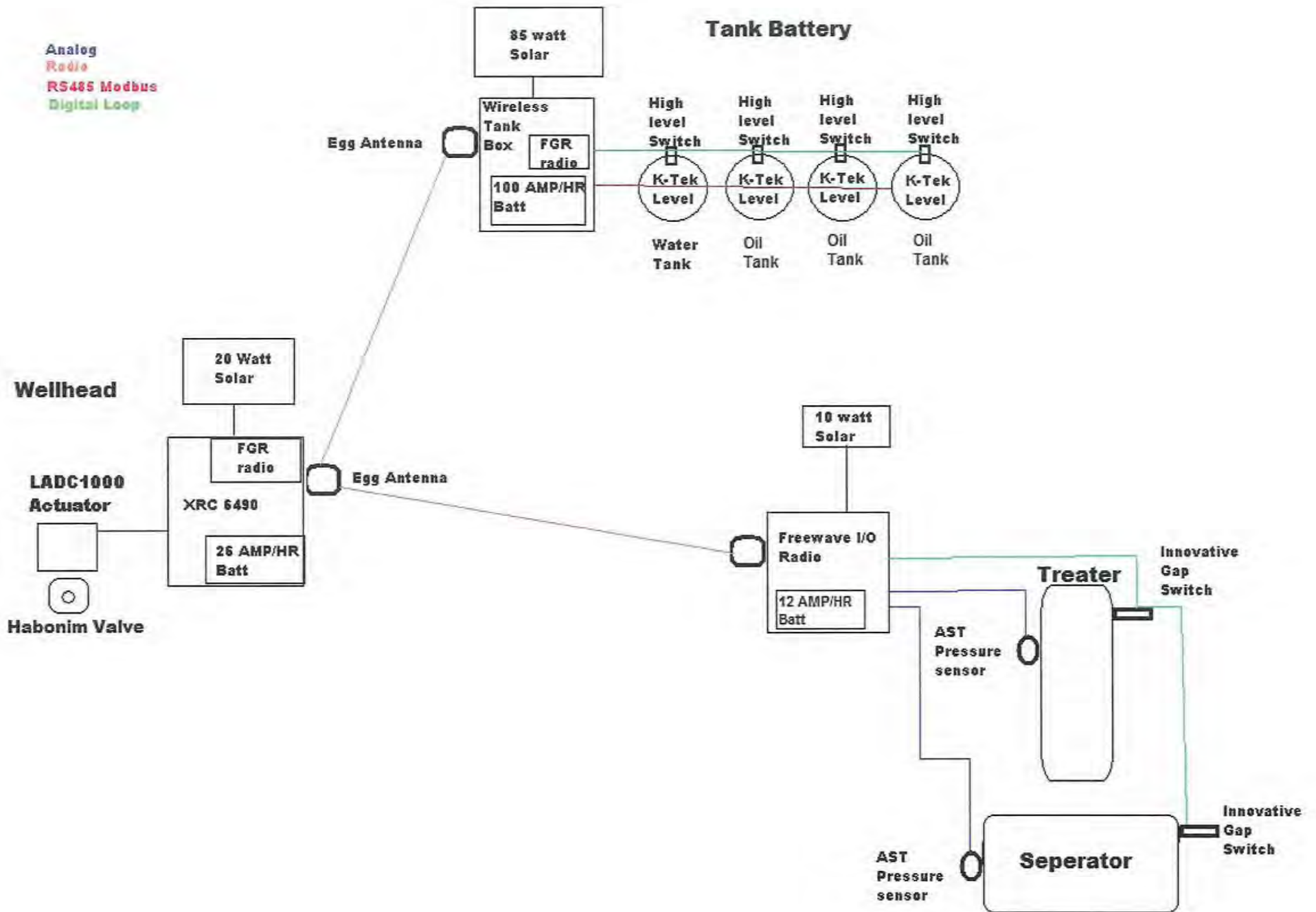


Sincerely,

CONTINENTAL RESOURCES, INC.

Becky Barnes
Regulatory Compliance Specialist

Analog
Radio
RS485 Modbus
Digital Loop





U003 Gap Switch and AST 4600 Transducer monitor pressures and liquid content of Flare and Sales Lines, transmitted to XRC via FreeWave Radio.

ABB TotalFlow XRC 6490

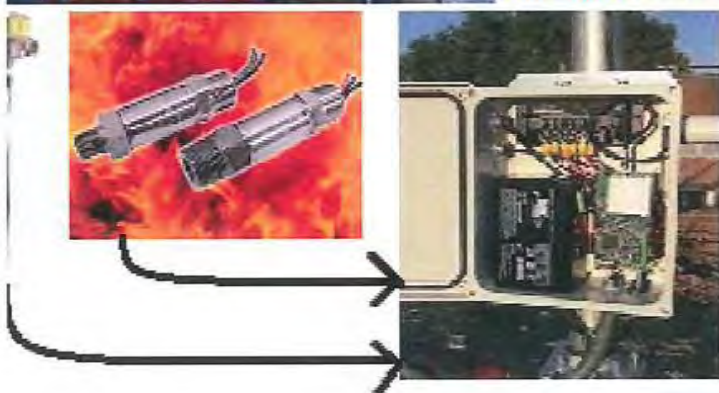


All well information is passed to your SCADA system via FreeWave Network (Future)

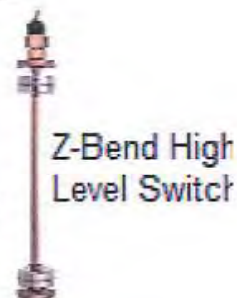
Winn-Marion's Well Head Kit with FGRIO Radio and Power Supply



Tank Level information is passed from Tank Battery to XRC via FreeWave Radio



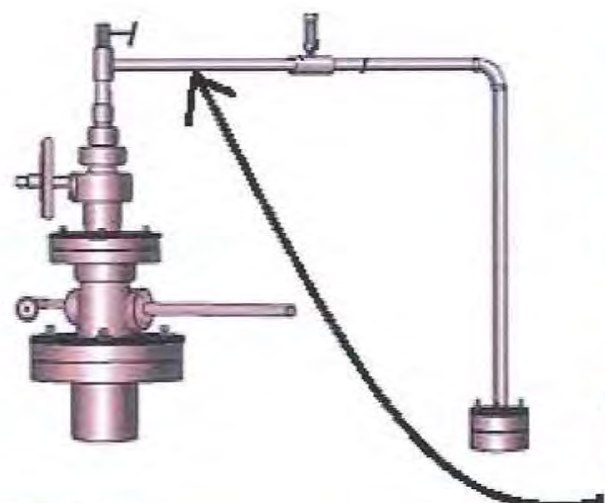
K-Tek MT5100 Guided Wave Radar



Z-Bend High Level Switch



ESD Valve Package
Standard Port Ball Valve
rated to 6000 psi topped
with a 12 VDC Actuator
w/ Battery Backup



Winn-Marion, Inc.

BECKY BARNES
NOTARY
11006023
EXP. 7/5/2015
PUBLIC
STATE OF OKLAHOMA

CLR Spill Trailer Inventory

(To be Checked After Each Use)

Supplies	Quantity	Actual	Supplies	Quantity	Actual
Personal Protection			Miscellaneous		
Trauma/1st Aid Kit	1		EnviroClean (5-gal units of concentrate)	2	
Eye Wash	1		Duct Tape (Case)	3	
Hand Cleaner	2		Flashlights	6	
Nitrile Gloves (L & XL Case)	2		Flood Lights	2	
FRC Rain Coat - Extra Large	3		Extension Cord 50' 12-gauge	5	
FRC Rain Coat - Large	3		55-gal. Drums w/lids	2	
Rubber Safety Toed Boots - Size 10	2		Large Trash Cans	2	
Rubber Safety Toed Boots - Size 11	2		HD Drum Liners - boxes	2	
Rubber Safety Toed Boots - Size 12	2		Hoses - Kit (Blue & Green)	5	
FRC Tyvex Suits - Case XL	1		Plastic Buckets	5	
Neoprene Chest Waders - L	1		Propane Cylinders - 20-lb.	2	
Neoprene Chest Waders - L	1		Propane Weed Burner W/Hose	1	
Containment			Pump - Trash	2	
Absorbent (sphag)	10		Pump - 115V Water Transfer	2	
Absorbent Boom 3" x 10'	2		Gas Powered Generator (3-5K Watt)	1	
Absorbent Boom 5" x 10'	10		Misc. Ratchet Straps	6	
Absorbent Boom 8" x 10'	8		Rope 1/2" x 100'	2	
Containment Boom - Fast Water	3		Rope 1/4" x 50'	4	
Absorbent Pads (Hydrocarbon)	10		Rope 3/8" x 100'	2	
Absorbent Pads (Universal)	5		Shop Towels - box	2	
Absorbent Pillows 18" x 18" box	3		Caulking Gun	2	
Absorbent Pom Pom Cube	7		Silicon Tubes	10	
Absorbent Sweep - 16" x 100' - Bag	5		Metal Stakes/Spikes	8	
Miscellaneous			Metal T-Posts	6	
Antifreeze	2		Bungee Cords	3	
Push Broom	2		Wire - 25' roll - smooth	1	
Shovels	2		Fire Extinguisher	1	
Rake	5		Equipment Hooks	6	
Squeegees	2		Shelving	4	
Scoop	2		Drawers	1	
Spark Resistant Scoop	1		Misc. Building Supplies	1	

Continental Resources, Inc. (CRI) respectfully submits the following information concerning the drilling of the Columbus Federal 1-16H, 2-16H and 3-16H and the Tallahassee 2-16H and 3-16H, Sec. 16, T153N, R101W, Williams County, North Dakota.

CRI would like to propose the following safeguards and precautions to be taken while drilling the Columbus Federal-Tallahassee wells to prevent any contamination to freshwater sources during the drilling and completion of the wells:

- 1) During construction of the location, the entire location will be constructed per NDIC permit stipulations, ensuring any spills or runoff which occur on location do not penetrate the fresh ground water and are contained on the surface of the location. These modifications include, but are not limited to, the inclusion of a liner being placed under the location, and cementing of the rathole and mousehole.
- 2) Drainage will be re-routed to avoid the location, and erosion controls will be employed, as appropriate, around the site to reduce erosion and the resulting sediment contamination to freshwater runoff due to weather events.
- 3) The earthen berm, constructed to keep any freshwater runoff off the location, will also eliminate any spills from leaving the location
- 4) No reserve pit or dry cuttings pit will be utilized on location.
- 5) The conductor will be drilled to a depth of 80', and 20" pipe will be run to depth and cemented to surface.
- 6) During drilling operations, a freshwater protection string of 13-3/8" 48# H40 casing will be set to a depth of 500' and cemented to surface to protect the shallow freshwater zones. Standard 9-5/8" 36# J-55 surface casing will be set 100' into the Pierre Shale, to a depth of 1940', and cemented to surface.
- 7) A frac string will be used to protect the intermediate casing during hydraulic fracturing of the well.
- 8) CRI is submitting a comprehensive rig specific Spill Prevention Containment and Countermeasure Plan to prepare for any event which may occur during drilling and completion operations. A spill trailer will be located on location for spill response, if necessary.

CRI believes adequate planning and precautions are being taken to prevent any contamination to ground water and surface waters.


Sarah Madden, Project Development Engineer
Continental Resources, Inc.

STATE OF OKLAHOMA)
)ss:
COUNTY OF OKLAHOMA)

On the 7th day of March 2013, before me, a Notary Public in and for said County and State, personally appeared **Sarah Madden**, known to me to be a Project Development Engineer of Continental Resources, Inc., the Corporation that executed the within instrument, and acknowledged to me that such Corporation executed the same.


Notary Public

Oklahoma County, Oklahoma

My Commission Expires: 7/5/2015

Commission No.: 11006023



**SPILL PREVENTION CONTAINMENT
AND COUNTERMEASURE PLAN**

FOR

**CYCLONE DRILLING, INC.
P.O. BOX 908
GILLETTE, WYOMING 82717-908**

PREPARED BY;

**TOP LINE ENGINEERING, LLC
12635 HWY 200, P.O. BOX 884
SIDNEY, MONTANA 59270
PH; (701)570-2844**

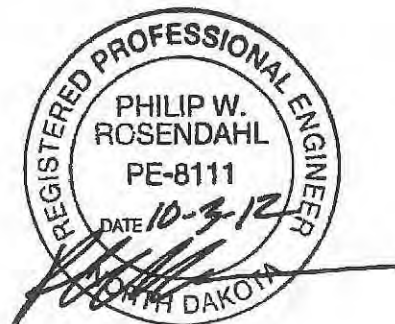


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APPENDIX

TYPICAL DRILL RIG LOCATION PLAN

DISCHARGE REPORTING FORM

DISCHARGE SOURCE INSPECTION RECORD

SPILL PREVENTION CONTAINMENT AND COUNTERMEASURE PLAN (SPCC)

PER 40 CFR 112.7 DATED: SEPTEMBER 14, 2012

This plan will follow the guidelines and format of 40 CFR part 112.7 and will replace the self-certified plan currently in affect.

112.7(a)(1) Discussion of facilities' conformance with the requirements listed in this part. The storage tanks on site store petroleum based drilling mud, diesel fuel, the invert tanks being the base for the drilling mud, salt water tanks, fresh water tanks, crude oil storage tanks and empty tanks for emergency storage. In addition to the storage tanks, other operations that can produce contaminants are the drilling operation itself and the cuttings drying operation. It is the purpose of this plan to define the prevention containment and countermeasure plan for discharges from any of these sources on the site. All the actions taken in this regard are per approved policies of this part of 40 CFR 112. Universally and in setting up the drilling operation on the site prevention measures are taken by grading the site to contain discharges, building in effective valves and other related operational and control mechanisms to prevent discharges. This equipment is installed and tested prior to beginning drilling operations. The tanks and equipment are installed within secondary containment berms and containment trenches and sumps are installed around the facilities prior to beginning drilling operations with capacity to contain with freeboard a rupture of the largest vessel associated with the containment. In the event of a discharge the sites are equipped with vacuums that are capable of picking up the spilled material and loading it into containers where it can be transferred to an approved disposal location. In the event of a discharge larger than can be handled by Cyclone and other manpower on location or that leaks offsite and potentially into navigable waters of the US, the operator, under his directions will arrange for trained contract personnel to immediately contain and clean-up the discharge.

112.7(a)(2) This plan will comply with all applicable items of this part. With no deviations anticipated.

112.7(a)(3) The attached sketch shows a typical layout of the drill rig sites depicting the location, size and contents of the tanks. There are no underground tanks nor are there any mobile or portable containers associated with these drilling locations. The site is prepared in advance of the drill rig coming onto the site. The sites are designed and constructed to contain spills and tank ruptures on the site. The drill rig and associated tanks and equipment are laid out in an efficient and effective manner to streamline the drilling operation. Frack tanks are located in such a manner that they have easy access to the mud tanks with valves located on both ends of transfer lines. The drilling mud tanks are plumbed to the drill rig with valves appropriately located to

stop flow from the tanks to the rig or from the rig to the tanks. The cuttings from the drilling operation are screened from the recycled drilling mud and directed to a container where they are dried using fly ash or some other equally effective absorbent material and are then transferred to the lined cuttings disposal pit onsite. The fuel tank is tied into the fuel lines providing fuel flow to the generators. Blowout preventers are placed over the drill hole casing and tested prior to beginning the drilling.

112.7(a)(3)(i) The type of oil in each fixed container and its storage capacity. No mobile or portable containers anticipated.

SOURCE	MAJOR TYPE OF FAILURE	TOTAL CAP. (BBLs.) (Type Fluid)	RATE (BBLs/MIN)	DIRECTION OF FLOW	SECONDARY CONTAINMENT
Diesel Fuel Tank	Leak or rupture	14,000 Maximum (Diesel)	Depends on size	Dams and/or Trenches	Site Perimeter Containment
Pumps & Engines	Leakage	½ - gal/hr. (lube oil)	½ - 1 gal/hr.	Drip Pans or Trenches	Site Perimeter Containment
Crude Oil Tanks	Rupture or Leak	400 BBLs	Depends on Type of Failure	Contained within Berm	Tank Battery Berm
Mud Tanks, Vibrator Rotary Hoses	Leak or rupture	450 BBLs Maximum per tank (Petroleum based drilling fluid)	Varies	Drip Pans for Hoses, Trench & Sump or Berms for Tanks	Site Perimeter Containment
Wellhead Blowout Preventer	Blowout or kicking well	Depends on severity (Well bore fluids)	Will vary	Away from well head to Trench and Sump	Site Perimeter Containment

112.7(a)(3)(ii) Discharge Prevention Measures including procedures for routine handling of products (loading, unloading and facility transfers, etc.); Before the drill rig moves on, the site is prepared for discharges associated with tank ruptures, as the site is graded to contain all discharges and storm runoff. When the rig is moved onto the site and set up additional discharge prevention and containment measures are taken. Equipment inspection records

are reviewed and the equipment is inspected following written procedures, the written procedures are attached in the appendix of the plan. A dated record of the inspection, signed by the appropriate supervisor or inspector is recorded and filed with time sheets or safety meeting minutes, when the drill rig changes locations and notes of deficiencies or acceptance are included. Operation and maintenance of equipment is designed to prevent discharges. Personnel are trained and made aware of the discharge prevention procedures and applicable pollution control laws, rules and regulations. The tool pusher in charge of the drill rig will have in his possession a copy of the Cyclone Drilling SPCC Plan and it is his responsibility to instruct the employees and make them aware of the discharge prevention measures contained in the plan. Scheduled briefings and refreshers are to be conducted with all the employees prior to and during rig up operations, such briefings are to be held during regularly scheduled safety meetings.

Employees are assigned to make routine inspections of valves, hose connections and other fluid connections for leaks. These inspections are made and documented on a monthly basis at a minimum. These leaks are provided drip pans and reported to the appropriate personnel for repair. If repairs cannot be readily made then regular emptying of the drip pans is required until the repairs can be made. Inspections of trenches and sumps are made on a regular basis to insure that they are free flowing and functional. The containment berms are also inspected periodically to insure their stability and function.

112.7(a)(3)(iii) Discharge or drainage controls such as secondary containment around containers and other structures, equipment, and procedures for the control of the discharge; Containment berms are placed around the tanks containing crude oil, the invert drilling fluid, salt water tanks and the empty tanks to be used for emergency storage, as secondary containment. The containment berms are of adequate height to contain a rupture and total failure of the largest tank contained within it amounting to 400 BBLS requiring a containment capacity of 3,000 cu. ft. Similarly, berms are built around the crude oil storage tanks on the site and requiring the same containment volume. Berms or trenches and sumps are constructed around the drill rig itself, the drilling mud tanks and diesel fuel tank, the sumps and trenches are sized to contain the volume of the largest tank in the case of a rupture and complete failure. Additionally, drip pans are placed under leaking valves, hose joints and other sources of small leaks until the item can be replaced or repaired to not leak. Regular inspections of these facilities are made to insure that they will function per plan.

112.7(a)(3)(iv) Countermeasures for discharge discovery, response, and cleanup (both the facility's capability and those that might be required of a contractor); Regular inspections may reveal a discharge such as a leaking valve or hose joint. These instances can readily be cleaned up by employees of Cyclone Drilling using on site vacuums and containers. Larger spills up to and

beyond the reporting limits can also be handled by Cyclone Drilling employees. Spills that exceed the limitations of the onsite cleanup equipment or in the event of a spill migrating offsite, will require the tool pushers' notification of the operator/owner of the site and they will contact the clean-up contractor they have contracted with to contain and clean-up such events.

112.7(a)(3)(v) Methods of disposal of recovered materials in accordance with applicable legal requirements: When spills occur the fluids will be either soaked up with absorptive material and placed in a container for transporting to an approved disposal site or vacuumed up and placed in a storage container for transportation to an approved disposal site. It is understood by all Cyclone Drilling employees that there is not an approved disposal site on or around the drilling rig to dispose of these recovered materials or fluids, other than drying drilling cuttings and disposing of them in the cuttings pit.

112.7(a)(3)(vi) Contact list and phone numbers for the facility response coordinator, National Response Center, cleanup contractors with whom operator has an agreement for response, and all appropriate Federal, State, and local agencies who must be contacted in the event of a discharge as described in § 112.1(b). Facility Response Coordinator for Cyclone Drilling HSE Dept. (Ph. 307-682-4161) is responsible for contacting the operators environmental group so they can contact the appropriate Federal, State and Local personnel in the event of a discharge as described in § 112.1(b).

112.7(a)(4) Information and procedures to enable a person reporting a discharge as described in § 112.1(b) to relate information on the exact address or location and phone number of the facility; the date and time of the discharge, the type of material discharged; estimates of the total quantity discharged; estimates of the quantity discharged as described in § 112.1(b); the source of the discharge: a description of all affected media; the cause of the discharge; any damages or injuries caused by the discharge; actions being used to stop, remove, and mitigate the effects of the discharge; whether an evacuation may be needed; and, the names of individuals and/or organizations who have also been contacted. In the event of a discharge as described in §112.1(b) it is the responsibility of the tool pusher to relate the above information to the operators environmental group so they can contact, any emergency organizations and provide the reporting necessary to the Federal, State and Local agencies that require reporting of such a discharge. The tool pusher on the rig is the person in charge and either he or the drilling superintendent have the responsibility of reporting the answers to the above questions. These questions are spelled out on a "Discharge Reporting Form" that each has at his disposal in the appendix of "Cyclone Drilling Oil Spill Contingency Plan" In reporting a discharge a list of all individuals and or organizations that received the report will be listed.

112.7(a)(5) Plan procedures to be used when a discharge occurs:

- 1) Immediately extinguish any heater or fire that may ignite the spill.
- 2) No smoking during spill control operations.
- 3) Close all associated valves.
- 4) Direct discharge to ditches or drains that will carry the discharge to a safe holding sump or reserve pit.
- 5) Distribute hill, fibertex, gel, barite, or any other absorptive material available as required to contain the discharge not entering the ditch, sump or reserve pit.
- 6) Inspect area to ensure that all of the discharge is contained in ditches, sumps or reserve pits. Add ditches or diversion structures as required to contain the discharge onsite.
- 7) Start jet or sump pumps and transfer discharged material from sumps to reserve pit or holding tanks.
- 8) After discharge is stopped, collect all used hulls, fibertex and similar absorptive material for disposal per instruction from the drilling superintendent or the tool pusher. No oil is to remain in the ditches or sumps that may create a fire hazard.
- 9) Hold collected discharged material for hauling and disposal in an approved location.
- 10) If discharge migrates from the drill rig location Cyclone Drilling personnel will use these same procedures stop and recover the discharge. Additionally the tool pusher will notify the owners representative to notify the contract clean-up company for clean-up and mitigation of the offsite discharge migration.
- 11) Fill out Site Discharge Reporting Form and distribute to appropriate agencies and personnel. (Form included in Appendix of Plan)

112.7(a)(5)(b) Flow diagram of potential discharges from significant sources as a result of facilities failure: Flow diagrams and quantities included in appendix of the plan.

112.7(a)(5)(c) Site discharge containment system: The site discharge containment system for the Cyclone Drilling Rigs is a combination of containment berms, confinement trenches, sumps and reserve pits. Drip pans will be used for small discharges until repairs are made or equipment replaced. (See typical site plan included in appendix.)

112.7(a)(5)(d) Not applicable as all measures spelled out in this part are practical as describe in each section.

112.7(a)(5)(e) Inspections, tests and records: Written procedures are provided on the "Discharge Source Inspection Record", these inspections are recorded at least on a monthly basis or whenever a leak or discharge is detected and reported, using the "Discharge Source Inspection Record" in the appendix. These records are signed by the tool pusher onsite and maintained by Cyclone Drilling for a period of 3 years.

112.7(a)(5)(f) Personnel Training and Discharge Prevention Procedures:

The personnel working on the drilling crews are trained in the maintenance and operation of all the equipment to prevent discharges, the discharge procedures and general facility operations. They are also made aware of the contents of the SPCC Plan. The tool pusher on each crew is the person on the location who is accountable for discharge prevention and who reports to the Cyclone Drilling Management. The tool pusher is responsible for conducting discharge prevention briefings and assuring an adequate understanding of the Cyclone Drilling SPCC Plan for the facility and any recent developments of new precautionary measures due to failures or malfunctions are implemented.

112.7(a)(5)(g) Site Security: All persons entering the site are required to check in with the tool pusher of the on duty crew. Unauthorized persons are not allowed on the site. Unauthorized visitors are easily detected, questioned regarding their presents and appropriately escorted to take care of their business.

112.7(a)(5)(h) Facility Tank Truck Loading/Unloading: The tank truck loading and unloading areas on the Cyclone Drilling Sites are located adjacent to the frack tank or crude oil tank batteries the area adjacent to the tank battery, where tank trucks are loaded and unloaded will be graded to drain into the containment trenches and associated sump, surrounding the drill rig, mud tanks, fuel tank, etc. Wheel chocks are used to prevent the truck from moving while connected to the tanks during loading/unloading operations. Before departure the vehicle is inspected for leaks in the lowermost drains and outlets and if they are any detected the associated valves are tightened or adjusted to prevent discharge while in transit. All above ground containers are checked and evaluated for risk of discharge or failure and as necessary appropriate action shall be taken.

APPENDIX



June 12, 2012

Industrial Commission of North Dakota
Oil & Gas Division
600 East Boulevard, Dept 405
Bismarck, North Dakota 58505

Re: Columbus Federal 2-16H

Continental Resources, Inc., would like to request all filings and information regarding the above captioned well be considered "Tight Hole".

Please charge the Continental Resources, Inc., credit card that is on file with your agency for the application fee of this well.

Thank you for your prompt attention to this matter. If you have any questions, you may contact me at 580-548-5139 or email the following Terry.Olson@clr.com.

Sincerely,

CONTINENTAL RESOURCES, INC.


Terry L. Olson
Regulatory Compliance Specialist